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(54) **MEDICAMENT FOR TREATMENT AND/OR PREVENTION OF CANCER**

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ABSTRACT

The present invention relates to a medicament for treatment and/or prevention of cancer, comprising an antibody or a fragment thereof having an immunological reactivity with CAPRIN-1 protein, and a MAPK pathway inhibitor together or separately in combination.

Foreign Application Priority Data

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Specification includes a Sequence Listing.

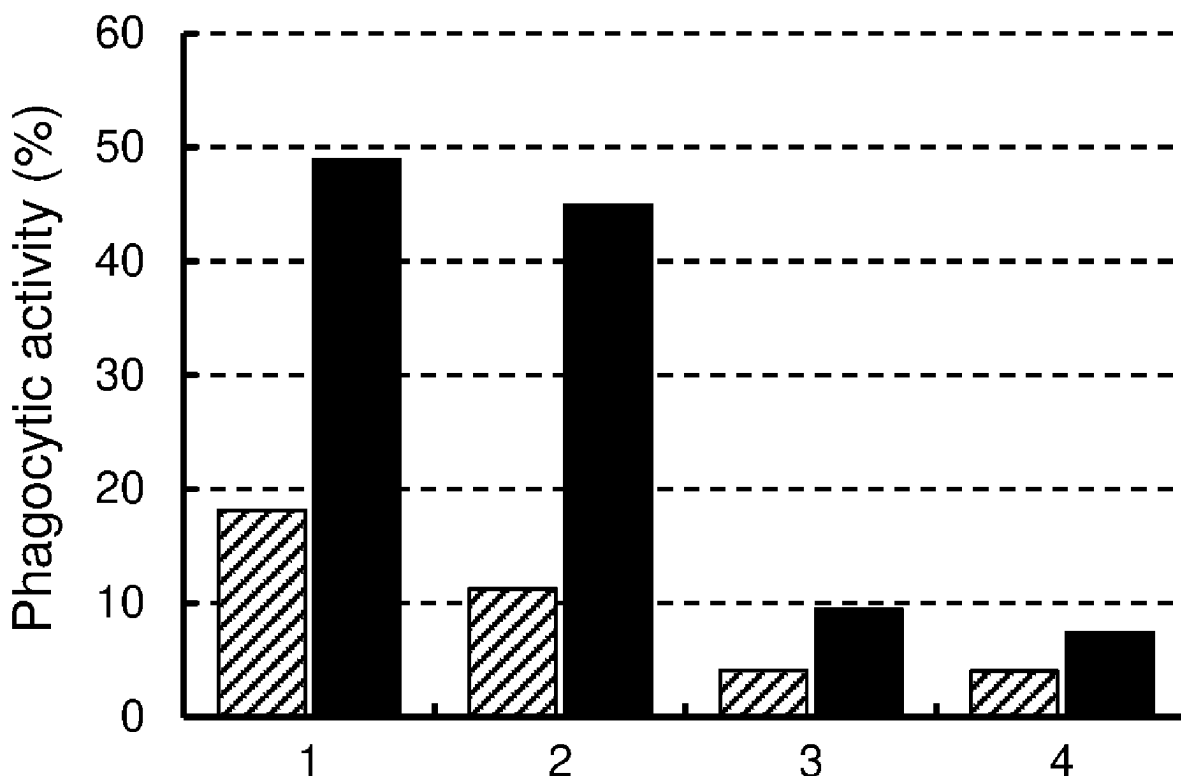


Fig. 1

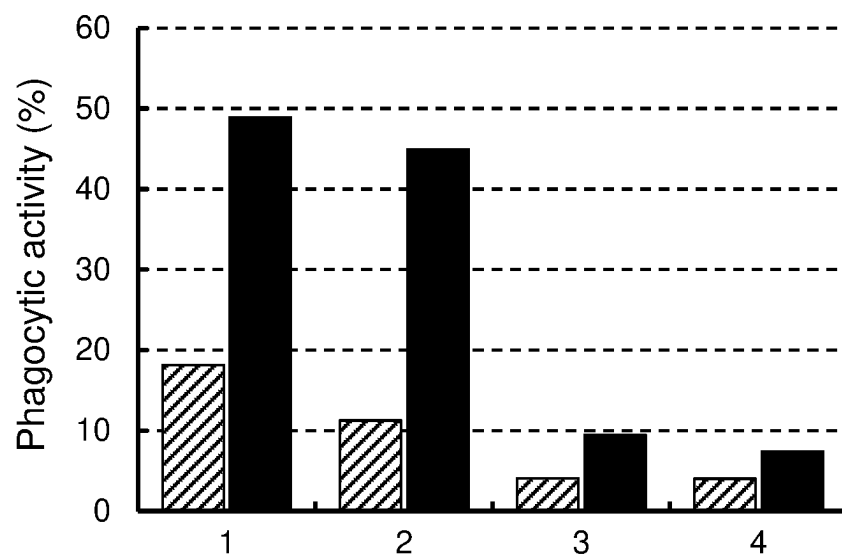


Fig. 2

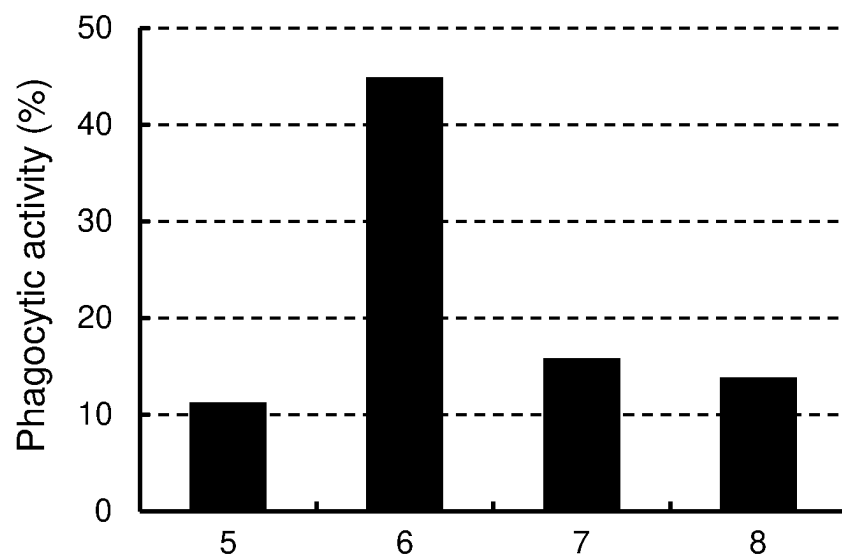


Fig. 3

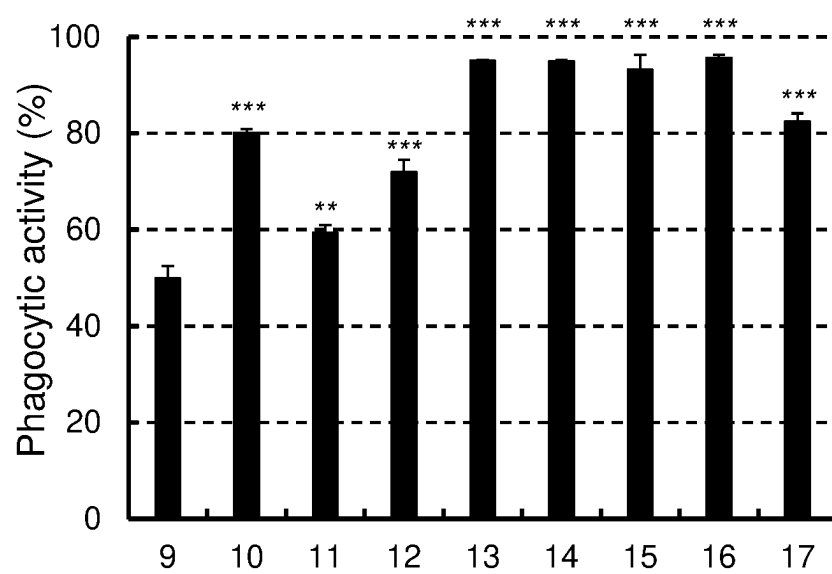


Fig. 4

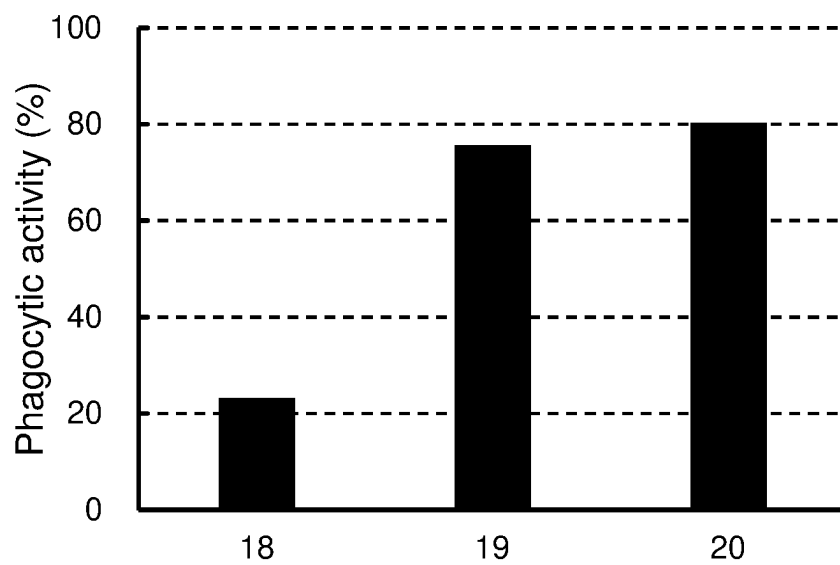
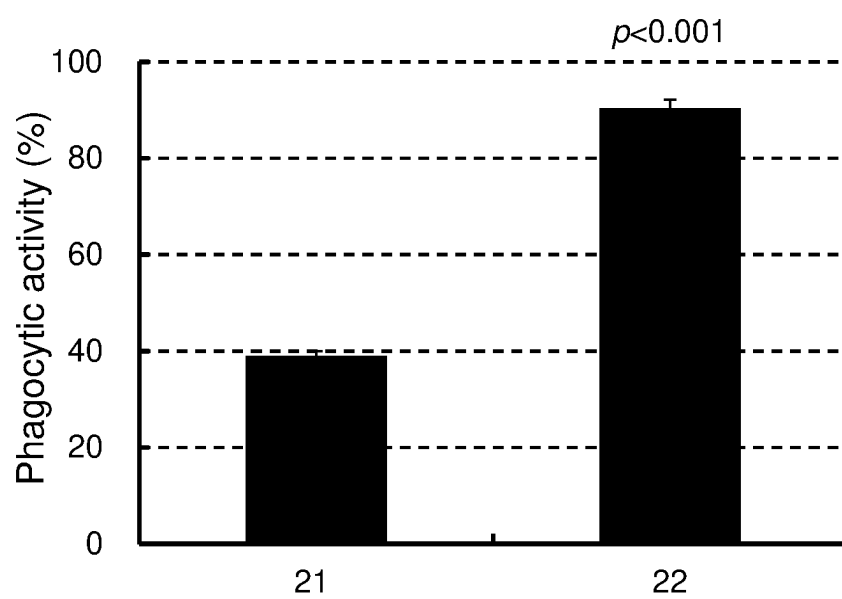


Fig. 5



MEDICAMENT FOR TREATMENT AND/OR PREVENTION OF CANCER

REFERENCE TO ELECTRONIC SEQUENCE LISTING

[0001] The application contains a Sequence Listing which has been submitted electronically in .XML format and is hereby incorporated by reference in its entirety. Said .XML copy, created on Aug. 6, 2024, is named “1254-0689PUS1.xml” and is 388,623 bytes in size. The sequence listing contained in this .XML file is part of the specification and is hereby incorporated by reference herein in its entirety.

TECHNICAL FIELD

[0002] The present invention relates to a medicament for treatment and/or prevention of a cancer, comprising an antibody against CAPRIN-1 protein, or a fragment thereof, and a MAPK pathway inhibitor.

BACKGROUND ART

[0003] Various antibody medicines targeting specific antigen proteins on cancer cells are applied as therapeutic agents for cancers with fewer side effects to cancer treatment because of their cancer specificity. For example, cytoplasmic-activation and proliferation-associated protein 1 (CAPRIN-1) is expressed on cell membrane surfaces of many solid cancers. Antibodies against this CAPRIN-1 protein are known to be promising in pharmaceutical uses for treatment and/or prevention of cancers (Patent Literature 1).

[0004] In recent years, treatment methods using combinations of pluralities of therapeutic agents for cancer have been clinically used as standard treatment methods in order to enhance the effectiveness of the therapeutic agents for cancers. In general, for example, breast cancer is treated by a treatment method using a combination of doxorubicin and cyclophosphamide or using a combination of a plurality of anticancer agents such as paclitaxel, trastuzumab, and pertuzumab. Therapeutic agents for cancers comprising anti-CAPRIN-1 antibodies as active ingredients have also been confirmed to have therapeutic effects on the cancers by combinations with chemotherapeutics (Patent Literature 2). However, treatment of a cancer by a combination of chemotherapeutics is not effective for every cancer to which the treatment is applied, and few combinations of chemotherapeutics synergistically drastically enhance therapeutic effects, though some combinations additively enhance therapeutic effects.

[0005] MAPK (mitogen-activated protein kinase) signaling is a signaling pathway that plays a central role in the control of cell proliferation, differentiation, and survival, and this pathway is constituted by four types of proteins RAS/RAF/MEK/ERK (Non Patent Literature 1). RAS activates RAF, and the activated RAF functions as MAP kinase kinase (MAPKKK or MAP3K). Then, the RAF phosphorylates and activates MAP kinase kinase (MAPKK). This MAP kinase kinase (MAPKK) in this pathway is called MEK (MAPK or ERK kinase). MEK phosphorylates and activates the third and final enzyme in this pathway, i.e., MAP kinase (MAPK) called ERK (extracellular signal-regulated kinase). The ERK, when activated, can be translocated into the nucleus, and the ERK in this nucleus phosphorylates a transcriptional factor, thereby controlling

its activity in an important cell process such as gene control, cell cycle transition or cell differentiation (Non Patent Literature 2).

[0006] MAPK signaling is important for physiological functions of normal cells, whereas its abnormal activation or appearance of mutant proteins are closely related to the onset or progression of cancers and approximately 85% of all cancers reportedly have abnormal MAPK signaling (Non Patent Literature 1). Against this backdrop, the MAPK signaling is regarded as an important target for cancer treatment. For example, an inhibitor targeting MEK located downstream in the MAPK signaling has been put into practical use as an anticancer agent (Non Patent Literature 1). However, a problem of the MEK inhibitor is that cancer cells acquire unresponsiveness to the inhibitor due to further abnormalities in the constituent factors of the MAPK signaling, activation of other signaling pathways that bypass the MAPK signaling, or the like so that the MEK inhibitor alone fails to completely cure the cancer, though high response is exhibited at an early stage from the start of treatment (Non Patent Literature 3). Likewise, although a RAF inhibitor exhibits marked clinical activity in patients with BRAF V600E or BRAF V600K melanoma, the drug efficacy of the RAF inhibitor alone is limited due to the emergence of drug resistance (Non Patent Literature 4).

CITATION LIST

Patent Literature

- [0007]** Patent Literature 1: WO2010/016526
[0008] Patent Literature 2: WO2011/096535

Non Patent Literature

- [0009]** Non Patent Literature 1: J. Hematol. Oncol., 2020, 17; 13(1): 113
[0010] Non Patent Literature 2: Biol Cell., 2001, 93(1-2): 53-62.
[0011] Non Patent Literature 3: Int. J. Mol. Sci., 2020, 21(3): 1102
[0012] Non Patent Literature 4: Nat Med. 2013, 19(11): 1401-9.

SUMMARY OF INVENTION

Object to be Achieved

[0013] An object of the present invention is to provide a medicament for treatment and/or prevention of a cancer specifically expressing CAPRIN-1 protein on a cell surface.

Solution to Achieve Object

[0014] As a result of intensive studies, the present inventors have found that a combination (combined use) of an antibody against CAPRIN-1 protein, or a fragment thereof, having an immunological reactivity with cancer cells, and a MAPK pathway inhibitor exerts a very strong antitumor effect. On the basis of these findings, the present invention has been completed.

[0015] Specifically, the present invention relates to the following embodiments (1) to (18):

[0016] (1) A medicament for treatment and/or prevention of cancer, comprising an antibody or a fragment thereof having an immunological reactivity with CAPRIN-1 pro-

tein, and a MAPK (mitogen-activated protein kinase) pathway inhibitor together or separately in combination.

[0017] (2) The medicament according to (1), wherein the MAPK pathway inhibitor comprises a MEK inhibitor, a RAS inhibitor, a RAF inhibitor, an ERK inhibitor, or a combination of any two or more thereof.

[0018] (3) The medicament according to (2), wherein the MEK inhibitor is at least one selected from the group consisting of trametinib, cobimetinib, selumetinib, refametinib, SL-327, and their derivatives.

[0019] (4) The medicament according to (2), wherein the RAS inhibitor is at least one selected from the group consisting of sotorasib, lonafarnib, salirasib, and their derivatives.

[0020] (5) The medicament according to (2), wherein the RAF inhibitor is at least one selected from the group consisting of dabrafenib, ZM 336372, and their derivatives.

[0021] (6) The medicament according to (2), wherein the ERK inhibitor is at least one selected from the group consisting of bosutinib, SCH772984, and their derivatives.

[0022] (7) The medicament according to any of (1) to (6), wherein the antibody or the fragment has an immunological reactivity with CAPRIN-1 protein having an amino acid sequence shown by any one of the even numbered SEQ ID NOs: 2 to 30, or an amino acid sequence having 80% or more sequence identity with the amino acid sequence.

[0023] (8) The medicament according to any of (1) to (7), wherein the antibody or the fragment thereof has an immunological reactivity with an extracellular region of CAPRIN-1 protein present on a cancer cell surface.

[0024] (9) The medicament according to any of (1) to (8), wherein the antibody or the fragment thereof has an immunological reactivity with a partial polypeptide of CAPRIN-1 protein, the partial polypeptide having an amino acid sequence shown by any one of SEQ ID NOs: 31 to 35, 296 to 299, 308 and 309, or an amino acid sequence having 80% or more sequence identity with the amino acid sequence.

[0025] (10) The medicament according to any of (1) to (9), wherein the antibody is a monoclonal antibody or a polyclonal antibody.

[0026] (11) The medicament according to any of (1) to (10), wherein the antibody or the fragment thereof is any one of the following (A) to (M):

(A) an antibody or a fragment thereof comprising a heavy-chain variable region comprising complementarity determining regions of SEQ ID NOs: 36, 37 and 38 (CDR1, CDR2 and CDR3, respectively) and a light-chain variable region comprising complementarity determining regions of SEQ ID NOs: 40, 41 and 42 (CDR1, CDR2 and CDR3, respectively), and having an immunological reactivity with CAPRIN-1 protein;

(B) an antibody or a fragment thereof comprising a heavy-chain variable region comprising complementarity determining regions of SEQ ID NOs: 44, 45 and 46 (CDR1, CDR2 and CDR3, respectively) and a light-chain variable region comprising complementarity determining regions of SEQ ID NOs: 48, 49 and 50 (CDR1, CDR2 and CDR3, respectively), and having an immunological reactivity with CAPRIN-1 protein;

(C) an antibody or a fragment thereof comprising a heavy-chain variable region comprising complementarity determining regions of SEQ ID NOs: 52, 53 and 54 (CDR1, CDR2 and CDR3, respectively) and a light-chain variable region comprising complementarity determining regions of

SEQ ID NOs: 56, 57 and 58 (CDR1, CDR2 and CDR3, respectively), and having an immunological reactivity with CAPRIN-1 protein;

(D) an antibody or a fragment thereof comprising a heavy-chain variable region comprising complementarity determining regions of SEQ ID NOs: 60, 61 and 62 (CDR1, CDR2 and CDR3, respectively) and a light-chain variable region comprising complementarity determining regions of SEQ ID NOs: 64, 65 and 66 (CDR1, CDR2 and CDR3, respectively), and having an immunological reactivity with CAPRIN-1 protein;

(E) an antibody or a fragment thereof comprising a heavy-chain variable region comprising complementarity determining regions of SEQ ID NOs: 170, 171 and 172 (CDR1, CDR2 and CDR3, respectively) and a light-chain variable region comprising complementarity determining regions of SEQ ID NOs: 173, 174 and 175 (CDR1, CDR2 and CDR3, respectively), and having an immunological reactivity with CAPRIN-1 protein;

(F) an antibody or a fragment thereof comprising a heavy-chain variable region comprising complementarity determining regions of SEQ ID NOs: 176, 177 and 178 (CDR1, CDR2 and CDR3, respectively) and a light-chain variable region comprising complementarity determining regions of SEQ ID NOs: 179, 180 and 181 (CDR1, CDR2 and CDR3, respectively), and having an immunological reactivity with CAPRIN-1 protein;

(G) an antibody or a fragment thereof comprising a heavy-chain variable region comprising complementarity determining regions of SEQ ID NOs: 182, 183 and 184 (CDR1, CDR2 and CDR3, respectively) and a light-chain variable region comprising complementarity determining regions of SEQ ID NOs: 185, 186 and 187 (CDR1, CDR2 and CDR3, respectively), and having an immunological reactivity with CAPRIN-1 protein;

(H) an antibody or a fragment thereof comprising a heavy-chain variable region comprising complementarity determining regions of SEQ ID NOs: 188, 189 and 190 (CDR1, CDR2 and CDR3, respectively) and a light-chain variable region comprising complementarity determining regions of SEQ ID NOs: 191, 192 and 193 (CDR1, CDR2 and CDR3, respectively), and having an immunological reactivity with CAPRIN-1 protein;

(I) an antibody or a fragment thereof comprising a heavy-chain variable region comprising complementarity determining regions of SEQ ID NOs: 146, 147 and 148 (CDR1, CDR2 and CDR3, respectively) and a light-chain variable region comprising complementarity determining regions of SEQ ID NOs: 149, 150 and 151 (CDR1, CDR2 and CDR3, respectively), and having an immunological reactivity with CAPRIN-1 protein;

(J) an antibody or a fragment thereof comprising a heavy-chain variable region comprising complementarity determining regions of SEQ ID NOs: 272, 273 and 274 (CDR1, CDR2 and CDR3, respectively) and a light-chain variable region comprising complementarity determining regions of SEQ ID NOs: 275, 276 and 277 (CDR1, CDR2 and CDR3, respectively), and having an immunological reactivity with CAPRIN-1 protein;

(K) an antibody or a fragment thereof comprising a heavy-chain variable region comprising complementarity determining regions of SEQ ID NOs: 290, 291 and 292 (CDR1, CDR2 and CDR3, respectively) and a light-chain variable region comprising complementarity determining regions of

SEQ ID NO: 112 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 113;

(ab) an antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 114 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 115;

(ac) an antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 116 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 117;

(ad) an antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 118 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 119;

(ae) an antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 120 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 121;

(af) an antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 122 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 123;

(ag) an antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 124 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 125;

(ah) an antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 126 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 127;

(ai) an antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 128 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 129;

(aj) an antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 130 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 131;

(ak) an antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 132 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 133; and

(al) an antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 300 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 304.

[0028] (13) The medicament according to any of (1) to (12), wherein the antibody is a human antibody, a humanized antibody, a chimeric antibody or a single chain antibody.

[0029] (14) The medicament according to any of (1) to (13), wherein the cancer is a cancer expressing CAPRIN-1 protein on a cell membrane surface.

[0030] (15) The medicament according to any of (1) to (14), wherein the cancer is melanoma, lung cancer, thyroid cancer, colon cancer, prostate cancer, ovarian cancer, pancreatic cancer, kidney cancer, breast cancer, gastric cancer, bile duct cancer, renal cell cancer, Hodgkin's lymphoma, head and neck cancer, mesothelioma, colorectal cancer, esophageal cancer, gastroesophageal cancer, hepatocellular cancer, glioblastoma, urothelial cancer, urinary bladder cancer, uterus cancer, primary central nervous system lymphoma, primary testicular lymphoma, biliary tract cancer, brain tumor, leukemia, lymphoma, liver cancer, sarcoma, fibrosarcoma, mastocytoma, adrenal cortex cancer, Ewing's

tumor, multiple myeloma, testicle cancer, basal cell cancer, Paget's disease or skin cancer.

[0031] (16) An agent increasing drug efficacy of a pharmaceutical composition for treatment and/or prevention of cancer comprising an antibody or a fragment thereof having an immunological reactivity with CAPRIN-1 protein as an active ingredient, wherein the agent comprises MAPK pathway inhibitor as an active ingredient.

[0032] (17) An agent increasing drug efficacy of a pharmaceutical composition for treatment and/or prevention of cancer comprising MAPK pathway inhibitor as an active ingredient, wherein the agent comprises an antibody or a fragment thereof having an immunological reactivity with CAPRIN-1 protein as an active ingredient.

[0033] (18) A method for treating and/or preventing cancer, comprising administering an antibody or a fragment thereof having an immunological reactivity with CAPRIN-1 protein, and MAPK pathway inhibitor together or separately to a subject.

[0034] The present specification encompasses the contents disclosed in Japanese Patent Application No. 2021-122072 on which the priority of the present application is based.

Advantageous Effects of Invention

[0035] The combination of an antibody against CAPRIN-1 protein, or a fragment thereof and a MAPK pathway inhibitor according to the present invention exerts a stronger antitumor effect than that of the antibody against CAPRIN-1 protein alone and an existing chemotherapeutic alone. Thus, the combination of the antibody against CAPRIN-1 protein and the MAPK pathway inhibitor is effective for treatment or prevention of cancer.

BRIEF DESCRIPTION OF DRAWINGS

[0036] FIG. 1 is a diagram showing human monocytic cell (THP-1)-mediated phagocytic activity of a combination of an anti-CAPRIN-1 antibody and trametinib against a human cancer cell line. Reference number 1 depicts the phagocytic activity against a human melanoma cell line (A375), reference number 2 depicts the phagocytic activity against a human colon cancer cell line (HCT116), reference number 3 depicts the phagocytic activity against a human prostate cancer cell line (DU145), and reference number 4 depicts the phagocytic activity against a human lung cancer cell line (A549). Shaded bar graph: a test group without a drug used in combination (DMSO was added instead of the drug). Filled bar graph; a trametinib combination test group.

[0037] FIG. 2 is a diagram showing human monocytic cell (THP-1)-mediated phagocytic activity of a combination of an anti-CAPRIN-1 antibody and any of various drugs against a human colon cancer cell line HCT116. Reference number 5 depicts the phagocytic activity in a test group without a drug used in combination (DMSO was added instead of the drug to the cells), reference number 6 depicts the phagocytic activity in a trametinib combination test group (50 nM), reference number 7 depicts the phagocytic activity in a fluorouracil (5-FU) combination test group (0.4 µg/mL), and reference number 8 depicts the phagocytic activity in an irinotecan combination test group (15 µM).

[0038] FIG. 3 is a diagram showing human monocytic cell (THP-1)-mediated phagocytic activity of a combination of an anti-CAPRIN-1 antibody and any of various drugs against a human colon cancer cell line HCT116. Reference

number 9 depicts the phagocytic activity in a test group without a drug used in combination, reference number 10 depicts the phagocytic activity in a lonafarnib combination test group (10 μ M), reference number 11 depicts the phagocytic activity in a sotorasib combination test group (10 μ M), reference number 12 depicts the phagocytic activity in a salirasib combination test group (100 μ M), reference number 13 depicts the phagocytic activity in a selumetinib combination test group (10 μ M), reference number 14 depicts the phagocytic activity in a refametinib combination test group (1000 nM), reference number 15 depicts the phagocytic activity in a SL-327 combination test group (10 μ M), reference number 16 depicts the phagocytic activity in a cobimetinib combination test group (800 nM), and reference number 17 depicts the phagocytic activity in a bosutinib combination test group (10 μ M). Error bars depict standard deviation (S.D.). As a result of conducting Student's t test, the phagocytic activities of drug combination test groups were significantly higher than that of the test group without a drug used in combination (significance level: 5%). **, $p < 0.01$; ***, $p < 0.001$.

[0039] FIG. 4 is a diagram showing human monocytic cell (THP-1)-mediated phagocytic activity of a combination of an anti-CAPRIN-1 antibody and any of various drugs against a human colon cancer cell line HCT116. Reference number 18 depicts the phagocytic activity in a test group without a drug used in combination, reference number 19 depicts the phagocytic activity in a dabrafenib combination test group (10 μ M), and reference number 20 depicts the phagocytic activity in a ZM 336372 combination test group (100 μ M).

[0040] FIG. 5 is a diagram showing human monocytic cell (THP-1)-mediated phagocytic activity of a combination of an anti-CAPRIN-1 antibody and any of various drugs against a human colon cancer cell line HCT116. Reference number 21 depicts the phagocytic activity in a test group without a drug used in combination, and reference number 22 depicts the phagocytic activity in a SCH772984 combination test group (10 μ M). Error bars depict standard deviation (S.D.). As a result of conducting Student's t test, the phagocytic activity of the SCH772984 combination test group was significantly higher than that of the test group without a drug used in combination ($p < 0.001$; significance level: 5%).

DESCRIPTION OF EMBODIMENTS

[0041] The antitumor effect of the combination of an antibody against CAPRIN-1 protein or a fragment thereof (hereinafter, referred to as an "anti-CAPRIN-1 antibody") and a MAPK pathway inhibitor, used in the present invention is preferably evaluated by examining in vitro the phagocytic activity against cancer cells by immunocytes in coculturing the cancer cells and the immunocytes as mentioned later. The immunocyte used in evaluating the antitumor effect in vitro may be any cell as long as the immunocyte is a blood cell having phagocytic activity, and preferably, is a human monocytic cell (THP-1 or U937). When an antibody binds to a cancer cell, this is recognized by an immunocyte so that the cancer cell is killed via phagocytic activity by the immunocyte. Therefore, an in vivo antitumor effect can be predicted by evaluating the in vitro antitumor effect.

[0042] The term "combination" (or "combined use") described herein refers to simultaneous administration or

addition or administration or addition in a predetermined interval of the anti-CAPRIN-1 antibody and the MAPK pathway inhibitor as independent active ingredients to the same organism or cell. The interval may be simultaneous administration or may be 30 minutes later, 1 hour later, 3 hours later, 6 hours later, 12 hours later, 1 day later, 2 days later, 3 days later, 5 days later, 7 days later, 2 weeks later, 3 weeks later, or 4 weeks later. During one of the anti-CAPRIN-1 antibody or the MAPK pathway inhibitor exhibits its antitumor effect, the other can be administered or added.

[0043] The term "comprising . . . together or separately in combination" described herein refers to comprising a plurality of drugs in a form that allows the drugs to be administered simultaneously or separately to a patient. The form may be, for example, the form of a so-called mixed formulation in which a plurality of drugs are mixed, or may be the form of a so-called kit formulation (pharmaceutical kit) comprising a plurality of drugs as individual formulations. The form also encompasses the form of a kit formulation comprising a plurality of drugs in any combination in two or more of formulations.

[0044] Such a kit formulation according to the present invention may be, for example, a kit formulation comprising: a formulation (or a pharmaceutical composition) comprising the anti-CAPRIN-1 antibody and a formulation (or a pharmaceutical composition) comprising the MAPK pathway inhibitor.

[0045] The anti-CAPRIN-1 antibody according to the present invention may be a monoclonal antibody or a polyclonal antibody and is preferably a monoclonal antibody. The antibody of the present invention may be any type of antibody as long as it can exhibit antitumor activity. The antibody may be a recombinant antibody, a human antibody, a humanized antibody, a chimeric antibody, or a non-human animal antibody.

[0046] Subjects being subjected to treatment and/or prevention of cancer according to the present invention are mammals such as primates, pet animals, livestock animals, or sport animals and preferably humans, dogs and cats, and more preferably humans.

[0047] Medicaments comprising an anti-CAPRIN-1 antibody and a MAPK route inhibiting agent as active ingredients, and methods for treating and/or preventing cancers, related to the present invention, will be explained below.

Anti-CAPRIN-1 Antibody

[0048] Among CAPRIN-1 proteins having an amino acid sequence shown by any one of the even numbered SEQ ID NOs: 2 to 30, which are specific examples of antigens having an immunological reactivity with the anti-CAPRIN-1 antibody used in the present invention, the amino acid sequences shown by SEQ ID NOs: 6, 8, 10, 12 and 14 are amino acid sequences of canine CAPRIN-1 proteins; the amino acid sequences shown by SEQ ID NOs: 2 and 4 are amino acid sequences of human CAPRIN-1 proteins; the amino acid sequence shown by SEQ ID NO: 16 is an amino acid sequence of a bovine CAPRIN-1 protein; the amino acid sequence shown by SEQ ID NO: 18 is an amino acid sequence of a horse CAPRIN-1 protein; the amino acid sequences shown by SEQ ID NOs: 20, 22, 24, 26 and 28 are amino acid sequences of mouse CAPRIN-1 proteins; and the amino acid sequence shown by SEQ ID NO: 30 is an amino acid sequence of a chicken CAPRIN-1 protein.

[0049] The anti-CAPRIN-1 antibody used in the present invention may have an immunological reactivity with a CAPRIN-1 protein variant having 80% or more, preferably 90% or more, more preferably 95% or more, and further preferably 99% or more sequence identity to the amino acid sequence shown by any one of the even numbered SEQ ID NOs: 2 to 30. The term “% sequence identity” as used herein means a percentage (%) of the number of identical amino acids (or nucleotides) to the total number of amino acids (or nucleotides) in the case that two sequences are aligned such that maximum similarity can be achieved with or without introduction of gaps.

[0050] In the present invention, the anti-CAPRIN-1 antibody refers to an antibody or a fragment (antigen binding fragment) thereof having an immunological reactivity with a full-length CAPRIN-1 protein or a fragment thereof. The term “immunological reactivity” used herein indicates the characteristics of an antibody specifically binding in vivo to a CAPRIN-1 protein or a partial polypeptide thereof.

[0051] The anti-CAPRIN-1 antibody used in the present invention may be a monoclonal antibody or a polyclonal antibody.

[0052] Polyclonal antibodies having an immunological reactivity with a full-length CAPRIN-1 protein or a fragment thereof (anti-CAPRIN-1 polyclonal antibodies) can be obtained, for example, in a manner described below. Serum is obtained after mice, human antibody-producing mice, rats, rabbits, chickens, or the like are immunized using a naturally occurring CAPRIN-1 protein or the protein fused with GST or the like, or a partial peptide thereof. The obtained serum is purified via ammonium sulfate precipitation, protein A, protein G, DEAE ion-exchange columns, affinity columns to which a CAPRIN-1 protein or a partial peptide is coupled, or the like.

[0053] Nucleotide sequences and amino acid sequences of CAPRIN-1 and homologs thereof used in the immunization can be obtained by, for example, accessing the website of GenBank (NCBI, USA) and using the BLAST or FASTA algorithm (Karlin and Altschul, Proc. Natl. Acad. Sci. USA, 90:5873-5877, 1993; and Altschul et al., Nucleic Acids Res. 25:3389-3402, 1997). Methods for producing CAPRIN-1 protein can be obtained with reference to WO2014/012479 or may employ cells or the like expressing CAPRIN-1 protein.

[0054] Monoclonal antibodies having an immunological reactivity with a full-length CAPRIN-1 protein or a fragment thereof (anti-CAPRIN-1 monoclonal antibodies) can be obtained, for example, in a manner described below. Breast cancer cells SK-BR-3 expressing CAPRIN-1, a full-length CAPRIN-1 protein or a fragment thereof, or the like is administered to mice for immunization. Splenocytes separated from the mice are fused with myeloma cells. Clones capable of producing anti-CAPRIN-1 monoclonal antibodies can be selected from the obtained fusion cells (hybridomas) to obtain these antibodies. The antibodies produced from the selected hybridomas can be obtained in the same way as the aforementioned method for purifying polyclonal antibodies.

[0055] The antibody used in the present invention includes human antibodies, humanized antibodies, chimeric antibodies, non-human animal antibodies, and single chain antibodies.

[0056] For human antibodies, human lymphocytes infected with EB virus are sensitized with a protein, protein-

expressing cells, or a lysate thereof. The sensitized lymphocytes are fused with human-derived myeloma cells such as U266 cells. Antibodies having an immunological reactivity with a full-length CAPRIN-1 protein or a fragment thereof can be obtained from the obtained fusion cells.

[0057] A humanized antibody is a modified antibody, and it is sometimes referred to as a reshaped human antibody. It is known that a humanized antibody is constructed by transplanting complementarity determining regions of an immunized animal-derived antibody into complementarity determining regions of a human antibody. In addition, a general gene recombinant technique therefor is well known. Specifically, a DNA sequence designed in a manner that allows complementarity determining regions of mouse or rabbit antibody to be ligated to human antibody framework regions is synthesized by the PCR method using several oligonucleotides prepared in such a manner that the oligonucleotides have portions overlapping each other at an end of each thereof. A humanized antibody can be obtained by ligating the above obtained DNA to DNA encoding a human antibody constant region, incorporating the resultant into an expression vector, and introducing the vector into a host for antibody production (see EP-A-239400 and WO96/02576). Framework regions of human antibody ligated to each other via complementarity determining regions are selected on the assumption that complementarity determining regions can form a good antigen binding site. If necessary, amino acids in framework regions of an antibody variable region may be substituted in such a manner that complementarity determining regions in a reshaped human antibody form an appropriate antigen binding site (Sato K. et al., Cancer Research 1993, 53: 851-856). In addition, the framework regions may be substituted with framework regions from a different human antibody (see WO99/51743).

[0058] In general, antibodies are heteromultimeric glycoproteins each comprising at least two heavy chains and two light chains. Antibodies each comprise two identical light chains and two identical heavy chains. Each heavy chain has a heavy-chain variable region at one end thereof, to which some constant regions are bound in series. Each light chain has a light-chain variable region at one end thereof to which some constant regions are bound in series. Variable regions have specific variable regions, which are called complementarity determining regions (CDRs) and imparts binding specificity to an antibody. A relatively conserved portion in a variable region is called a framework region (FR). A complete heavy-chain or light-chain variable region comprises 4 FRs connected to each other via 3 CDRs (CDR1 to CDR3).

[0059] Sequences of human-derived heavy-chain and light-chain constant regions and variable regions can be obtained from, for example, NCBI (USA; GenBank, UniGene, etc.). For example, for a human IgG1 heavy-chain constant region, see registration No. J00228; for a human IgG2 heavy-chain constant region, see registration No. J00230; for a human light chain κ constant region, see sequences such as registration Nos. V00557, X64135, and X64133; and for a human light chain λ constant region, see sequences such as registration Nos. X64132 and X64134.

[0060] A chimeric antibody is an antibody produced by combining sequences from different animals. An example thereof is an antibody consisting of mouse antibody heavy-chain and light-chain variable regions and constant regions of human antibody heavy-chain and light-chain variable

regions. Such a chimeric antibody can be produced by a known method. For example, it can be obtained by ligating DNA encoding an antibody V region to DNA encoding a human antibody C region, incorporating the resultant into an expression vector, and introducing the vector into a host for antibody production.

[0061] Non-human animal antibodies are obtained by immunizing non-human animals with sensitizing antigens according to a known method or by intraperitoneally, intracutaneously, or subcutaneously injecting sensitizing antigens into animals such as mice as a general method. For injecting sensitizing antigens, an appropriate amount of various adjuvants including CFA (Freund's complete adjuvant) is mixed therewith and the mixture is administered to animals several times. After immunization of animals and confirmation of an anti-CAPRIN-1 antibody contained in serum, the serum is obtained and the non-human animal antibody can be obtained by purification via ammonium sulfate precipitation, protein A, protein G, DEAE ion-exchange columns, affinity columns to which a CAPRIN-1 protein or a partial peptide thereof is coupled, or the like, as mentioned above. In the case of obtaining monoclonal antibodies from non-human animals, a monoclonal antibody is obtained by collecting immunocytes from the immunized animals and subjected to cell fusion with myeloma cells. The cell fusion of immunocytes with myeloma cells can be carried out according to a known method (see Kohler, G. and Milstein, C. *Methods Enzymol.* (1981) 73, 3-46).

[0062] The antibody used in the present invention can also be obtained as a gene recombinant antibody produced by cloning an antibody gene from a hybridoma, incorporating the clone into an adequate vector, introducing the vector into a host, and producing the antibody by using a gene recombinant technique (see Carl, A. K. Borrebaeck, James, W. Larrick, *THERAPEUTIC MONOCLONAL ANTIBODIES*, Published in the United Kingdom by MACMILLAN PUBLISHERS LTD, 1990).

[0063] Amino acids in a variable region (e.g., FR) or a constant region in the anti-CAPRIN-1 antibody used in the present invention may be substituted with different amino acids. The amino acid substitution is a substitution of 1 or more, for example, less than 15, less than 10, not more than 8, not more than 6, not more than 5, not more than 4, not more than 3, or not more than 2 amino acids, preferably 1 to 9 amino acids. A substituted antibody should have characteristics of specifically binding to the antigen and binding affinity for the antigen equivalent to or more than those of an unsubstituted antibody and should be an antibody that causes no rejection when applied to humans. The amino acid substitution is preferably a conservative amino acid substitution, which is a substitution between amino acids having similar characteristics in terms of charge, side chains, polarity, aromaticity, and the like. For example, characteristically similar amino acids can be classified into the following types: basic amino acids (arginine, lysine, and histidine); acidic amino acids (aspartic acid and glutamic acid); uncharged polar amino acids (glycine, asparagine, glutamine, serine, threonine, cysteine, and tyrosine); nonpolar amino acids (leucine, isoleucine, alanine, valine, proline, phenylalanine, tryptophan, and methionine); branched-chain amino acids (threonine, valine, isoleucine); and aromatic amino acids (phenylalanine, tyrosine, tryptophan, and histidine).

[0064] The anti-CAPRIN-1 antibody used in the present invention is expected to have a stronger antitumor effect when having higher binding affinity for CAPRIN-1 protein on cancer cell surfaces. Association constant (affinity constant) K_a (kon/koff) is preferably at least 10^7 M^{-1} , at least 10^8 M^{-1} , at least 5×10^8 M^{-1} , at least 10^9 M^{-1} , at least 5×10^9 M^{-1} , at least 10^{10} M^{-1} , at least 5×10^{10} M^{-1} , at least 10^{11} M^{-1} , at least 5×10^{11} M^{-1} , at least 10^{12} M^{-1} , or at least 10^{13} M^{-1} .

[0065] The anti-CAPRIN-1 antibody used in the present invention may be chemically modified. Examples of such an antibody modifier can include antibodies bound to various molecules such as polyethylene glycol (PEG) and antitumor compounds (for example, antitumor agents listed below). Regarding antibody modifiers of the present invention, substances that bind to an antibody are not limited. Such an antibody modifier can be obtained by chemically modifying an obtained antibody. Methods of such modification have been already established in the field related to the present invention.

[0066] The binding strength of the anti-CAPRIN-1 antibody used in the present invention against effector cells can be improved by substituting 1, 2 or several amino acids in the heavy-chain constant region of the antibody or by removing fucose bound to N-acetylglucosamine in a N-glycoside-linked sugar chain bound to the heavy-chain constant region. The anti-CAPRIN-1 antibody described above may have the amino acid substitution alone or may be a composition with an antibody bound to fucose.

[0067] Antibodies in which 1, 2 or several amino acids in the heavy-chain constant region have been substituted can be produced with reference to, for example, WO2004/063351, WO2011/120135, U.S. Pat. No. 8,388,955, WO2011/005481, U.S. Pat. No. 6,737,056, and WO2005/063351.

[0068] Antibodies in which fucose bound to N-acetylglucosamine in a N-glycoside-linked sugar chain in the heavy-chain constant region has been removed, or producing cells thereof can be produced with reference to U.S. Pat. No. 6,602,684, EP U.S. Pat. No. 1,914,244, and U.S. Pat. No. 7,579,170. Compositions of antibodies in which fucose bound to N-acetylglucosamine in a N-glycoside-linked sugar chain bound to the heavy-chain constant region has been removed, with antibodies bound to fucose, or producing cells thereof can be produced with reference to, for example, U.S. Pat. No. 8,642,292.

[0069] The anti-CAPRIN-1 polyclonal antibody and the anti-CAPRIN-1 monoclonal antibody used in the present invention, methods for producing or purifying antibodies and methods for producing a CAPRIN-1 protein or partial polypeptide thereof used in immunization can be obtained with reference to WO2010/016526, WO2011/096517, WO2011/096528, WO2011/096519, WO2011/096533, WO2011/096534, WO2011/096535, WO2013/018886, WO2013/018894, WO2013/018892, WO2013/018891, WO2013/018889, WO2013/018883, WO2013/125636, WO2013/125654, WO2013/125630, WO2013/125640, WO2013/147169, WO2013/147176 and WO2015/020212.

[0070] Specific examples of the anti-CAPRIN-1 antibody according to the present invention include anti-CAPRIN-1 antibodies described in WO2010/016526, WO2011/096517, WO2011/096528, WO2011/096519, WO2011/096533, WO2011/096534, WO2011/096535, WO2013/018886, WO2013/018894, WO2013/018892, WO2013/018891,

WO2013/018889, WO2013/018883, WO2013/125636, WO2013/125654, WO2013/125630, WO2013/125640, WO2013/147169, WO2013/147176 and WO2015/020212 mentioned above. Preferred examples of the anti-CAPRIN-1 antibody include the following.

[0071] An antibody or a fragment thereof having an immunological reactivity with a partial polypeptide of a CAPRIN-1 protein having the amino acid sequence shown by SEQ ID NO: 2 or SEQ ID NO: 4 or an amino acid sequence having 80% or more (preferably 85% or more, more preferably 90% or more, further preferably 95% or more, and still further preferably 99% or more) sequence identity to the amino acid sequence.

[0072] An antibody or a fragment thereof having an immunological reactivity with a partial polypeptide of CAPRIN-1 protein, the partial polypeptide having the amino acid sequence shown by SEQ ID NO: 31 or an amino acid sequence having 80% or more (preferably 85% or more, more preferably 90% or more, and further preferably 95% or more) sequence identity to the amino acid sequence, preferably an antibody or a fragment thereof comprising a heavy-chain variable region comprising complementarity determining regions of SEQ ID NOs: 36, 37 and 38 (CDR1, CDR2 and CDR3, respectively) and a light-chain variable region comprising complementarity determining regions of SEQ ID NOs: 40, 41 and 42 (CDR1, CDR2 and CDR3, respectively), and having an immunological reactivity with CAPRIN-1 protein, an antibody or a fragment thereof comprising a heavy-chain variable region comprising complementarity determining regions of SEQ ID NOs: 140, 141 and 142 (CDR1, CDR2 and CDR3, respectively) and a light-chain variable region comprising complementarity determining regions of SEQ ID NOs: 143, 144 and 145 (CDR1, CDR2 and CDR3, respectively), and having an immunological reactivity with CAPRIN-1 protein, or an antibody or a fragment thereof comprising a heavy-chain variable region comprising complementarity determining regions of SEQ ID NOs: 164, 165 and 166 (CDR1, CDR2 and CDR3, respectively) and a light-chain variable region comprising complementarity determining regions of SEQ ID NOs: 167, 168 and 169 (CDR1, CDR2 and CDR3, respectively), and having an immunological reactivity with CAPRIN-1 protein, and more preferably an antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 39 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 43, an antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 70 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 71, or an antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 78 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 79.

[0073] An antibody or a fragment thereof having an immunological reactivity with a partial polypeptide of CAPRIN-1 protein, the partial polypeptide having the amino acid sequence shown by SEQ ID NO: 33 or an amino acid sequence having 80% or more (preferably 85% or more, more preferably 90% or more, and further preferably 95% or more) sequence identity to the amino acid sequence, preferably an antibody or a fragment thereof comprising a heavy-chain variable region comprising complementarity

determining regions of SEQ ID NOs: 60, 61 and 62 (CDR1, CDR2 and CDR3, respectively) and a light-chain variable region comprising complementarity determining regions of SEQ ID NOs: 64, 65 and 66 (CDR1, CDR2 and CDR3, respectively), and having an immunological reactivity with CAPRIN-1 protein, and more preferably an antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 63 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 67.

[0074] An antibody or a fragment thereof having an immunological reactivity with a partial polypeptide of CAPRIN-1 protein, the partial polypeptide having the amino acid sequence shown by SEQ ID NO: 32 or an amino acid sequence having 80% or more (preferably 85% or more, more preferably 90% or more, and further preferably 95% or more) sequence identity to the amino acid sequence, preferably an antibody or a fragment thereof comprising a heavy-chain variable region comprising complementarity determining regions of SEQ ID NOs: 52, 53 and 54 (CDR1, CDR2 and CDR3, respectively) and a light-chain variable region comprising complementarity determining regions of SEQ ID NOs: 56, 57 and 58 (CDR1, CDR2 and CDR3, respectively), and having an immunological reactivity with CAPRIN-1 protein, and more preferably an antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 55 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 59.

[0075] An antibody or a fragment thereof having an immunological reactivity with a partial polypeptide of CAPRIN-1 protein, the partial polypeptide having the amino acid sequence shown by SEQ ID NO: 34 or an amino acid sequence having 80% or more (preferably 85% or more, more preferably 90% or more, and further preferably 95% or more) sequence identity to the amino acid sequence, preferably an antibody or a fragment thereof comprising a heavy-chain variable region comprising complementarity determining regions of SEQ ID NOs: 170, 171 and 172 (CDR1, CDR2 and CDR3, respectively) and a light-chain variable region comprising complementarity determining regions of SEQ ID NOs: 173, 174 and 175 (CDR1, CDR2 and CDR3, respectively), and having an immunological reactivity with CAPRIN-1 protein, or an antibody or a fragment thereof comprising a heavy-chain variable region comprising complementarity determining regions of SEQ ID NOs: 176, 177 and 178 (CDR1, CDR2 and CDR3, respectively) and a light-chain variable region comprising complementarity determining regions of SEQ ID NOs: 179, 180 and 181 (CDR1, CDR2 and CDR3, respectively), and having an immunological reactivity with CAPRIN-1 protein, and more preferably an antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 80 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 81, or an antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 82 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 83.

[0076] An antibody or a fragment thereof having an immunological reactivity with a partial polypeptide of CAPRIN-1 protein, the partial polypeptide having the amino acid sequence shown by SEQ ID NO: 35 or an amino acid

and CDR3, respectively), and having an immunological reactivity with CAPRIN-1 protein, and more preferably an antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 68 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 69.

[0083] An antibody or a fragment thereof having an immunological reactivity with a partial polypeptide of CAPRIN-1 protein, the partial polypeptide having the amino acid sequence shown by SEQ ID NO: 309 or an amino acid sequence having 80% or more (preferably 85% or more, more preferably 90% or more, and further preferably 95% or more) sequence identity to the amino acid sequence, preferably an antibody or a fragment thereof comprising a heavy-chain variable region comprising complementarity determining regions of SEQ ID NOs: 134, 135 and 136 (CDR1, CDR2 and CDR3, respectively) and a light-chain variable region comprising complementarity determining regions of SEQ ID NOs: 137, 138 and 139 (CDR1, CDR2 and CDR3, respectively), and having an immunological reactivity with CAPRIN-1 protein, and more preferably an antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 68 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 69.

[0084] In addition, the following anti-CAPRIN-1 antibodies are also preferably used.

[0085] An antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 68 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 69.

[0086] An antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 70 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 71.

[0087] An antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 72 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 73.

[0088] An antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 74 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 75.

[0089] An antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 76 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 77.

[0090] An antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 78 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 79.

[0091] An antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 80 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 81.

[0092] An antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid

sequence of SEQ ID NO: 82 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 83.

[0093] An antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 84 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 85.

[0094] An antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 86 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 87.

[0095] An antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 88 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 89.

[0096] An antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 90 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 91.

[0097] An antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 92 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 93.

[0098] An antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 94 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 95.

[0099] An antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 96 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 97.

[0100] An antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 98 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 99.

[0101] An antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 100 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 101.

[0102] An antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 102 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 103.

[0103] An antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 104 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 105.

[0104] An antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 106 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 107.

[0105] An antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 108 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 109.

[0106] An antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 110 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 111.

[0107] An antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 112 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 113.

[0108] An antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 114 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 115.

[0109] An antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 116 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 117.

[0110] An antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 118 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 119.

[0111] An antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 120 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 121.

[0112] An antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 122 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 123.

[0113] An antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 124 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 125.

[0114] An antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 126 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 127.

[0115] An antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 128 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 129.

[0116] An antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 130 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 131.

[0117] An antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid

sequence of SEQ ID NO: 132 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 133.

[0118] An antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 300 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 304.

[0119] In Examples mentioned later, the polyclonal antibody or the monoclonal antibody against a full-length CAP-RIN-1 protein or a polypeptide of a portion of a region expressed on cell membrane surfaces of cancer cells was confirmed to exhibit reactivity with cell membrane surfaces of a plurality of human cancer cells. Furthermore, results indicating the response in human cancer patients were obtained, and a marked antitumor effect was obtained by which a tumor completely disappeared at some cancer sites.

MAPK Pathway Inhibitor

[0120] The MAPK pathway refers to a signal transduction pathway that involves the activation of mitogen-activated protein kinase (MAPK) in the RAS/RAF/MEK/ERK pathway and links intracellular response to the binding of a growth factor to a cell surface receptor, and is also called MAPK signaling pathway, MAPK/ERK signal transduction pathway, RAF-MEK-ERK pathway, RAS/MAPK pathway, or RAS-RAF-MEK-ERK pathway in literatures. As a review of enzymes involved in the MAPK pathway, see, for example, *Biochim Biophys Acta.*, 2007, 1773 (8): 1263-84. The term "MAPK pathway" is directed to its many protein components and a kinase cascade which is a portion of this signal transduction pathway as well as various targets controlled by this pathway.

[0121] The MAPK pathway inhibitor refers to an inhibitor for the biological activity of wild-type or any mutant of any of enzymes involved in the MAPK pathway. Specific examples of the MAPK pathway inhibitor include, but are not limited to, a RAS inhibitor, a RAF inhibitor, a MEK inhibitor and an ERK inhibitor. The MAPK pathway inhibitor may be a therapeutically effective derivative of a substance known as the MAPK pathway inhibitor as long as it has MAPK pathway inhibitory action.

RAS Inhibitor

[0122] The RAS inhibitor is a drug having the action of inhibiting the activity of RAS, one of the proteins constituting the MAPK pathway. RAS has a molecular weight of 21 kDa and belongs to a large family as monomer GTPase. The RAS subfamilies constitutes the small G protein superfamily, which includes three classical RASs, HRAS, KRAS, and NRAS, as well as R-RAS, TC21 (R-RAS2), M-RAS (R-RAS3), Rap1A, Rap1B, Rap2A, Rap2B, RalA, and RalB. RAS shifts from an inactive GDP-bound state to an active GTP-bound state in response to various extracellular stimuli. This GDP/GTP exchange reaction is accelerated by GEF(s) (guanine-nucleotide exchange factors). The GTP-bound RAS returns to a GDP-bound form by endogenous GTPase. This reaction is accelerated by GTPase-activating proteins (GAPs). The activated GTP-bound RAS interacts with a wide range of downstream targets and activates downstream signaling. RAF kinase is known as one of the major targets. RAS is a cancer gene having a high mutation frequency whose mutation is detected in approximately 30%

of cancer patients. In recent years, a drug specifically inhibiting the proliferation of some cancers having a KRAS gene mutation (KRAS G12C mutation) has been approved, and other RAS inhibitors are also under medicinal development.

[0123] Specific examples of the RAS inhibitor include adagrasib (MRTX849), garsorasib, lonafarnib (SCH66336), salirasib, sotorasib (AMG510), sotorasib racemate, (-)-rasfonin, (Rac)-antineoplaston A10, 6H05 (TFA), antineoplaston A10, Antitumor agent-60, APS6-45, ARS-1620, ARS-853 (ARS853), ASP2453, AZD4625, BAY-293, BI-2852, BI-3406, CASIN, CMC2.24, Deltarasin, Deltarasin hydrochloride, Diazepinomicin, Farnesyl thiosalicic acid, FTI-276 trifluoroacetate salt, FTI-277, FTI-277 hydrochloride, GDC-6036, GGTI-286, GGTI-286 hydrochloride, GGTI298 trifluoroacetate, JDQ-443 (WO2021/120890), K20, Kobe0065, Kobe2602, KRAS mutant protein inhibitor 1, L-744,832 Dihydrochloride, LC-2, Manumycin A, MCP110, ML 210, MLS-573151, MRTX0902, MRTX1133, MRTX1133 formic, MRTX-1257, MRTX849 ethoxypropanoic acid, MRTX-EX185, NSC-658497, Oncrasin-1, Pan-RAS-IN-1, PHT-7.3, PROTAC K-Ras Degradar-1, PROTAC SOS1 degrader-2, RAS GTPase inhibitor 1, RAS inhibitor Abd-7, Ras Inhibitory Peptide, Ras/Rac Transformation Blocker, SCH 51344, RAS/RAS-RAF-IN-1, Rasarfin, RM-018, RMC-0331, RTIL 13, SAH-SOS1A, SAH-SOS1A TFA, SCH 51344, SOS1-IN-10 (WO2022/017519; compound 8), SOS1-IN-11, SOS1-IN-12, SOS1-IN-13, SOS1-IN-3 (WO2019/122129A1; compound 1-1), SOS1-IN-4 (WO2021/228028; example 65), SOS1-IN-5 (WO2021/203768; compound 4), SOS1-IN-6, SOS1-IN-7, SOS1-IN-8 (WO2022/017339; compound 2), SOS1-IN-9 (WO2022/028506; compound 302), XRP44X, KRpep-2d, KY1220, NAV-2729, zoledronic acid (ZOL 446), BQU57, CID-1067700, Fendiline hydrochloride, ARS-1323 (WO2015/054572), ARS-1630 (WO2015/054572), KRas G12C inhibitor 1 (US2018/0072723), KRas G12C inhibitor 2 (WO2021/118877), KRas G12C inhibitor 3 (US2018/0072723 A1), KRAS G12C inhibitor 13 (WO2018/143315A1; compound 30), KRAS G12C inhibitor 14 (WO2019/110751A1; compound 17), KRAS G12C inhibitor 15 (WO2019/110751; compound 22), KRAS G12C inhibitor 16 (WO2019/110751; compound 39), KRAS G12C inhibitor 17 (WO2019/110751A1; compound 82), KRAS G12C inhibitor 18, KRAS G12C inhibitor 19 (WO2021/118877), KRAS G12C inhibitor 20 (CN112694475A; example 1), KRAS G12C inhibitor 21 (WO2021/219090; example 7), KRAS G12C inhibitor 22 (WO2021/219072A1; example 120), KRAS G12C inhibitor 23 (WO2021/218939; compound 1), KRAS G12C inhibitor 24 (CN113563323A; compound 1), KRAS G12C inhibitor 25 (WO2021/216770; compound 3), KRAS G12C inhibitor 26 (WO2021/109737), KRAS G12C inhibitor 27 (WO2021/109737), KRAS G12C inhibitor 28 (WO2021/113595; Example 1), KRAS G12C inhibitor 29 (WO2021/252339), KRAS G12C inhibitor 30 (WO2021/252339A1; compound 2), KRAS G12C inhibitor 31 (WO2021/252339), KRAS G12C inhibitor 32, KRAS G12C inhibitor 33 (WO2021/244603A1; compound 1), KRAS G12C inhibitor 34 (WO2021/239058), KRAS G12C inhibitor 35 (CN112920183A; compound 3), KRAS G12C inhibitor 36 (WO2021/121367; compound 1-2), KRAS G12C inhibitor 37 (WO2018/143315; compound 65), KRAS G12C inhibitor 38 (WO2021/129820; compound 171), KRAS G12C

inhibitor 39 (WO2019/099524; compound 494), KRas G12C inhibitor 4 (WO2020/233592; compound 2), KRAS G12C inhibitor 40 (WO2021/129824; compound 70), KRAS G12C inhibitor 41 (WO2021/129824; compound 121), KRAS G12C inhibitor 42 (WO2020/146613; compound 10), KRAS G12C inhibitor 43, KRAS G12C inhibitor 44, KRAS G12C inhibitor 45, KRAS G12C inhibitor 46, KRAS G12C inhibitor 47, KRAS G12C inhibitor 48, KRAS G12C inhibitor 49, KRAS G12C inhibitor 5, KRAS G12C inhibitor 50, KRAS G12C inhibitor 51, KRAS G12C inhibitor 52, KRAS G12C inhibitor 53, KRAS G12C inhibitor 54, KRAS G12C inhibitor 55, K-Ras G12C-IN-1, K-Ras G12C-IN-2, K-Ras G12C-IN-3, K-Ras G12C-IN-4, KRAS G12D inhibitor 1, KRAS G12D inhibitor 10 (WO2021/108683; compound 34), KRAS G12D inhibitor 11 (WO2021/108683; compound 52), KRAS G12D inhibitor 12 (WO2021/108683; compound 134), KRAS G12D inhibitor 13 (WO2021/108683; compound 142), KRAS G12D inhibitor 14, KRAS G12D inhibitor 15 (WO2022/042630; compound 243), KRAS G12D inhibitor 3 TFA (WO2022/002102; compound 146), KRAS G12D inhibitor 3 (WO2022/002102; compound 146), KRAS G12D inhibitor 5, KRAS G12D inhibitor 6 (WO2021/108683; compound 112), KRAS G12D inhibitor 7 (WO2021/108683; compound 114), KRAS G12D inhibitor 8 (WO2021/107160; compound 2), KRAS G12D inhibitor 9 (WO2021/108683; compound 20), KRAS inhibitor-10 (WO2021/005165), KRAS inhibitor-11, KRAS inhibitor-12, KRAS inhibitor-13, KRAS inhibitor-14, KRAS inhibitor-15, KRAS inhibitor-16, KRAS inhibitor-18, KRAS inhibitor-4, KRAS inhibitor-6 (WO2017/087528; compound A), KRAS inhibitor-7 (WO2017/087528; compound B), KRAS inhibitor-8 (WO2017/087528; compound C), KRAS inhibitor-9, K-Ras (G12C) inhibitor 12, K-Ras (G12C) inhibitor 6, K-Ras (G12C) inhibitor 9, K-Ras-IN-1 and their pharmaceutically acceptable (known) salts or (known) derivatives. The RAS inhibitor is preferably adagrasib, garsorasib, lonafarnib, salirasib, sotorasib, zoledronic acid (ZOL 446), antineoplaston A10, or a pharmaceutically acceptable (known) salt or (known) derivative thereof, and more preferably lonafarnib, salirasib, sotorasib, or a pharmaceutically acceptable (known) derivative thereof.

[0124] Lonafarnib inhibits farnesyltransferase which activates H-ras, K-ras-4B and N-ras. Lonafarnib has a CAS number represented by 193275-84-2, an IUPAC name of 4-[2-[4-[(2R)-6,15-dibromo-13-chloro-4-azatricyclo[9.4.0.0^{3,8}]pentadeca-1(11),3(8),4,6,12,14-hexaen-2-yl]piperidin-1-yl]-2-oxoethyl]piperidine-1-carboxamide, a molecular formula of C₂₇H₃₁Br₂C₁N₄O₂, and a molecular weight of 638.8.

[0125] Salirasib (also known as farnesylthiosalicic acid or FTS) inhibits the methylation of RAS (active RAS). Salirasib has a CAS number represented by 162520-00-5, an IUPAC name of 2-[(2E,6E)-3,7,11-trimethyldodeca-2,6,10-trienyl]sulfanylbzoic acid, a molecular formula of C₂₂H₃₀O₂S, and a molecular weight of 358.5.

[0126] Sotorasib inhibits KRAS G12C. Sotorasib has a CAS number represented by 2252403-56-6, an IUPAC name of 6-fluoro-7-(2-fluoro-6-hydroxyphenyl)-1-(4-methyl-2-propan-2-ylpyridin-3-yl)-4-[(2S)-2-methyl-4-prop-2-enoylpiperazin-1-yl]pyrido[2,3-d]pyrimidin-2-one, a molecular formula of C₃₀H₃₀F₂N₆O₃, and a molecular weight of 560.59.

RAF Inhibitor

[0127] The RAF inhibitor is a drug having the action of inhibiting the activity of RAF, one of the proteins constituting the MAPK pathway. RAF kinase is a family consisting of three serine/threonine kinases associated with retrovirus cancer genes, and is constituted by three members ARAF, BRAF, and CRAF (RAF1). BRAF has a structure central to signal transduction, called activation loop. When BRAF is activated, this loop usually contributes to the stabilization of an active form through the phosphorylation of a plurality of sites. A particular amino acid (residue position: 600) in this loop is a location of an oncogenic mutation. When the position of this loop is shifted from a position in a dormant state due to change in amino acid at this position from valine to glutamic acid (called V600E mutation wherein V is valine, and E is glutamic acid), the enzyme becomes always in an active state so that cancer cells can proliferate without undergoing any usual control. The BRAF mutation is found in various cancers including melanoma and colon cancer. A CRAF mutation is found in a plurality of cancer types, and its activation is known to participate in the resistance of malignant melanoma to BRAF inhibitors. The activation of the RAF kinase requires dephosphorylation, and phosphorylation by protein tyrosine kinase of the SRC family and other protein serine/threonine kinases, in addition to interaction with RAS-GTP, and also requires dimerization of RAF. A RAF dimer in an active state then phosphorylates and activates MEK.

[0128] Specific examples of the RAF inhibitor include agerafenib, agerafenib hydrochloride, belvarafenib (HM95573/GDC-5573/RG6185), dabrafenib (GSK2118436A/GSK2118436), encorafenib-13C,d3, naporafenib (LXH254), rineterkib, tovorafenib (MLN2480), vemurafenib (PLX4032/RG7204/RO5185426), (Z)-GW 5074, Antitumor agent-60, AZ 628, AZ304, BI-882370, B-Raf IN 1, B-Raf IN 2 (WO2021/116055), B-Raf IN 5, B-Raf IN 6, B-Raf IN 7, B-Raf IN 8, B-Raf IN 9, BRAF inhibitor (WO2011/103196; Compound P-0850), BRAF V600E/CRAF-IN-1, BRAF V600E/CRAF-IN-2, CCT196969, Doramapimod (BIRB 796), Encorafenib-13C, d3, GDC-0879, GNE-9815, GSK-114, GW 5074, HG6-64-1 (HMSL 10017-101-1, WO2011/090738; example 9 (XI-1)), L-779450, LXH254, LY3009120 (DP-4978), MCP110, ML786 dihydrochloride, Pan-RAF kinase inhibitor 1 (WO2021/110141; compound 16B), PLX-4720, PLX-4720-d7, PLX7904, PLX7922, PLX8394, PROTAC B-Raf degrader 1, Raf inhibitor 1, Raf inhibitor 1 dihydrochloride, Raf inhibitor 2 (EP1003721B1), RAF mutant-IN-1 (WO2019107987A1), RAF709, RAF-IN-1, RAS/RAS-RAF-IN-1, Ro 5126766 (CH5126766), RRD-251, SB-590885, SB-682330A, SHR902275, TAK-580 (MLN 2480/BIIB-024), TAK-632, TBAP-001 (WO2015/075483), ZM 336372 and their pharmaceutically acceptable (known) salts or (known) derivatives. The RAF inhibitor is preferably agerafenib, belvarafenib, dabrafenib, naporafenib, rineterkib, tovorafenib, vemurafenib, or a pharmaceutically acceptable (known) salt or (known) derivative thereof, and more preferably dabrafenib (GSK2118436A/GSK2118436), ZM 336372, or a pharmaceutically acceptable (known) derivative thereof.

[0129] Dabrafenib (GSK2118436A/GSK2118436) inhibits the kinase activity of BRAF mutants (V600E, V600K and V600D). Dabrafenib has a CAS number represented by 1195765-45-7, an IUPAC name of N-[3-[5-(2-aminopyrimi-

din-4-yl)-2-tert-butyl-1,3-thiazol-4-yl]-2-fluorophenyl]-2,6-difluorobenzenesulfonamide, a molecular formula of $C_{23}H_{20}F_3N_5O_2S_2$, and a molecular weight of 519.6. Dabrafenib is also used as a mesylate salt thereof, which has a CAS number represented by 1195768-06-9, an IUPAC name of N-[3-[5-(2-aminopyrimidin-4-yl)-2-tert-butyl-1,3-thiazol-4-yl]-2-fluorophenyl]-2,6-difluorobenzenesulfonamide;methanesulfonic acid, a molecular formula of $C_{23}H_{20}F_3N_5O_2S_2 \cdot CH_4SO_3$, and a molecular weight of 615. 7. Dabrafenib simply described herein is a generic name for a free form and a mesylate salt of dabrafenib.

[0130] ZM 336372 (also known as Zinc00581684) inhibits the kinase activity of CRAF. ZM 336372 has a CAS number represented by 208260-29-1, an IUPAC name of 3-(dimethylamino)-N-[3-[(4-hydroxybenzoyl)amino]-4-methylphenyl]benzamide, a molecular formula of $C_{23}H_{23}N_3O_3$, and a molecular weight of 389.4.

MEK Inhibitor

[0131] The MEK inhibitor is a drug having the action of inhibiting the activity of MEK (MAPK/ERK kinase), one of the proteins constituting the MAPK pathway. MEK is a bispecific kinase (enzyme having both serine/threonine kinase and tyrosine kinase activities) that is activated by phosphorylation downstream of RAF in the MAPK signaling pathway. A typical phosphorylation substrate of MEK is ERK (extracellular signal-regulated kinase), and one of the main actions of the MEK inhibitor is the suppression of ERK phosphorylation. ERK in an inactive state is typically localized in the cytoplasm, while translocated into the nucleus when activated by phosphorylation, and phosphorylates various transcriptional factors including c-Jun, c-Myc, c-Fos, ETS1, and the like. ERK regulates the expression of many genes via such activity control of the transcriptional factors. MEK has a plurality of isoforms, among which, particularly, MEK1 and MEK2 play an important role in the development or progression of cancers and are responsible for signal transduction in the MAPK pathway. Therefore, the MEK inhibitor is preferably a MEK1 and/or MEK2 inhibitor.

[0132] Specific examples of the MEK inhibitor include trametinib (GSK1120212/JTP-74057), cobimetinib (GDC-0973/XL518), binimetinib (MEK162/ARRY-162/ARRY-438162), selumetinib (AZD6244/ARRY-142886), mirdametininib, pelitinib (EKB-569/AS703026/MSK1936369B/SAR 245509), refametinib (BAY 869766/RDEA119), zapnometininib (PD0184264/ATR-002), pimasertib (AS703026/MSK1936369B/SAR 245509), (S,R,S)-AHPC-Me-C10-Br, 1-Allylcyclopropane-1-sulfonyl Chloride, APS-2-79, APS-2-79 hydrochloride, APS-2-79 hydrochloride, Arctigenin, ARRY142866, ARRY438162, AZD6244, AZD8330 (ARRY-424704/ARRY-704), Balamapimod (MKI 833), BAY 869766, BI-847325, C16-PAF, CH4987655, CH5126766, CHMFL-EGFR-202, CI-1040 (PD 184352), CIF, ClnQ-03, CKI27, CS-3006, E6201 (ER-806201), EBI-1051, Epiberberine chloride, G-573, GDC-0623 (RG 7421/MEK inhibitor 1), GDC0973, GSK1120212, Hypothemycin, Isorhamnetin, JTP-74057, Lidocaine, Lidocaine hydrochloride, Lidocaine-d10, Lidocaine-d10 hydrochloride, Lidocaine-d10 N-Oxide, MAP855, MEK Inhibitor Set, MEK Inhibitor VIII, MEK inhibitor, MEK/PI3K-IN-1, MEK/PI3K-IN-2, MEK1/2 Inhibitor IV, MEK1/2-IN-2, MEK162 (ARRY438162), MEK-IN-1 (WO2008/076415), MEK-IN-5, MS432, MSC-2015103B, MSC-1936369,

Myricetin, PD 184352, PD 198306, PD0325901 (MEK1/2 Inhibitor III), PD0325901-O-C2-dioxolane, PD184161, PD318088, PD-334581, PD98059, RDEA 119, RG7167, RG7304, RG7420, RGB-286638, RGB-286638 free base, Ro 5126766 (CH5126766), RO4987655 (CH4987655), SHR-7390, SL-327, TAK-733, TCS PIM-1 1 (SC 204330), trans-Zeatin, trans-Zeatin-d5, U0124, U0126, U0126-EtOH, XL518 and their pharmaceutically acceptable (known) salts or (known) derivatives. The MEK inhibitor is preferably trametinib, cobimetinib, binimetinib, selumetinib, mirdametinib, pelitinib, refametinib, zapnometinib, pimasertib, SL-327, or a pharmaceutically acceptable (known) salt or (known) derivative thereof, and more preferably trametinib, cobimetinib, selumetinib, refametinib, SL-327, or a pharmaceutically acceptable (known) derivative thereof.

[0133] Trametinib (GSK1120212/JTP-74057) has a CAS number represented by 871700-17-3, an IUPAC name of N-[3-[3-cyclopropyl-5-(2-fluoro-4-iodoanilino)-6,8-dimethyl-2,4,7-trioxypyrido[4,3-d]pyrimidin-1-yl]phenyl]acetamide, a molecular formula of $C_{26}H_{23}FIN_5O_4$, and a molecular weight of 615.4. Trametinib is poorly soluble in itself and as such, is also used as DMSO solvate thereof. The trametinib DMSO solvate has a CAS number represented by 1187431-43-1, an IUPAC name of N-[3-[3-cyclopropyl-5-(2-fluoro-4-iodoanilino)-6,8-dimethyl-2,4,7-trioxypyrido[4,3-d]pyrimidin-1-yl]phenyl]acetamide;methylsulfinylmethane, a molecular formula of $C_{28}H_{29}FIN_5O_5S$, and a molecular weight of 693.5. Trametinib simply described herein is a generic name for a free form and DMSO solvate of trametinib.

[0134] Cobimetinib inhibits the kinase activity of MEK1 and MEK2 and has a CAS number represented by 934660-93-2, an IUPAC name of [3,4-difluoro-2-(2-fluoro-4-iodoanilino)phenyl]-[3-hydroxy-3-[(2S)-piperidin-2-yl]azetidin-1-yl]methanone, a molecular formula of $C_{21}H_{21}F_3IN_3O_2$, and a molecular weight of 531.3. Cobimetinib is also used as a fumarate salt thereof, which has a CAS number represented by 1369665-02-0, an IUPAC name of (E)-but-2-enedioic acid;[3,4-difluoro-2-(2-fluoro-4-iodoanilino)phenyl]-[3-hydroxy-3-[(2S)-piperidin-2-yl]azetidin-1-yl]methanone, a molecular formula of $(C_{21}H_{21}F_3IN_3O_2)_2 \cdot C_4H_4O_4$, and a molecular weight of 1178.7. Cobimetinib simply described herein is a generic name for a free form and fumarate salt of cobimetinib.

[0135] Selumetinib inhibits the kinase activity of MEK. Selumetinib has a CAS number represented by 606143-52-6, an IUPAC name of 6-(4-bromo-2-chloroanilino)-7-fluoro-N-(2-hydroxyethoxy)-3-methylbenzimidazole-5-carboxamide, a molecular formula of $C_{17}H_{15}BrClFN_4O_3$, and a molecular weight of 457.7. Selumetinib is also used as a sulfate salt thereof, which has a CAS number represented by 943332-08-9, an IUPAC name of 6-(4-bromo-2-chloroanilino)-7-fluoro-N-(2-hydroxyethoxy)-3-methylbenzimidazole-5-carboxamide;sulfuric acid, a molecular formula of $C_{17}H_{15}BrClFN_4O_3 \cdot H_2SO_4$, and a molecular weight of 555.8. Selumetinib simply described herein is a generic name for a free form and sulfate salt of selumetinib.

[0136] Refametinib inhibits the kinase activities of MEK1 and MEK2. Refametinib has a CAS number represented by 923032-37-5, an IUPAC name of N-[3,4-difluoro-2-(2-fluoro-4-iodoanilino)-6-methoxyphenyl]-1-[(2S)-2,3-dihydroxypropyl]cyclopropane-1-sulfonamide, a molecular formula of $C_{19}H_{20}F_3IN_2O_5S$, and a molecular weight of 572.3.

[0137] SL-327 inhibits the kinase activities of MEK1 and MEK2. SL-327 has a CAS number represented by 305350-87-2, an IUPAC name of (Z)-3-amino-3-(4-aminophenyl)sulfanyl-2-[2-(trifluoromethyl)phenyl]prop-2-enenitrile, a molecular formula of $C_{16}H_{12}F_3N_3S$, and a molecular weight of 335.3.

ERK Inhibitor

[0138] The ERK inhibitor is a drug having the action of inhibiting the activity of ERK, one of the proteins constituting the MAPK signaling pathway. ERK (extracellular signal-regulated kinase) is a serine/threonine kinase and generally refers to isoforms of ERK1 of 44 kDa and ERK2 of 42 kDa. When low-molecular-weight G protein RAS is activated, RAF and MEK are sequentially activated. The activated MEK phosphorylates tyrosine and threonine in the activation loop of ERK so that the kinase activity of the ERK itself is turned on. This series of signal transduction events is a mainstream of the MAP kinase pathway and called RAS-RAF-MEK-ERK signal transduction cascade. The activated ERK is known to phosphorylate diverse substrates and thereby participate in the exertion of various cell functions such as cell proliferation, differentiation, or survival, cell motility, or new tumor formation.

[0139] Specific examples of the ERK inhibitor include tizaterkib (WO2017/080979A1), alobresib (GS-5829), bortezomib (PS-341), bosutinib (SKI-606), edaxeterkib, ravoxertinib (GDC-0994), rineterkib, temuterkib (LY3214996), ulixertinib (BVD-523), ERK1/2 inhibitor 4 (WO2020/238776), ERK1/2 inhibitor 5 (WO2020/238776), ERK1/2 inhibitor 6 (WO2021/063335; compound 1), ERK1/2 inhibitor 7 (WO2021/110168; WX006), ERK1/2 inhibitor 8 (WO2021/110168; WX007), ERK1/2 inhibitor 3 (WO2021/218912; compound 1), (E)-Osmundacetone, (rel)-AR234960, 2,5-Dihydroxyacetophenone, ACA-28, Adjudin, AG126, AKT-IN-11, Astragaloside IV, ASTX-029, AZD0364 (ATG-017), Bohemine, C16-PAF, Cafestol, CC-90003, Cearoin, Chicanine, CHPG, CHPG sodium salt, CKLF1-C27 TFA, CKLF1-C27, Corynoxene, Cucurbitacin IIb, DEL-22379, Deltonin, DMU-212, EF24, Enniatin A1, Enniatin B, Enniatin B1, epi-Eriocalyxin A, ERK1/2 inhibitor 1, ERK1/2 inhibitor 2, ERK2 IN-1, ERK-IN-2, ERK-IN-3, ERK-IN-3 benzenesulfonate, FR 180204, Gypenoside L, HIOC, Hirsutenone, HI-TOPK-032, Honokiol, I-191, KO-947, Lidocaine, LM22B-10, Longdaysin, Loureirin B, Magnolin, Methylthiouracil, MK-8353 (SCH900353), Mogrol, Motixafortide (BL-8040), MRTX-1257, Mulberroside A, Nitidine chloride, NMDAR/TRPM4-IN-2 free base, Omtripitolide, Pachymic acid, Pamoic acid disodium, Pamoic acid, PD98059, Piperlongumine, Pluripotin (SC1), SCH772984, Tauroursodeoxycholate, TBHQ, TCS ERK 11e, Tenuifolioside A, TIC10 (ONC201), trans-Zeatin, trans-Zeatin-d5, Urolithin B, VX-11e, Withanolide B, Xantocillin, Yoda 1, ZM 241385 and their pharmaceutically acceptable (known) salts or (known) derivatives. The ERK inhibitor is preferably tizaterkib, alobresib, bortezomib, bosutinib, edaxeterkib, ravoxertinib, rineterkib, rineterkib hydrochloride, temuterkib, ulixertinib, SCH772984, or a pharmaceutically acceptable (known) salt or (known) derivative thereof, and more preferably, bosutinib, SCH772984, or a pharmaceutically acceptable (known) derivative thereof.

[0140] Bosutinib (SKI-606) is a Src/Abl inhibitor and also inhibits MAPK/ERK. Bosutinib has a CAS number represented by 380843-75-4, an IUPAC name of 4-(2,4-dichloro-

5-methoxyanilino)-6-methoxy-7-[3-(4-methylpiperazin-1-yl)propoxy]quinoline-3-carbonitrile, a molecular formula of $C_{26}H_{29}Cl_2N_5O_3$, and a molecular weight of 530.4. Bosutinib is also used as a hydrate thereof, which has a CAS number represented by 918639-08-4, an IUPAC name of 4-(2,4-dichloro-5-methoxyanilino)-6-methoxy-7-[3-(4-methylpiperazin-1-yl)propoxy]quinoline-3-carbonitrile;hydrate, a molecular formula of $C_{26}H_{29}Cl_2N_5O_3 \cdot H_2O$, and a molecular weight of 548.5. Bosutinib simply described herein is a generic name for a free form and hydrate of bosutinib.

[0141] SCH772984 inhibits the kinase activity of ERK1/2 and has a CAS number represented by 942183-80-4, an IUPAC name of (3R)-1-[2-oxo-2-[4-(4-pyrimidin-2-ylphenyl)piperazin-1-yl]ethyl]-N-(3-pyridin-4-yl-1H-indazol-5-yl)pyrrolidine-3-carboxamide, a molecular formula of $C_{33}H_{33}N_9O_2$, and a molecular weight of 587.7.

Other Drugs

[0142] The medicament of the present invention may comprise, as an active ingredient, an antitumor agent known in literatures, etc., in addition to the anti-CAPRIN-1 antibody and the MAPK pathway inhibitor, as long as it does not inhibit the effects as the medicament of the present invention. Specific examples of known antitumor agents include, but are not particularly limited to, 5-fluorouracil, irinotecan, oxaliplatin, carboplatin, cisplatin, nedaplatin, gemcitabine, paclitaxel, nabpaclitaxel, imiquimod, immune checkpoint inhibitors, doxorubicin, daunorubicin, cyclophosphamide, methotrexate, thiopeta, busulfan, improsulfan, piposulfan, benzodopa, carboquone, meturedopa, uredopa, alretamine, triethylenemelamine, triethylenephosphoramide, triethylenethiophosphoramide, trimethylolomelamine, bullatacin, bullatacinone, camptothecin, bryostatin, calystatin, cryptophycin 1, cryptophycin 8, dolastatin, duocarmycin, eleutherobin, pancratistatin, sarcodictyin, spongistatin, chlorambucil, chlornaphazine, cholophosphamide, estramustine, ifosfamide, mechlorethamine, mechlorethamine oxide hydrochloride, melphalan, novembichin, phenesterine, prednimustine, trofosfamide, uracil mustard, carmustine, chlorozotocin, fotemustine, lomustine, nimustine, ranimustine, calicheamicin, dynemicin, clodronate, esperamicin, aclacinomycin, actinomycin, authramycin, azaserine, bleomycin, cactinomycin, carabacin, carminomycin, carzinophillin, chromomycin, dactinomycin, detorubicin, 6-diazo-5-oxo-L-norleucine, adriamycin, epirubicin, esorubicin, idarubicin, marcellomycin, mitomycin C, mycophenolic acid, nogalamycin, olivomycin, peplomycin, potfiromycin, puro-mycin, quelamycin, rodorubicin, streptonigrin, streptozocin, tubercidin, ubenimex, zinostatin, zorubicin, denopterin, pteropterin, trimetrexate, fludarabine, 6-mercaptopurine, thiamiprine, thioguanine, ancitabine, azacitidine, 6-azauridine, carmofur, cytarabine, dideoxyuridine, doxifluridine, enocitabine, floxuridine, androgens such as calusterone, dromostanolone propionate, epitostanol, mepitostane, testosterone, aminoglutethimide, mitotane, trilostane, frolinic acid, aceglutone, aldophosphamide glycoside, aminolevulinic acid, eniluracil, amsacrine, bestabucil, bisantrene, edatraxate, defofamine, demecolcine, diaziquone, elfornithine, elliptinium acetate, epothilone, etoglucid, lentinan, lonidamine, maytansine, ansamitocine, mitogazone, mitoxantrone, mopidanmol, nitraerine, pentostatin, phenamet, pirarubicin, losoxantrone, podophyllinic acid, 2-ethylhydrazide, procarbazine, razoxane, rhizoxin, schizophyllan, spirogermanium, tenuazonic acid, triaziquone, roridine A,

anguidine, urethane, vindesine, dacarbazine, mannomustine, mitobronitol, mitolactol, pipobroman, gacytosine, docetaxel, chlorambucil, gemcitabine, 6-thioguanine, mercaptopurine, vinblastine, etoposide, ifosfamide, mitoxantrone, vincristine, vinorelbine, novantrone, teniposide, edatraxate, daunomycin, aminopterin, xeloda, ibandronate, difluoromethylornithine (DMFO), topoisomerase inhibitors, retinoic acid, folinic acid, and their pharmaceutically acceptable (known) salts or (known) derivatives.

Antitumor Effect of Present Invention

[0143] The antitumor effect of the combination of the anti-CAPRIN-1 antibody and the MAPK pathway inhibitor according to the present invention can be evaluated in vivo or in vitro. The in vivo antitumor effect can be evaluated by administering the anti-CAPRIN-1 antibody and the MAPK pathway inhibitor to an organism having cancer, measuring the size of a tumor after the administration, and examining the size of the cancer over time. Also, the in vivo antitumor effect can be evaluated by examining a survival rate of organisms. Alternatively, the in vivo antitumor effect may be evaluated by examining the ability to produce cytokines or chemokines. The in vivo antitumor effect can be further evaluated by examining prevention of cancer, prevention of metastasis or prevention of recurrence.

[0144] The in vitro antitumor effect can be evaluated by examining the cytotoxic activity or phagocytic activity against cancer cells by immunocytes in coculturing the cancer cells and the immunocytes. Thus, the antitumor effect of the combination of the anti-CAPRIN-1 antibody and the MAPK pathway inhibitor can be evaluated by adding the anti-CAPRIN-1 antibody and the MAPK pathway inhibitor in combination to a coculture system of cancer cells and immunocytes, and examining the cytotoxic activity or phagocytic activity against the cancer cells by the immunocytes. The immunocyte for use here may be any cell as long as the immunocyte is a blood cell having cytotoxic activity or phagocytic activity. A human NK cell is preferred for evaluating the cytotoxic activity, and human monocytic cells (THP-1 or U937) are preferred for evaluating the phagocytic activity. When an antibody binds to a cancer cell, this is recognized by an immunocyte so that the cancer cell is killed via cytotoxic activity or phagocytic activity by the immunocytes. Therefore, an in vivo antitumor effect can be predicted by evaluating the in vitro antitumor effect.

[0145] An ability of an anti-CAPRIN-1 antibody used in the present invention to bind to CAPRIN-1 can be determined via binding assay using, for example, ELISA, a Western blot method, immunofluorescence, or flowcytometry analysis.

[0146] The combination of the anti-CAPRIN-1 antibody and the MAPK pathway inhibitor according to the present invention increases an in vitro antitumor effect as compared with the anti-CAPRIN-1 antibody alone. The rate of increase is, for example, 1.1 or more times, preferably 1.5 or more times, more preferably 2 or more times, and further preferably 3 or more times.

Medicament for Treatment and/or Prevention of Cancer

[0147] A medicament of the present invention is aimed at treating and/or preventing cancer. A cancer targeted by the medicament of the present invention is not particularly

limited as long as it is a cancer (cell) expressing CAPRIN-1 protein, particularly, a cancer (cell) expressing CAPRIN-1 protein on a cell membrane surface.

[0148] The term “treatment” used herein refers to treatment of cancer based on an antitumor effect mentioned above. The term “prevention” used herein refers to not only prevention of development of cancer but prevention of metastasis or recurrence of cancer.

[0149] Both the terms “tumor” and “cancer” used herein refer to malignant neoplasm, and thus they are used in an exchangeable manner.

[0150] Cancer that can be a target in the present invention is any cancer as long as the cancer expresses CAPRIN-1 protein on a cell membrane surface. The cancer is preferably melanoma, lung cancer, thyroid cancer, colon cancer, prostate cancer, ovarian cancer, pancreatic cancer, kidney cancer, breast cancer, gastric cancer, bile duct cancer, renal cell cancer, Hodgkin’s lymphoma, head and neck cancer, mesothelioma, colorectal cancer, esophageal cancer, gastroesophageal cancer, hepatocellular cancer, glioblastoma, urothelial cancer, urinary bladder cancer, uterus cancer, primary central nervous system lymphoma, primary testicular lymphoma, biliary tract cancer, brain tumor, leukemia, lymphoma, liver cancer, sarcoma, fibrosarcoma, mastocytoma, adrenal cortex cancer, Ewing’s tumor, multiple myeloma, testicle cancer, basal cell cancer, Paget’s disease or skin cancer, and more preferably melanoma, non-small cell lung cancer, thyroid cancer, colon cancer, or prostate cancer. These cancers may be primary cancers, metastatic cancers, cancers that have metastasized, cancers that have recurred, postoperative cancers, or unresectable cancers. These cancers may have a specific gene mutation. Examples of the gene mutation include BRAF gene mutations, KRAS gene mutations, and EGFR gene mutations. The term melanoma is often used interchangeably with malignant melanoma.

[0151] More specifically, examples of the cancer include, but are not limited to, for example, Bowen’s disease, prickle cell cancer, extramammary Paget’s disease, mycosis fungoides, Sezary’s syndrome, cutaneous T/NK-cell lymphoma, T-cell leukemia or lymphoma having a lesion only in the skin, cutaneous B-cell lymphoma (indolent group), cutaneous T-cell lymphatic or mammary adenoma, complex mammary adenoma, malignant mixed tumor of mammary gland, intraductal papillary carcinoma of mammary gland, lung adenocarcinoma, squamous cell cancer, small cell cancer, large cell cancer, glioma which is a neuroepithelial tissue tumor, glioblastoma, neuroblastoma, ependymoma, neuronal tumor, neuroectodermal tumor, neurilemoma, neurofibromatosis, meningioma, chronic lymphocytic leukemia, lymphoma, alimentary lymphoma, gastrointestinal lymphoma, small to medium cell lymphoma, cecal cancer, ascending colon cancer, descending colon cancer, transverse colon cancer, sigmoid colon cancer, rectal cancer, ovarian epithelial cancer, germ cell tumor, interstitial cell tumor, pancreatic duct cancer, invasive pancreatic duct cancer, adenocarcinoma of pancreatic cancer, acinar cell cancer, adenosquamous cancer, giant cell tumor, intraductal papillary mucinous tumor, mucinous cystadenocarcinoma, pancreatoblastoma, pancreatic islet cell tumor, Frantz’s tumor, serous cystadenocarcinoma, solid pseudopapillary cancer, gastrinoma, glucagonoma, insulinoma, multiple endocrine neoplasia type-1 (Wermer syndrome), nonfunctional islet cell tumor, somatostatinoma, VIPoma, uterine cervical can-

cer, uterine body cancer, fibrosarcoma, osteosarcoma, joint sarcoma, Ewing sarcoma, Wilms’s tumor, hepatoblastoma, soft tissue sarcoma, acute leukemia, chronic leukemia, spinal cord tumor, malignant soft tissue tumor, tumors of teratoma group, and head and neck cancer including hypopharynx cancer, oropharynx cancer, tongue cancer, nasopharyngeal cancer, oral cavity cancer, lip cancer, sinus cancer, voice box cancer, cancer of the renal pelvis and ureter, urinary bladder cancer, urethra cancer, testicular tumor, malignant pleural mesothelioma, malignant bone tumor, uterine body cancer (postoperative chemotherapy, chemotherapy at the time of metastasis or relapse), and pediatric malignant solid tumor (rhabdomyosarcoma, neuroblastoma, hepatoblastoma, medulloblastoma, nephroblastoma, retinoblastoma, central nervous system germ cell tumor, and Ewing sarcoma family of tumors). The cancer also includes a palpable cancer, a subcutaneously existing cancer, an intracutaneously existing cancer, a superficial cancer, cancer existing in the dermis or cancer existing in a non-parenchymal organ, and a progressive cancer which originate from the cancers described above. The cancer also includes a palpable cancer, a subcutaneously existing cancer, an intracutaneously existing cancer, a superficial cancer, cancer existing in the dermis, and cancer existing in a non-parenchymal organ, which originate from the cancers described above and have metastasized and recurred.

[0152] A preferable subject (patient) that can be a target is a mammal and is, for example, a mammal including primates, pet animals, livestock animals, and sport animals. Humans, dogs and cats are particularly preferable.

[0153] A medicament of the present invention can be formulated by a method known to persons skilled in the art. For instance, the medicament of the present invention can be parenterally used in the form of a parenteral injection of: an aseptic solution comprising water or a pharmacologically acceptable non-water solution; or a suspension liquid. For each formulation or pharmaceutical composition in the medicament of the present invention, the active ingredient (at least one of the anti-CAPRIN-1 antibody and the MAPK pathway inhibitor) may be appropriately combined with, for example, a pharmacologically acceptable carrier, medium or additive; specifically, sterilized water, physiological saline, an isotonic solution, a buffer (buffer solution, etc.), plant oil, an oily solution, an antioxidant, a solubilizer, an emulsifier, a suspension, a surfactant, a stabilizer, a fragrance, an excipient, or a binder, and preferably, may be formulated by mixing with them in a unit dosage form required for a generally acceptable pharmaceutical formulation. An amount of an active ingredient in a formulation is determined such that an appropriate dosage within an indicated range can be achieved.

[0154] An aseptic composition for injection can be prepared in accordance with general formulation practice using a vehicle such as distilled water for injection. An aqueous solution for injection purposes includes, for example, physiological saline or isotonic solutions comprising glucose and other adjuvants such as D-sorbitol, D-mannose, D-mannitol, and sodium chloride. Such solution may be used with an appropriate dissolution aid. Such dissolution aid includes, for example, alcohols such as ethanol and polyalcohol, such as propylene glycol, polyethylene glycol, or nonion surfactants such as polysorbate 80™ and HCO-60. Oily liquid includes, for example, sesame oil or soybean oil. Such oily liquid may be used in combination with a dissolution aid

such as benzyl benzoate or benzyl alcohol. In addition, it may be mixed with a buffering agent such as a phosphate buffer solution or a sodium acetate buffer solution, a soothing agent such as procaine hydrochloride, a stabilizer such as benzyl alcohol or phenol, or an antioxidant. In general, a formulated injection solution is introduced into an adequate ample.

[0155] The above pharmaceutical composition is orally or parenterally administered. Preferably, it is parenterally administered. Specifically, dosage forms include injectable agents, intranasally-administered agents, transpulmonarily-administered agents, and percutaneously-administered agents. For example, injectable agents can be systemically or locally administered via intravenous injection, intramuscular injection, intraperitoneal injection, subcutaneous injection, or intratumoral injection. The percutaneously-administered agents include, for example, agents called liniments and external medicines. The external medicines include, for example, solid agents, solutions, sprays, ointments, creams, and gels.

[0156] The administration method can be appropriately determined depending on age, weight, gender, and symptoms of a patient. A single dose of a pharmaceutical composition comprising at least one of the anti-CAPRIN-1 antibody and the MAPK pathway inhibitor can be selected within a range of, for example, 0.0001 mg to 1000 mg per kg of body weight as the amount of each active ingredient. Alternatively, the dose of respective active ingredients can be selected within a range of, for example, 0.001 to 100000 mg per patient's body, or 0.1 to 300 mg or 1 mg to 30 mg per kg of patient's body weight; however, it is not necessarily limited thereto. The dose and the administration method are changed depending on patient age, weight, gender, and symptoms. However, persons skilled in the art can appropriately select the dose and the method.

Administration Method

[0157] Treatment and/or prevention of cancer with a medicament for treatment and/or prevention of cancer of the present invention includes various modes, in addition to administration as a medicament mentioned above. For example, respective active ingredients in a medicament of the present invention can be administered simultaneously, concurrently, or individually in a staggered manner. As a specific example, active ingredients can be administered within a time interval up to approximately 3 weeks, i.e., the second active ingredient can be administered from immediately up to approximately 3 weeks after administration of the first active ingredient. These administrations may be carried out subsequently to a surgical procedure, or a surgical procedure may be carried out between the administrations of the first and second drugs. In addition, a medicament for treatment and/or prevention of cancer of the present invention may be administered according to a plurality of administration cycles. For example, in the case of carrying out simultaneous administration of respective active ingredients in the medicament for treatment and/or prevention of cancer of the present invention, a pharmaceutical composition comprising both active ingredients is administered once per approximately 2 days to approximately 3 weeks as one cycle. Then, this treatment cycle may be repeated, if necessary, according to the judgment of a physician in charge. Likewise, in the case of scheduling a formulation in a staggered manner, respective administration periods of indi-

vidual agents are adjusted so as to span the same period. The interval between cycles can vary from 0 to 2 months. Respective doses of the active ingredients in the medicament for treatment and/or prevention of cancer of the present invention can be set in the same way as in the respective doses of the active ingredients in the pharmaceutical composition.

Pharmaceutical Kit

[0158] A medicament for treatment and/or prevention of cancer of the present invention may be in the form of a pharmaceutical kit. The pharmaceutical kit is a package for using active ingredients in the form of separate pharmaceutical compositions (formulations) in a method for treating and/or preventing cancer. The package may comprise an instruction for administering each of the active ingredients. The respective active ingredients in the pharmaceutical compositions for treatment and/or prevention of cancer contained in the pharmaceutical kit can be in the form of pharmaceutical compositions each formulated as described above such that the active ingredients can be administered together or separately. Further, the pharmaceutical kit comprises active ingredients in amounts sufficient for one or more doses such that the active ingredients can be administered according to the administration method described above.

Treatment and/or Prevention Method

[0159] On the basis of the contents specifically described above, the present invention further provides a method for treating and/or preventing cancer, comprising administering the medicament of the present invention, or the anti-CAPRIN-1 antibody and the MAPK pathway inhibitor according to the present invention to a subject (patient). For example, the present invention further provides a method for treating and/or preventing cancer, comprising administering the medicament, etc. of the present invention to a subject (patient) having cancer or suspected of having cancer. In this embodiment, for example, the anti-CAPRIN-1 antibody (antibody or fragment thereof) and the MAPK pathway inhibitor according to the present invention, and an optional antitumor agent contained in the medicament can be administered simultaneously or separately to the subject (patient).

EXAMPLES

[0160] The present invention is hereafter described in detail with reference to the following examples, although the scope of the present invention is not limited thereto.

(Example 1) Production of Anti-CAPRIN-1 Antibody

[0161] One hundred (100) μ g of a human CAPRIN-1 recombinant protein produced according to Example 3 of WO2010/016526 was mixed with a MPL+TDM adjuvant (Sigma) in an amount equivalent to that of the recombinant protein. The mixture was used as an antigen solution per mouse. The antigen solution was administered intraperitoneally to 6-week-old Balb/c mice (Japan SLC Inc.) and then further administered 3 times and 24 times every week for completion of immunization. A spleen was removed on day 3 after the final immunization and then ground between two sterilized glass slides. Spleen cells were obtained by repeating the following procedure three times: washing with PBS

(-) (Nissui Corporation), centrifuging at 1500 rpm for 10 minutes, and removing supernatant. The obtained spleen cells were mixed with mouse myeloma cell SP2/0 (purchased from ATCC) at a ratio of 10:1. The PEG solution prepared by mixing 200 μ l of RPMI1640 medium containing 10% FBS heated at 37° C. and 800 μ l of PEG1500 (Boehringer) was added to the cells. The solution was incubated for 5 minutes for cell fusion. Centrifugation was performed at 1700 rpm for 5 minutes to remove supernatants. Cells were suspended in 150 ml of RPMI1640 medium (HAT selective medium) containing 15% FBS, to which 2% equivalent of HAT solution (Gibco) had been added, and then seeded onto fifteen 96-well plates (Nunc) at 100 μ l per well. Cells were cultured for 7 days under conditions of 37° C. and 5% CO₂, so that hybridomas resulting from fusion of spleen cells to myeloma cells were obtained. Hybridomas were selected using binding affinity to CAPRIN-1 protein of the antibody produced by the prepared hybridomas as an indicator. The CAPRIN-1 protein solution (1 μ g/ml) was added at 100 μ l per well of 96-well plates and then incubated at 4° C. for 18 hours. After each well was washed 3 times with PBS-T, 0.5% Bovine Serum Albumin (BSA) solution (Sigma) was added at 400 μ l per well, and then the plates were incubated at room temperature for 3 hours. The solution was removed and then each well was washed 3 times with 400 μ l of PBS-T. Each culture supernatant of the hybridomas obtained above was added at 100 μ l per well and then incubated at room temperature for 2 hours. After each well was washed 3 times with PBS-T, an HRP-labeled anti-mouse IgG (H+L) antibody (Invitrogen) diluted 5000-fold with PBS was added at 100 μ l per well and then incubated at room temperature for 1 hour. After each well was washed 3 times with PBS-T, a TMB substrate solution (Thermo) was added at 100 μ l per well and then incubated for 15-30 minutes, so that a color reaction was performed. After color development, 1 N sulfuric acid was added at 100 μ l per well to stop the reaction. Absorbance at 450 nm and absorbance at 595 nm were measured using an absorption spectrometer. As a result, a plurality of hybridomas producing antibodies with high absorbances were selected. The selected hybridomas were added at 0.5 hybridomas per well of 96-well plates and then cultured. After 1 week, hybridomas forming single colonies in wells were observed. Cells in these wells were further cultured. Hybridomas were selected using binding affinity to CAPRIN-1 protein of the antibody produced by cloned hybridomas as an indicator. The CAPRIN-1 protein solution (1 μ g/ml) was added at 100 μ l per well of 96-well plates and then incubated at 4° C. for 18 hours. Each well was washed 3 times with PBS-T, a 0.5% BSA solution was added at 400 μ l per well, and then incubated at room temperature for 3 hours. The solution was removed and then each well was washed 3 times with 400 μ l of PBS-T. Each culture supernatant of the hybridomas obtained above was added at 100 μ l per well and then incubated at room temperature for 2 hours. Each well was washed 3 times with PBS-T, an HRP-labeled anti-mouse IgG (H+L) antibody (Invitrogen) diluted 5000-fold with PBS was added at 100 μ l per well and then incubated at room temperature for 1 hour. Each well was washed 3 times with PBS-T, a TMB substrate solution (Thermo) was added at 100 μ l per well and then incubated for 15-30 minutes, so that a color reaction was performed. After color development, 1 N sulfuric acid was added at 100 μ l per well to stop the reaction. Absorbance at 450 nm and absorbance at 595 nm

were measured using an absorption spectrometer. As a result, a plurality of mouse monoclonal antibodies exerting reactivity with CAPRIN-1 protein were obtained.

[0162] Reactivity of each monoclonal antibody with human cancer cells confirmed to express CAPRIN-1 protein on cell membrane surfaces was further confirmed by flow cytometry. A mouse IgG control antibody exhibiting no reactivity with the cancer cells was used as a negative control. As a result of confirmation, several monoclonal antibodies were obtained which had stronger fluorescence intensity against the cancer cells than that of the mouse IgG control antibody and reacted strongly with the cell membrane surfaces of the cancer cells expressing CAPRIN-1 on the cell membrane surfaces. From among them, a monoclonal antibody against CAPRIN-1 described in WO2013/125630, which was an antibody comprising the amino acid sequence of a heavy-chain variable region shown by SEQ ID NO: 114 and the amino acid sequence of a light-chain variable region shown by SEQ ID NO: 115, was selected as a monoclonal antibody exhibiting reactivity with CAPRIN-1 protein.

[0163] CDR1 to CDR3 of the heavy-chain variable region of the antibody selected were identified. A nucleotide sequence was designed so as to be able to express a heavy-chain variable region having framework regions comprising a human antibody sequence. This nucleotide sequence was inserted to a vector for mammalian expression having an insert of a human IgG1 heavy-chain constant region. Likewise, CDR1 to CDR3 of the light-chain variable region were identified. A nucleotide sequence was designed so as to be able to express a light-chain variable region having framework regions comprising a human antibody sequence. This nucleotide sequence was inserted to a vector for mammalian expression having an insert of a human IgG1 light-chain constant region. These two recombinant expression vectors were introduced to mammalian cells according to a general method and then a culture supernatant containing humanized monoclonal antibody #1 (humanized antibody #1) against CAPRIN-1 was obtained. The amino acid sequences of heavy chain CDR1, CDR2, and CDR3 of the humanized monoclonal antibody #1 are shown by SEQ ID NOs: 272, 273, and 274, respectively. The amino acid sequences of light chain CDR1, CDR2, and CDR3 of the humanized monoclonal antibody #1 are shown by SEQ ID NOs: 275, 276, and 277, respectively.

[0164] The obtained culture supernatant containing the obtained humanized anti-CAPRIN-1 monoclonal antibody #1 was purified using Hitrap Protein A Sepharose FF (GE Healthcare Bio-Sciences) according to a general method, replaced with PBS(-), and filtered through a 0.22 μ m filter (Millipore) for preparation of the filtrate.

[0165] The specific reactivity of the anti-CAPRIN-1 antibody to CAPRIN-1 protein was detected and confirmed by ELISA using CAPRIN-1 protein immobilized on a plate.

[0166] The reactivity of the anti-CAPRIN-1 antibody with cancer cells without permeation treatment of cell membranes was examined by flow cytometry to confirm that a portion of CAPRIN-1 was expressed on cell membrane surfaces of cancer cells as shown in Examples given below.

[0167] It was confirmed by flow cytometry that, against all of breast cancer cells (BT-474), colon cancer cells (HT-29 and HCT116), lung cancer cells (QG56, H1650, and A549), gastric cancer cells (NCI-N87), uterus cancer cells (HEC-1-A), prostate cancer cells (22Rv1 and DU145), pancreatic

cancer cells (Panc10.5), liver cancer cells (Hep3B), ovarian cancer cells (SKOV3), kidney cancer cells (Caki-2), brain tumor cells (U-87 MG), urinary bladder cancer cells (T24), esophageal cancer cells (OE33), leukemia cells (OCI-AML5), lymphoma cells (Ramos), gallbladder cancer cells (TGBC14TKB), fibrosarcoma cells (HT-1080), and melanoma cells (G-361 and A375), which are human cancer cells confirmed to express CAPRIN-1 gene, and mouse kidney cancer cells (Renca) and mouse breast cancer cells (4T1) confirmed to express CAPRIN-1 gene, the humanized antibody #1 had stronger fluorescence intensity than that of a human IgG control antibody and rabbit IgG antibody serving as negative controls exhibiting no reactivity with the cancer cells and reacted strongly with the cell membrane surfaces of the cancer cells expressing CAPRIN-1.

[0168] Likewise, the anti-CAPRIN-1 antibodies described in WO2010/016526, WO2011/096517, WO2011/096528, WO2011/096519, WO2011/096533, WO2011/096534, WO2011/096535, WO2013/018886, WO2013/018894, WO2013/018892, WO2013/018891, WO2013/018889, WO2013/018883, WO2013/125636, WO2013/125654, WO2013/125640, WO2013/147169, WO2013/147176 and WO2015/020212 were also confirmed to react strongly with the cancer cell membrane surfaces.

(Example 2) In Vitro Antitumor Effect of
Combination of Anti-CAPRIN-1 Antibody and
Trametinib

[0169] Antitumor effects of combinations of an anti-CAPRIN-1 antibody and a MEK inhibitor trametinib were evaluated in vitro. Antitumor effects of combinations of an anti-CAPRIN-1 antibody and an existing chemotherapeutic fluorouracil (5-FU) or irinotecan (CPT-11) were evaluated simultaneously therewith and compared with the antitumor effect of the combination of anti-CAPRIN-1 antibody and trametinib. The trametinib used was Trametinib (Cat. No. CS-0060) purchased from ChemScene LLC.

[0170] Specifically, in each combination test group (a trametinib combination test group, a fluorouracil combination test group and an irinotecan combination test group), the indicated drug used in combination was applied to human cancer cells, and then the cancer cells were cocultured with human monocytic cells (THP-1) in the presence of the anti-CAPRIN-1 antibody, and the antibody-mediated phagocytic activity against the cancer cells by THP-1 was evaluated. The human-derived cancer cells used were melanoma cell line A375 (used only in the trametinib combination test group), colon cancer cell line HCT116, prostate cancer cell line DU145 (used only in the trametinib combination test group), and lung cancer cell line A549 (used only in the trametinib combination test group). On a 6-well plate, the human-derived cancer cells were cultured for 2 days in the presence of trametinib for the trametinib combination test group, and cultured for 2 days in the presence of the each drug used in combination for the fluorouracil combination test group and the irinotecan combination test group. The concentration of the drug used in combination was set to a concentration that decreased the proliferation rate of the cancer cells by about half at the completion of culture. Specifically, for HCT116, trametinib was added at a concentration of 50 nM, fluorouracil was added at a concentration of 0.4 $\mu\text{g/mL}$, and irinotecan was added at a concentration of 15 μM ; trametinib was added to A375 at a concentration of 50 nM; trametinib was added to DU145 at

a concentration of 250 nM; and trametinib was added to A549 at a concentration of 250 nM. As a control, a test group without a drug used in combination was established. In the control group, only DMSO (dimethyl sulfoxide), a solvent for the drug used in combination, was added to the cancer cells at a final concentration of 0.1%, and then the cancer cells were cultured for 2 days.

[0171] Each cancer cell line after the culture was dissociated with TrypLE Express (Thermo), and the cancer cells were stained by the addition of calcein-AM at a final concentration of 0.02 $\mu\text{g/mL}$ and incubation at 37° C. for 30 minutes. Next, the cells were dispensed at 1×10^4 cancer cells per well to a 96-well plate, and the anti-CAPRIN-1 antibody (final concentration: 5 $\mu\text{g/mL}$) and 1×10^5 THP-1 cells were added to each well, followed by culture at 37° C. for 2.5 hours under a condition of 5% CO_2 . Then, the cells were washed with PBS (phosphate buffer solution) containing 1% FBS (fetal bovine serum), and the human monocytic cells were stained through a reaction with an APC (allophycocyanin)-labeled anti-human CD45 antibody (final concentration: 0.25 $\mu\text{g/mL}$). Further, dead cells were stained with propidium iodide (PI) (final concentration: 0.1 $\mu\text{g/mL}$), followed by the measurement of fluorescence from the cells by flow cytometry. PI-positive dead cells were excluded from analysis. The human monocytic cells THP-1 recognize an antibody bound onto cancer cells and phagocytize the cancer cells. At that time, if the cancer cells have been stained with calcein-AM, the monocytic cells ingesting the cancer cells by phagocytosis also become positive to calcein-AM. Thus, the phagocytic activity in this evaluation system was calculated as the ratio (%) of APC-positive/calcein-AM-positive cells to the whole calcein-AM-positive population.

[0172] As a result of evaluation using the anti-CAPRIN-1 antibody prepared in Example 1 (anti-CAPRIN-1 humanized antibody #1) as an anti-CAPRIN-1 antibody, phagocytic activity against cancer cells of 18%, 11%, 4%, and 4% was observed for A375, HCT116, DU145, and A549, respectively, in the test group without a drug used in combination. By contrast, phagocytic activity against cancer cells of 49%, 45%, 9%, and 7% was observed for A375, HCT116, DU145, and A549, respectively, in the trametinib combination test group. Thus, the antitumor effect of the anti-CAPRIN-1 antibody on these four human cancer cells was increased by approximately 1.8 to 4 times by using the anti-CAPRIN-1 antibody in combination with trametinib (FIG. 1).

[0173] For HCT116, phagocytic activity against cancer cells was 45% in the trametinib combination test group, whereas the phagocytic activity against cancer cells was 16% and 14% in the fluorouracil combination test group and the irinotecan combination test group, respectively. Thus, the combination of the anti-CAPRIN-1 antibody and trametinib strongly increased the phagocytic activity against cancer cells as compared with the combinations of the anti-CAPRIN-1 antibody and the existing chemotherapeutics described above (FIG. 2).

[0174] Combinations of any anti-CAPRIN-1 antibody described in WO2010/016526, WO2011/096517, WO2011/096528, WO2011/096519, WO2011/096533, WO2011/096534, WO2011/096535, WO2013/018886, WO2013/018894, WO2013/018892, WO2013/018891, WO2013/018889, WO2013/018883, WO2013/125636, WO2013/125654, WO2013/125640, WO2013/147169, WO2013/147176 and WO2015/020212 and trametinib exhibit similar

increase in phagocytic activity against the above human cancer cells as in the combination of the anti-CAPRIN-1 humanized antibody #1 and trametinib.

(Example 3) In Vitro Antitumor Effect of
Combination of Anti-CAPRIN-1 Antibody and
RAS Inhibitor

[0175] Antitumor effects of combinations of anti-CAPRIN-1 antibodies and RAS inhibitors were evaluated in vitro. The RAS inhibitors used were lonafarnib (Cat. No. S2797), sotorasib (Cat. No. S8830), and salirasib (Cat. No. S7684) purchased from Selleck Chemicals LLC.

[0176] Specifically, in each RAS inhibitor combination test group, the indicated drug used in combination was applied to human cancer cells, and then the cancer cells were cocultured with human monocytic cells (THP-1) in the presence of the anti-CAPRIN-1 antibody, and the antibody-mediated phagocytic activity against the cancer cells by THP-1 was evaluated.

[0177] The human-derived cancer cells used were a colon cancer cell line HCT116. On a 6-well plate, the human-derived cancer cells were cultured for 2 days in the presence of the indicated drug used in combination in the lonafarnib combination test group, the sotorasib combination test group and the salirasib combination test group. Specifically, for HCT116, lonafarnib was added at a concentration of 10 μ M, sotorasib was added at a concentration of 10 μ M, and salirasib was added at a concentration of 100 μ M. As a control, a test group without a drug used in combination was established. In the control group, the cancer cells were cultured beforehand for 2 days in a state without addition of the drug used in combination.

[0178] Each cancer cell line after the culture was dissociated with TrypLE Express (Thermo), and the cancer cells were stained by the addition of calcein-AM at a final concentration of 0.04 μ g/mL and incubation at 37° C. for 10 minutes. Next, the cells were dispensed at 5×10^3 cancer cells per well to a 96-well plate, and the anti-CAPRIN-1 antibody (final concentration: 1 μ g/mL) and 1.25×10^5 THP-1 cells were added to each well, followed by culture at 37° C. for 1 hour under a condition of 5% CO₂. Then, the cells were washed with PBS (phosphate buffer solution) containing 1% FBS (fetal bovine serum), and the human monocytic cells were stained through a reaction with an APC (allophycocyanin)-labeled anti-human CD45 antibody (final concentration: 0.25 μ g/mL). Further, dead cells were stained with propidium iodide (PI) (final concentration: 0.1 μ g/mL), followed by the measurement of fluorescence from the cells by flow cytometry. PI-positive dead cells were excluded from analysis. The human monocytic cells THP-1 recognize an antibody bound onto cancer cells and phagocytize the cancer cells. At that time, if the cancer cells have been stained with calcein-AM, the monocytic cells ingesting the cancer cells by phagocytosis also become positive to calcein-AM. Thus, the phagocytic activity in this evaluation system was calculated as the ratio (%) of APC-positive/calcein-AM-positive cells to the whole calcein-AM-positive population.

[0179] As a result of evaluation using the anti-CAPRIN-1 antibody prepared in Example 1 (anti-CAPRIN-1 humanized antibody #1) as an anti-CAPRIN-1 antibody, phagocytic activity against cancer cells of 50% was observed for HCT116 in the test group without a drug used in combination. By contrast, phagocytic activity against cancer cells of

59% to 80% was observed for all the lonafarnib, sotorasib, and salirasib combination test groups. As a result of conducting Student's t test, the phagocytic activity of the lonafarnib, sotorasib, and salirasib combination test groups was significantly higher than that of the test group without a drug used in combination ($p < 0.001$, $p < 0.01$, and $p < 0.001$, respectively; significance level: 5%). Thus, the antitumor effect of the anti-CAPRIN-1 antibody on the human cancer cells was significantly increased by approximately 1.2 to 1.6 times by using the anti-CAPRIN-1 antibody in combination with lonafarnib, sotorasib, or salirasib (FIG. 3).

[0180] Combinations of any anti-CAPRIN-1 antibody described in WO2010/016526, WO2011/096517, WO2011/096528, WO2011/096519, WO2011/096533, WO2011/096534, WO2011/096535, WO2013/018886, WO2013/018894, WO2013/018892, WO2013/018891, WO2013/018889, WO2013/018883, WO2013/125636, WO2013/125654, WO2013/125640, WO2013/147169, WO2013/147176 and WO2015/020212 and lonafarnib, sotorasib, or salirasib exhibit similar increase in phagocytic activity against human cancer cells as in the combination of the anti-CAPRIN-1 humanized antibody #1 and lonafarnib, sotorasib, or salirasib.

(Example 4) In Vitro Antitumor Effect of
Combination of Anti-CAPRIN-1 Antibody and
RAF Inhibitor

[0181] Antitumor effects of combinations of anti-CAPRIN-1 antibodies and RAF inhibitors were evaluated in vitro. The RAF inhibitors used were dabrafenib (Cat. No. CS-0692) and ZM 336372 (Cat. No. CS-0693) purchased from ChemScene LLC.

[0182] Specifically, in each RAF inhibitor combination test group, the indicated drug used in combination was applied to human cancer cells, and then the cancer cells were cocultured with human monocytic cells (THP-1) in the presence of the anti-CAPRIN-1 antibody, and the antibody-mediated phagocytic activity against the cancer cells by THP-1 was evaluated.

[0183] The human-derived cancer cells used were a colon cancer cell line HCT116. On a 6-well plate, the human-derived cancer cells were cultured for 2 days in the presence of the indicated drug used in combination in the dabrafenib combination test group and the ZM 336372 combination test group. Specifically, for HCT116, dabrafenib was added at a concentration of 10 μ M and ZM 336372 was added at a concentration of 100 μ M. As a control, a test group without a drug used in combination was established. In the control group, the cancer cells were cultured beforehand for 2 days in a state without addition of the drug used in combination.

[0184] Each cancer cell line after the culture was dissociated with TrypLE Express (Thermo), and the cancer cells were stained by the addition of calcein-AM at a final concentration of 0.04 μ g/mL and incubation at 37° C. for 10 minutes. Next, the cells were dispensed at 5×10^3 cancer cells per well to a 96-well plate, and the anti-CAPRIN-1 antibody (final concentration: 1 μ g/mL) and 1.25×10^5 THP-1 cells were added to each well, followed by culture at 37° C. for 1 hour under a condition of 5% CO₂. Then, the cells were washed with PBS (phosphate buffer solution) containing 1% FBS (fetal bovine serum), and the human monocytic cells were stained through a reaction with an APC (allophycocyanin)-labeled anti-human CD45 antibody (final concentration: 0.25 μ g/mL). Further, dead cells were stained with

propidium iodide (PI) (final concentration: 0.1 $\mu\text{g/mL}$), followed by the measurement of fluorescence from the cells by flow cytometry. PI-positive dead cells were excluded from analysis. The human monocytic cells THP-1 recognize an antibody bound onto cancer cells and phagocytize the cancer cells. At that time, if the cancer cells have been stained with calcein-AM, the monocytic cells ingesting the cancer cells by phagocytosis also become positive to calcein-AM. Thus, the phagocytic activity in this evaluation system was calculated as the ratio (%) of APC-positive/calcein-AM-positive cells to the whole calcein-AM-positive population.

[0185] As a result of evaluation using the anti-CAPRIN-1 antibody prepared in Example 1 (anti-CAPRIN-1 humanized antibody #1) as an anti-CAPRIN-1 antibody, phagocytic activity against cancer cells of 23% was observed for HCT116 in the test group without a drug used in combination. By contrast, phagocytic activity against cancer cells of 75% or more was observed in both the dabrafenib and ZM 336372 combination test groups. Thus, the antitumor effect of the anti-CAPRIN-1 antibody on the human cancer cells was increased by approximately 3.3 to 3.5 times by using the anti-CAPRIN-1 antibody in combination with dabrafenib or ZM 336372 (FIG. 4).

[0186] Combinations of any anti-CAPRIN-1 antibody described in WO2010/016526, WO2011/096517, WO2011/096528, WO2011/096519, WO2011/096533, WO2011/096534, WO2011/096535, WO2013/018886, WO2013/018894, WO2013/018892, WO2013/018891, WO2013/018889, WO2013/018883, WO2013/125636, WO2013/125654, WO2013/125640, WO2013/147169, WO2013/147176 and WO2015/020212 and dabrafenib or ZM 336372 exhibit similar increase in phagocytic activity against human cancer cells as in the combination of the anti-CAPRIN-1 humanized antibody #1 and dabrafenib or ZM 336372.

(Example 5) In Vitro Antitumor Effect of
Combination of Anti-CAPRIN-1 Antibody and
MEK Inhibitor-2

[0187] Antitumor effects of combinations of anti-CAPRIN-1 antibodies and MEK inhibitors other than trametinib were evaluated in vitro. The MEK inhibitors used were selumetinib (Cat. No. S1008), refametinib (Cat. No. S1089), and SL-327 (Cat. No. S1066) purchased from Selleck Chemicals LLC, and cobimetinib (Cat. No. 19563) purchased from Cayman Chemical Company.

[0188] Specifically, in each MEK inhibitor combination test group, the indicated drug used in combination was applied to human cancer cells, and then the cancer cells were cocultured with human monocytic cells (THP-1) in the presence of the anti-CAPRIN-1 antibody, and the antibody-mediated phagocytic activity against the cancer cells by THP-1 was evaluated.

[0189] The human-derived cancer cells used were a colon cancer cell line HCT116. On a 6-well plate, the human-derived cancer cells were cultured for 2 days in the presence of the indicated drug used in combination in the selumetinib combination test group, the refametinib combination test group, the SL-327 combination test group and the cobimetinib combination test group. Specifically, for HCT116, selumetinib was added at a concentration of 10 μM , refametinib was added at a concentration of 1000 nM, SL-327 was added at a concentration of 10 μM , and cobi-

metinib was added at a concentration of 800 nM. As a control, a test group without a drug used in combination was established. In the control group, the cancer cells were cultured beforehand for 2 days in a state without addition of the drug used in combination.

[0190] Each cancer cell line after the culture was dissociated with TrypLE Express (Thermo), and the cancer cells were stained by the addition of calcein-AM at a final concentration of 0.04 $\mu\text{g/mL}$ and incubation at 37° C. for 10 minutes. Next, the cells were dispensed at 5×10^3 cancer cells per well to a 96-well plate, and the anti-CAPRIN-1 antibody (final concentration: 1 $\mu\text{g/mL}$) and 1.25×10^5 THP-1 cells were added to each well, followed by culture at 37° C. for 1 hour under a condition of 5% CO_2 . Then, the cells were washed with PBS (phosphate buffer solution) containing 1% FBS (fetal bovine serum), and the human monocytic cells were stained through a reaction with an APC (allophycocyanin)-labeled anti-human CD45 antibody (final concentration: 0.25 $\mu\text{g/mL}$). Further, dead cells were stained with propidium iodide (PI) (final concentration: 0.1 $\mu\text{g/mL}$), followed by the measurement of fluorescence from the cells by flow cytometry. PI-positive dead cells were excluded from analysis. The human monocytic cells THP-1 recognize an antibody bound onto cancer cells and phagocytize the cancer cells. At that time, if the cancer cells have been stained with calcein-AM, the monocytic cells ingesting the cancer cells by phagocytosis also become positive to calcein-AM. Thus, the phagocytic activity in this evaluation system was calculated as the ratio (%) of APC-positive/calcein-AM-positive cells to the whole calcein-AM-positive population.

[0191] As a result of evaluation using the anti-CAPRIN-1 antibody prepared in Example 1 (anti-CAPRIN-1 humanized antibody #1) as an anti-CAPRIN-1 antibody, phagocytic activity against cancer cells of 50% was observed for HCT116 in the c test group without a drug used in combination. By contrast, phagocytic activity against cancer cells of 93% or more was observed for all the selumetinib, refametinib, SL-327, and cobimetinib combination test groups. Thus, the antitumor effect of the anti-CAPRIN-1 antibody on the human cancer cells was significantly increased by 1.8 times by using the anti-CAPRIN-1 antibody in combination with selumetinib, refametinib, SL-327, or cobimetinib (FIG. 3).

[0192] Combinations of any anti-CAPRIN-1 antibody described in WO2010/016526, WO2011/096517, WO2011/096528, WO2011/096519, WO2011/096533, WO2011/096534, WO2011/096535, WO2013/018886, WO2013/018894, WO2013/018892, WO2013/018891, WO2013/018889, WO2013/018883, WO2013/125636, WO2013/125654, WO2013/125640, WO2013/147169, WO2013/147176 and WO2015/020212 and selumetinib, refametinib, SL-327, or cobimetinib exhibit similar increase in phagocytic activity against human cancer cells as in the combination of the anti-CAPRIN-1 humanized antibody #1 and selumetinib, refametinib, SL-327, or cobimetinib.

(Example 6) In Vitro Antitumor Effect of
Combination of Anti-CAPRIN-1 Antibody and
ERK Inhibitor

[0193] Antitumor effects of combinations of anti-CAPRIN-1 antibodies and ERK inhibitors were evaluated in vitro. The ERK inhibitors used were bosutinib (Cat. No.

S1014) purchased from Selleck Chemicals LLC and SCH772984 (Cat. No. 19166) purchased from Cayman Chemical Company.

[0194] Specifically, in each ERK inhibitor combination test group, the indicated drug used in combination was applied to human cancer cells, and then the cancer cells were cocultured with human monocytic cells (THP-1) in the presence of the anti-CAPRIN-1 antibody, and the antibody-mediated phagocytic activity against the cancer cells by THP-1 was evaluated.

[0195] The human-derived cancer cells used were a colon cancer cell line HCT116. On a 6-well plate, the human-derived cancer cells were cultured for 2 days in the presence of the indicated drug used in combination in the SCH772984 combination test group and the bosutinib combination test group. Specifically, for HCT116, SCH772984 was added at a concentration of 10 μ M and bosutinib was added at a concentration of 10 μ M. As a control, a test group without a drug used in combination was established. In the control group, and the cancer cells were cultured beforehand for 2 days in a state without addition of the drug used in combination.

[0196] Each cancer cell line after the culture was dissociated with TrypLE Express (Thermo), and the cancer cells were stained by the addition of calcein-AM at a final concentration of 0.04 μ g/mL and incubation at 37° C. for 10 minutes. Next, the cells were dispensed at 5×10^3 cancer cells per well to a 96-well plate, and the anti-CAPRIN-1 antibody (final concentration: 1 μ g/mL) and 1.25×10^5 THP-1 cells were added to each well, followed by culture at 37° C. for 1 hour under a condition of 5% CO₂. Then, the cells were washed with PBS (phosphate buffer solution) containing 1% FBS (fetal bovine serum), and the human monocytic cells were stained through a reaction with an APC (allophycocyanin)-labeled anti-human CD45 antibody (final concentration: 0.25 μ g/mL). Further, dead cells were stained with propidium iodide (PI) (final concentration: 0.1 μ g/mL), followed by the measurement of fluorescence from the cells by flow cytometry. PI-positive dead cells were excluded from analysis. The human monocytic cells THP-1 recognize an antibody bound onto cancer cells and phagocytize the cancer cells. At that time, if the cancer cells have been stained with calcein-AM, the monocytic cells ingesting the

cancer cells by phagocytosis also become positive to calcein-AM. Thus, the phagocytic activity in this evaluation system was calculated as the ratio (%) of APC-positive/calcein-AM-positive cells to the whole calcein-AM-positive population.

[0197] As a result of evaluation using the anti-CAPRIN-1 antibody prepared in Example 1 (anti-CAPRIN-1 humanized antibody #1) as an anti-CAPRIN-1 antibody, phagocytic activity against cancer cells of 39% was observed for HCT116 in the test group without a drug used in combination. By contrast, phagocytic activity against cancer cells of 90% was observed for the SCH772984 combination test group (FIG. 5). As a result of conducting Student's t test, the phagocytic activity of the SCH772984 combination test group was significantly higher than that of the test group without a drug used in combination ($p < 0.001$; significance level: 5%). Also, phagocytic activity against cancer cells of 50% was observed for HCT116 in the test group without a drug used in combination. By contrast, phagocytic activity against cancer cells of 82% was observed in the bosutinib combination test group (FIG. 3). As a result of conducting Student's t test, the phagocytic activity of the bosutinib combination test group was significantly higher than that of the test group without a drug used in combination ($p < 0.001$; significance level: 5%). Thus, the antitumor effect of the anti-CAPRIN-1 antibody on the human cancer cells was significantly increased by approximately 1.7 to 2.3 times by using the anti-CAPRIN-1 antibody in combination with bosutinib or SCH772984.

[0198] Combinations of any anti-CAPRIN-1 antibody described in WO2010/016526, WO2011/096517, WO2011/096528, WO2011/096519, WO2011/096533, WO2011/096534, WO2011/096535, WO2013/018886, WO2013/018894, WO2013/018892, WO2013/018891, WO2013/018889, WO2013/018883, WO2013/125636, WO2013/125654, WO2013/125640, WO2013/147169, WO2013/147176 and WO2015/020212 and bosutinib or SCH772984 exhibit similar increase in phagocytic activity against human cancer cells as in the combination of the anti-CAPRIN-1 humanized antibody #1 and bosutinib or SCH772984.

[0199] All publications, patents and patent applications cited herein are incorporated herein by reference in their entirety.

SEQUENCE LISTING

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 organism = Homo sapiens

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 note = Canis lupus familiaris
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SEQ ID NO: 24 moltype = AA length = 698
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 source 1..698
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 24

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SEQ ID NO: 25 moltype = DNA length = 3548
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 organism = Mus musculus

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SEQ ID NO: 26      moltype = AA length = 692
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SEQUENCE: 29

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tcgggttcaga	ccgagggccat	gaagcagatc	ttgggagtg	tcgacaaaa	gctccgcaac	180
ctcgagaaga	aaaagagcaa	acttgacgat	taccaggaa	gaatgaacaa	gggggaacgt	240
ctaaatcaag	atcaactgga	tgcagtgtca	aaataccagg	aagtgcacaa	taacctggaa	300
ttcgctaaag	aactgcagag	gagctttatg	gcactgagcc	aagatatcca	gaaaaacaata	360
aaaaagacgg	ctcgagggga	gcagctgatg	agagaagagg	ctgagcagaa	gcgttttaaag	420
actgtgctag	agctgcagtt	cattttggac	aagttgggtg	acgatgaagt	gcgcagtgac	480
ttgaaacaag	gatcaaatgg	agtaccggta	ctgacagagg	aggaaactgac	aatgctggat	540
gaattttaca	agctagttta	ccctgaaaagg	gacatgaaca	tgaggttgaa	tgagcagtat	600
gagcaagcat	ctgttcacct	gtgggactta	ctggaaggga	aggaaaaacc	cgtttgtgga	660
acaaacctata	aagccctgaa	ggagggttgt	gaacgtattc	ttcaaaactag	ttactttgat	720
agcaccata	accatcagaa	cgggttatgt	gaggaagaag	aggcagcacc	cacacctgca	780
gtagaagaca	ctgtagcaga	agctgagcct	gatccagcag	aagaatttac	tgaacctact	840
gaagttgaat	cgactgagta	tgtaaacaga	caattcatgg	cagagactca	gttcagcagt	900
agtgagaagg	aacaggtaga	tgagtggaca	gttgaaacgg	ttgaggttgt	aaattcactg	960
gcgaacacaa	cacaagctac	atctctctca	gttctgtaac	ctcatacact	cactactgtg	1020
gctcaagcag	atcctcttgt	tagaagacag	agagtacagg	accttatggc	ccagatgcag	1080
ggtccatata	acttcacgca	ggactctatg	ctggagtttg	agaaccagac	acttgatcct	1140
gcatttgtat	ctccattcca	catgaatcca	gcacagaatt	tggacatgcc	gcaaatggtc	1200
tgccctccag	ttcatactga	gtcaagactt	gcccagccta	atcaagttcc	tgtgcaacca	1260
gaagctacgc	aggttcctct	ggtttcatct	acaagttagg	gatatacagc	ctcccagccc	1320
atgtatcagc	ctctatcaca	cacagagcaa	cggccacaga	aggaaatccat	tgaccagatt	1380
caggcttcaa	tgtcactgaa	tgcagaccag	accccgctcat	catcatcact	tcccactgca	1440
tcccagccgc	aagttttcca	agctggatct	agcaaacctt	tgcatagcag	cggaaatcaat	1500
gttaatgcag	ctccattcca	atccatgcaa	acagtattca	acatgaatgc	acctgttcct	1560
cctgttaatg	agccagaagc	ccttaagcaa	caaaatcagt	accaggccag	ttacaaccag	1620
agtttctcca	atcagccaca	ccaagtagaa	caatcagatc	ttcagcaaga	acagctccag	1680
acagtgtgtg	gtacttacc	tggttctccg	gaccagaccc	atcaagtggc	aggaaaccac	1740
cagcaaacctc	cccagcagaa	tactggattt	ccacgcaaca	gtcagcctta	ttacaacagt	1800
cggggagtg	ctcgtgtgtg	atcacgtggg	actcgtggat	tgatgaatgg	ttacagggga	1860
cctgcaaatg	gatttagagg	aggatatgat	ggctaccgtc	cttcattttc	caacactccg	1920
aacagtgtgt	acacgcagcc	ccaattta	gtctcctcag	attattcaaa	ctaccagcgg	1980
gatggatc	agcagaactt	caaacgtgtg	tctggacaaa	gtgggcctcg	gggagctcct	2040
cgaggtcgtg	gagggccccc	aagaccacaa	agagggatgc	ctcaaatgaa	cgctcagcaa	2100
gtgaattaa						2109

SEQ ID NO: 30 moltype = AA length = 702
 FEATURE Location/Qualifiers
 source 1..702
 mol_type = protein
 organism = Gallus gallus

SEQUENCE: 30

MPSATNGTMA	SSSGKAGPGG	NEQAPAAAAA	APQASGGSIT	SVQTEAMKQI	LGVIDKKLRN	60
LEKKKSKLDD	YQERMNKGRL	LNQDQLDAVS	KYQEVNTNNLE	FAKELQRSFM	ALSQDIQKTI	120
KKTARREQLM	REEAEQKRLK	TVLELQFILD	KLGDDEVRS	LKQGSNGVPV	LTEELTMLD	180
EFYKLVYPER	DMNRLNEQY	EQASVHLWDL	LEGKEKPVCG	TTYKALKEVV	ERILQTSYFD	240
STHNHQNGLC	EEEEEAPTPA	VEDTVAEAE	DPAEFEPTTE	EVETVEVNR	QFMAETQFSS	300
SEKEQVDEWT	VETVEVNSL	QQQTQATSP	VPEPHTLTTV	AQADPLVRRQ	RVQDLMAQMQ	360
GPYNFMQDSM	LEFENQTLDP	AIVSAQPMNP	AQNLDMPQMV	CPPVHTESRL	AQPNQVPVQ	420
EATQVPLVSS	TSEGYTASQ	MYQPSHTTEQ	RPQKESIDQI	QASMSLNADQ	TPSSSLPTA	480
SQPVQFQAGS	SKPLHSSGIN	VNAAPFQSMQ	TVFNMNAPVP	PVNEPEALKQ	QNQYQASYNQ	540
SFSNQPHQVE	QSDLQQEQLQ	TVVGTVHGSP	DQTHQVAGNH	QQQPQQNTGF	PRNSQPYNNS	600
RGVSRGSGRG	TRGLMNGYRG	PANGFRGGYD	GYRPSFSNTP	NSGYTQPFN	APRDYSNYQR	660
DGYQQNFKRG	SGQSGPRGAP	RGRGGPPRPN	RGMPQMNQVN			702

SEQ ID NO: 31 moltype = AA length = 63
 FEATURE Location/Qualifiers
 source 1..63

-continued

	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 31		
EYTEQSEVE STEYVNRQFM AETQFTSGEK EQVDEWTVET VEVVNSLQQQ PQAASPSVPE		60
PHS		63
SEQ ID NO: 32	moltype = AA length = 18	
FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 32		
VFNMNAPVPP VNEPETLK		18
SEQ ID NO: 33	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 33		
ATQVPLVSST SEGYYTA		16
SEQ ID NO: 34	moltype = AA length = 25	
FEATURE	Location/Qualifiers	
source	1..25	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 34		
QILGVIDKKL RNLEKKKGLK DDYQE		25
SEQ ID NO: 35	moltype = AA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 35		
PRGRGGPPRP NRGMPQMTQ QVN		23
SEQ ID NO: 36	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
source	1..5	
	mol_type = protein	
	organism = Gallus gallus	
SEQUENCE: 36		
SYQMN		5
SEQ ID NO: 37	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = Gallus gallus	
SEQUENCE: 37		
AINKFGNSTG HGAAVKG		17
SEQ ID NO: 38	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = protein	
	organism = Gallus gallus	
SEQUENCE: 38		
HAYGYCGSGT WCAAGEIDA		19
SEQ ID NO: 39	moltype = AA length = 128	
FEATURE	Location/Qualifiers	
source	1..128	
	mol_type = protein	
	organism = Gallus gallus	
SEQUENCE: 39		
AVTLDESGGG LQMSRGGSL VCKASGFDFS SYQMNWIRQA PGKGLEFVAA INKFGNSTGH		60
GAAVKGRVTI SRDNGQSTVR LQLNNLRAED TAIYFCTKHA YGYCGSGTWC AAGEIDAWGH		120
GTEIVVSS		128
SEQ ID NO: 40	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	

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SEQUENCE: 40	organism = Gallus gallus	
SGGGSYSYG		9
SEQ ID NO: 41	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = Gallus gallus	
SEQUENCE: 41		
NNKRPSD		7
SEQ ID NO: 42	moltype = AA length = 10	
FEATURE	Location/Qualifiers	
source	1..10	
	mol_type = protein	
	organism = Gallus gallus	
SEQUENCE: 42		
SGDSTDVAF		10
SEQ ID NO: 43	moltype = AA length = 108	
FEATURE	Location/Qualifiers	
source	1..108	
	mol_type = protein	
	organism = Gallus gallus	
SEQUENCE: 43		
QAASTQPSSV SANPGETVEI TCSGGGSYSY GWFQOKSPGS APVTVIYYNN KRPSDIPSRF		60
SGSKSGSTGT LTITGVQADD EAVYYCGSGD STDVAVFGAG TTLTVLGQ		108
SEQ ID NO: 44	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
source	1..5	
	mol_type = protein	
	organism = Oryctolagus cuniculus	
SEQUENCE: 44		
SHSLG		5
SEQ ID NO: 45	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = Oryctolagus cuniculus	
SEQUENCE: 45		
DIRSGGSAYY ANWAKG		16
SEQ ID NO: 46	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
source	1..13	
	mol_type = protein	
	organism = Oryctolagus cuniculus	
SEQUENCE: 46		
TNGPSDLTNR LDL		13
SEQ ID NO: 47	moltype = AA length = 121	
FEATURE	Location/Qualifiers	
source	1..121	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 47		
EQSLVESGGG LVQPGGSLRL SCAASGFSLs SHSLGWVRQA PGKGLEWIGD IRSGGSAYYA		60
NWAKGRFTIS RDNSKNTLYL QMNSLRAEDT AVYYCTRNG PSDLTNRLDL WQGTLVTVS		120
S		121
SEQ ID NO: 48	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
source	1..13	
	mol_type = protein	
	organism = Oryctolagus cuniculus	
SEQUENCE: 48		
QASQSLYNNE NLA		13
SEQ ID NO: 49	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = Oryctolagus cuniculus	

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SEQUENCE: 49 GASTLAS		7
SEQ ID NO: 50 FEATURE source	moltype = AA length = 13 Location/Qualifiers 1..13 mol_type = protein organism = Oryctolagus cuniculus	
SEQUENCE: 50 LGEFSCGSAD CFA		13
SEQ ID NO: 51 FEATURE source	moltype = AA length = 112 Location/Qualifiers 1..112 mol_type = protein organism = Homo sapiens	
SEQUENCE: 51 QVLTQSPSSL SASVGDRVTI NCQASQSLYN NENLAWFQOK PGKVPKRLLIY GASTLASGVS SRFSGSGSGT EFTLTISLQ CEDFAIYYCL GEFSCGSADC FAFGGGKVE IK		60 112
SEQ ID NO: 52 FEATURE source	moltype = AA length = 4 Location/Qualifiers 1..4 mol_type = protein organism = Gallus gallus	
SEQUENCE: 52 PDMG		4
SEQ ID NO: 53 FEATURE source	moltype = AA length = 17 Location/Qualifiers 1..17 mol_type = protein organism = Gallus gallus	
SEQUENCE: 53 QINDAGSRTW YATAVKG		17
SEQ ID NO: 54 FEATURE source	moltype = AA length = 12 Location/Qualifiers 1..12 mol_type = protein organism = Gallus gallus	
SEQUENCE: 54 GSGYVGAGAI DA		12
SEQ ID NO: 55 FEATURE source	moltype = AA length = 120 Location/Qualifiers 1..120 mol_type = protein organism = Gallus gallus	
SEQUENCE: 55 AVTLDESGGG LQTPGGGLSL VCKASGFTFS SFDMGWVRQA PGKGLEFVAQ INDAGSRTWY ATAVKGRATI SRDNGQTTVR LQNNLRAED TGTYYCTRGS GYVGAGAIDA WGHGTEVIVS		60 120
SEQ ID NO: 56 FEATURE source	moltype = AA length = 8 Location/Qualifiers 1..8 mol_type = protein organism = Gallus gallus	
SEQUENCE: 56 SGSGYYG		8
SEQ ID NO: 57 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = Gallus gallus	
SEQUENCE: 57 NDKRPSD		7
SEQ ID NO: 58 FEATURE source	moltype = AA length = 10 Location/Qualifiers 1..10 mol_type = protein organism = Gallus gallus	
SEQUENCE: 58 RYDSTDGIF		10

SEQ ID NO: 59	moltype = AA	length = 105	
FEATURE	Location/Qualifiers		
source	1..105		
	mol_type = protein		
	organism = Gallus gallus		
SEQUENCE: 59			
AALTQPSVS ANPGETVKIT	CSGGSGGYGW	YQQQKSPGSA	PVTVIYQNDK RPSDIPSRFS 60
GS GSGSTNTL TITGVQAEDE	AVYFCGRYDS	TDSGIFGAGT	TLTVL 105
SEQ ID NO: 60	moltype = AA	length = 6	
FEATURE	Location/Qualifiers		
source	1..6		
	mol_type = protein		
	organism = Oryctolagus cuniculus		
SEQUENCE: 60			
GSYYMS			6
SEQ ID NO: 61	moltype = AA	length = 17	
FEATURE	Location/Qualifiers		
source	1..17		
	mol_type = protein		
	organism = Oryctolagus cuniculus		
SEQUENCE: 61			
YIYIGDGVTA YANWAKG			17
SEQ ID NO: 62	moltype = AA	length = 4	
FEATURE	Location/Qualifiers		
source	1..4		
	mol_type = protein		
	organism = Oryctolagus cuniculus		
SEQUENCE: 62			
GNKL			4
SEQ ID NO: 63	moltype = AA	length = 112	
FEATURE	Location/Qualifiers		
source	1..112		
	mol_type = protein		
	organism = Oryctolagus cuniculus		
SEQUENCE: 63			
QSLEESGGDL VKPGASLTLT	CTASGFSPSG	SYYMSWVRQA	PGKGLEWIIAY IYIGDGVTA 60
ANWAKGRFTI SKASSTTVTL	QMTSLTAADT	ATYFCARGNK	LWGPGLTVTV SS 112
SEQ ID NO: 64	moltype = AA	length = 11	
FEATURE	Location/Qualifiers		
source	1..11		
	mol_type = protein		
	organism = Oryctolagus cuniculus		
SEQUENCE: 64			
QASQSISSYL A			11
SEQ ID NO: 65	moltype = AA	length = 7	
FEATURE	Location/Qualifiers		
source	1..7		
	mol_type = protein		
	organism = Oryctolagus cuniculus		
SEQUENCE: 65			
DASNLDS			7
SEQ ID NO: 66	moltype = AA	length = 14	
FEATURE	Location/Qualifiers		
source	1..14		
	mol_type = protein		
	organism = Oryctolagus cuniculus		
SEQUENCE: 66			
QCTAVSSATI YGNA			14
SEQ ID NO: 67	moltype = AA	length = 112	
FEATURE	Location/Qualifiers		
source	1..112		
	mol_type = protein		
	organism = Oryctolagus cuniculus		
SEQUENCE: 67			
DVVMQTQTPAS VEAAVGGTVT	IKCQASQSI	SYLAWYQQKP	GQPPKRLIYD ASNLD SGVPS 60
RFKGGSGGTD FTITISDLEC	ADAATYYCOC	TAVSSATIYG	NAFGGGTEVV VK 111

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SEQ ID NO: 68 moltype = AA length = 127
 FEATURE Location/Qualifiers
 source 1..127
 mol_type = protein
 organism = Gallus gallus

SEQUENCE: 68
 AVTLDESGGG LQTPGGALSL VCKASGFTFS GYDMLWVRQA PGKGLEWVAG IGSTGGGTDY 60
 GAAVKGRATI SRDNGQSTVR LQLNNLRAED TATYYCAKVA GGCNSGYCRD SPGSIDAWGH 120
 GTEVIVS 127

SEQ ID NO: 69 moltype = AA length = 107
 FEATURE Location/Qualifiers
 source 1..107
 mol_type = protein
 organism = Gallus gallus

SEQUENCE: 69
 AVTQQPASVS ANPGETVKIT CSGGGSRNYY GWYQQKSPGS VPVTVIYYDD QRPSNIPSRF 60
 SGALSGSTST LTITGVQADD EAVYFCGSAD SNTYEGSFGA GTTLTLVL 107

SEQ ID NO: 70 moltype = AA length = 148
 FEATURE Location/Qualifiers
 source 1..148
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 70
 MEWSGVFIPL LSGTAGVLSE VQLHQFGAEL VKPGASVKIS CKASGYTFTD YNMDWVKQSH 60
 GKSLEWIGDI NPNYDSTSYN QKPKGKATLT VDKSSSTAYM ELRSLTSEDV AVYYCARSRS 120
 YDYGFPAYWG QGTLVTVSAA KTTPPSVY 148

SEQ ID NO: 71 moltype = AA length = 105
 FEATURE Location/Qualifiers
 source 1..105
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 71
 GLFCSVERCH YQLQSSQNLL SIVNRYHYMS GNPPKLLVYP ALLIYEASIT KSCVPDRFTR 60
 SSGSTNFTLT INFVHADDLI FYYCQHNRRS FLPSSSVQVP RRRSN 105

SEQ ID NO: 72 moltype = AA length = 109
 FEATURE Location/Qualifiers
 source 1..109
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 72
 PRASLGVSET LLCTSGFTFT DYYMSWVRQP PGKALEWLGF IRNKANGYTT EYSASVKGRF 60
 TISRDNQSIS LYLQMNTLRA EDSATYYCAR ANWAFDYWGQ GTTIVTVSSK 109

SEQ ID NO: 73 moltype = AA length = 94
 FEATURE Location/Qualifiers
 source 1..94
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 73
 SGDRVSLSCR ASQISISNYLH WYQQKSHESP RLLIKYASQS ISGIPSRFSG SSGTDFTTLS 60
 INSVETEDFG MYFCQQNSW PYTFGGGTKL EIKQ 94

SEQ ID NO: 74 moltype = AA length = 118
 FEATURE Location/Qualifiers
 source 1..118
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 74
 AAELVRPGTS VKVCKASGY AFTNYLIVWI KQRPQGGLW IGVISPGSGG TNYNEKFKGK 60
 AILTADKSSS TAYMQLSLST SDEFAVYFCA REKIYDDYYE GYFDVWGAGP RHLLASLS 118

SEQ ID NO: 75 moltype = AA length = 107
 FEATURE Location/Qualifiers
 source 1..107
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 75
 GTRCDIRLTQ TTSSLASASLG DRVITSCSAS LGIGNYLNWY QQKPDGTVKL LIYYTSNLHS 60
 GVPSRFSGSG SGTDSLSTIS NLEPEDIATY YCQHYSKLPL TFGAGPS 107

SEQ ID NO: 76 moltype = AA length = 113
 FEATURE Location/Qualifiers

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source 1..113
mol_type = protein
organism = Mus musculus

SEQUENCE: 76
GAELVRSGAS VKMSCKASGY SFTDYNMYWV KQTPGQGLEW IGYIYPGNGG TNYNQKFKGK 60
ATLTADTSSS TAYMQISSLT SEDSAVYFCA RDYDDGGYAM DYWGQGTTVT VSS 113

SEQ ID NO: 77 moltype = AA length = 117
FEATURE Location/Qualifiers
source 1..117
mol_type = protein
organism = Mus musculus

SEQUENCE: 77
LLLWLTGARC DIQMTQSPAS LSASVGETVT ITCRASGNIH NYLTWYQQKQ GKSPQLLVYN 60
ATLTADGVPS RFGSGSGTQ YSLKINRLQP EDFGSYYCQH FWNIPWTFGG GTKLNSR 117

SEQ ID NO: 78 moltype = AA length = 114
FEATURE Location/Qualifiers
source 1..114
mol_type = protein
organism = Mus musculus

SEQUENCE: 78
DAELVKPGAS VKISCKASGY TFTDHSIHWV QQKPEQGLEW IGYISPGNGN IKYNEKFKGK 60
ATLTADKSSS TAYMQLNLSLT SEDSAVYFCK RSLGRGGPY FDYWGQGTTV TVSS 114

SEQ ID NO: 79 moltype = AA length = 108
FEATURE Location/Qualifiers
source 1..108
mol_type = protein
organism = Mus musculus

SEQUENCE: 79
DIVLTQAAPS LPVTPGESVS ISCRSSKSL LHSNGNTYLYW FLQRPQGSPQ LLIYRMSNLA 60
SGVPRDFSGS GSGTAFTLRI SRVEAEDVG VYCMQHREYP VTFGSGPN 108

SEQ ID NO: 80 moltype = AA length = 111
FEATURE Location/Qualifiers
source 1..111
mol_type = protein
organism = Mus musculus

SEQUENCE: 80
QVQLQQSGAE LMKPGASVKI SCKATGYTFS SYWIEWVKQR PGHGLEWIGE ILPGSGSTNY 60
NEKFKGKATF TADTSSNTAY MQLSLTSSED SAVYYCASY YWFDVWAQDH V 111

SEQ ID NO: 81 moltype = AA length = 109
FEATURE Location/Qualifiers
source 1..109
mol_type = protein
organism = Mus musculus

SEQUENCE: 81
IVMTQAAPSN PVTLGTSASI SCRSSKNLLH SNGITYLYWY LQRPQGSPQL LIYRVSNLAS 60
GVPNRFSGSE SGTDFTLRIS RVEAEDVG VYCAQLLELPY TSEGTRWE 109

SEQ ID NO: 82 moltype = AA length = 109
FEATURE Location/Qualifiers
source 1..109
mol_type = protein
organism = Mus musculus

SEQUENCE: 82
GGGLVKPGGS LKLSCAASGF AFSSYDMSWI RQTPEKRLEW VAYISSGAGS TYYPDVTVKGR 60
FTVSRDNAKN TLYLQMSSLK SEDTAMYCA RHFYRFDYWG QGTTVTVSS 109

SEQ ID NO: 83 moltype = AA length = 113
FEATURE Location/Qualifiers
source 1..113
mol_type = protein
organism = Mus musculus

SEQUENCE: 83
LLLCVSGAPG SIVMTQTPKF LLVSAGDRIT ITCASQSVS NDVAWYQQKP GQSPKLLIYY 60
ASNRYTGVPD RFTGSGYGT DFTFTISTVQA EDLAVYFCQQ DDRFPLTFGA GPS 113

SEQ ID NO: 84 moltype = AA length = 112
FEATURE Location/Qualifiers
source 1..112
mol_type = protein
organism = Mus musculus

SEQUENCE: 84

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QIQLVQSGPE LKKPGETVKI SCKASGYTFT NYGMNWKQA PGKGLKWMGW INTYTGEPTY 60
 ADDFKGRFAP SLETSASTAY LQINNLIKNE TATYFCATGA WFAWAKDSS RH 112

SEQ ID NO: 85 moltype = AA length = 107
 FEATURE Location/Qualifiers
 source 1..107
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 85
 GVEGLDIVMTQ SHKFMSTSVG DRVSITCKAS QDVGTAWAY QQKPGQSPKL LIYWASTRHT 60
 GVPDRFTGSG SGTDFTLTIS NVQSEDLADY FCQQYSSYPL TFGAGPS 107

SEQ ID NO: 86 moltype = AA length = 118
 FEATURE Location/Qualifiers
 source 1..118
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 86
 GGGLVQPGGS MKVSCVASGF SFIDFWMNWV RQSPEKGLEW VAEIRLKSNN YATHYAESVK 60
 GRFTISRDDS KSSVYLQMN LRPEDTGIY CTSLFYDYDG TSGFAYWGQG TTVTVLLK 118

SEQ ID NO: 87 moltype = AA length = 109
 FEATURE Location/Qualifiers
 source 1..109
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 87
 DIVMTQSPSS LTVTAGEKVT MHCKSSQSLL NSGDQKNYLT WYQKPGQPP KLLIYWASTR 60
 ESGVPDRFTG SGSGTDFTLT ISSVQAEDLA VYQCNDYDY PLTFGAGPS 109

SEQ ID NO: 88 moltype = AA length = 148
 FEATURE Location/Qualifiers
 source 1..148
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 88
 MEWSGVFIPL LSGTAGVLSE VQLHQFGAEL VKPGASVKIS CKASGYTFTD YNMDWVKQSH 60
 GKSLIEWIGDI NPNYDSTSYN QKFKGKATLT VDKSSSTAYM ELRSLTSED AVYYCARSRS 120
 YDIEGFAYWG QGTLVTVSAA KTTPPSVY 148

SEQ ID NO: 89 moltype = AA length = 139
 FEATURE Location/Qualifiers
 source 1..139
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 89
 MSVLTVQLGL LLLWLTGARC DIQMTQSPAS LSASVGETVT ITCRASGNIH NYLAWYQQKQ 60
 GKSPQLLVYN AKTLADGVPS RFGSGSGGTQ YSLKINSLOP EDPGSIYQCQ FWSTLTFGAG 120
 TKLELKRADA APTVSNPYD 139

SEQ ID NO: 90 moltype = AA length = 100
 FEATURE Location/Qualifiers
 source 1..100
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 90
 DILQASGYSF TGYTMNVKQ SHGKNLEWIG LINPYNGGTS YNQKFKGKAT LTVDKSSSTA 60
 YMELLSLTSE DSAVYYCARW GVWSAMDYWG QGTTVTVSSK 100

SEQ ID NO: 91 moltype = AA length = 90
 FEATURE Location/Qualifiers
 source 1..90
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 91
 DRVSITCKAS QNVRTAWAY QQKPRQSPKA LIYLASNRDT GLPDRFPGRG SGTDFTLNIT 60
 NVQSEDLADY FCLQHCNYPN EFRGCTKVPI 90

SEQ ID NO: 92 moltype = AA length = 116
 FEATURE Location/Qualifiers
 source 1..116
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 92
 LQESGAELAR PGASVKLSCK ASGYTFTSYW MQWVKQRPQG GLEWIGAIYP GDGDTRYTQK 60
 FKQKATLTAD KSSSTAYMQL SSLASEDSAV YYCARGEYGN YPAYWQGTT VTVSSN 116

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SEQ ID NO: 93 moltype = AA length = 100
 FEATURE Location/Qualifiers
 source 1..100
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 93
 TSDASLGERV TITCKASQDI NSYLSWFQOK PGKSPKTLIY RANRLVDGVP SRFSGSGSGQ 60
 DYSLTISLSE YEDMGIYYCL QYDEFPLTFG GGTKLEIKQK 100

SEQ ID NO: 94 moltype = AA length = 108
 FEATURE Location/Qualifiers
 source 1..108
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 94
 AWLSQLSCTA SGFNIKDTYM HWVKQRPEQG LEWIGRIDPA NGNTKYDPKF QGKATITADT 60
 SSNTAYLQLS SLTSEDVAVY YCARPIHYYY GSSLAYWGGG TTVTVSSK 108

SEQ ID NO: 95 moltype = AA length = 104
 FEATURE Location/Qualifiers
 source 1..104
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 95
 EFHAVSLGQR ATISCRASES VDSYGNSPFMH WYQQKPGQPP KLLIYRASNL ESGIPARFSG 60
 SGRSDFTLT INPVEADDVA TYCQQSNED PGRSEVVPWS RSNK 104

SEQ ID NO: 96 moltype = AA length = 109
 FEATURE Location/Qualifiers
 source 1..109
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 96
 PRASLGVSET LLCTSGFTFT DYYMSWVRQP PGKALEWLG IRNKANGYTT EYSASVKGRF 60
 TISRDNQS SI LYLQMNLT RA EDSATYYCAR ANWAFDYWGQ GTTVTVSSK 109

SEQ ID NO: 97 moltype = AA length = 94
 FEATURE Location/Qualifiers
 source 1..94
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 97
 SGDRVSLSCR ASQSISNYLH WYQQKSHEP RLLIKYASQS ISGIPSRFSG SGSGTDFTLS 60
 INSVETEDFG MYFCQQNSW PYTFGGGTLK EIKQ 94

SEQ ID NO: 98 moltype = AA length = 111
 FEATURE Location/Qualifiers
 source 1..111
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 98
 PACLPGGSLR LSCATSGFTF TDYYMSWVRQ PPGKALEWLG FIRNKANGYT TEYSASVKGR 60
 FTISRDNQS ILYLQMNLT RA AEDSATYYCA RAPLLYYAMD YWQGTTVTV S 111

SEQ ID NO: 99 moltype = AA length = 102
 FEATURE Location/Qualifiers
 source 1..102
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 99
 RLPPFYSLEQR ATISYRASKN VSTSGYSYMH WNQQKPGQPP KLLIYLVSNL ESGVPARFSG 60
 SGSGTDFTLN IHPVEEDAA TYCQHIREL TRSELVPSWK SN 102

SEQ ID NO: 100 moltype = AA length = 101
 FEATURE Location/Qualifiers
 source 1..101
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 100
 VSCKASGYTF TSYWMHWVKQ RPKGGLIEWIG MIDPSNSETR LNQKFKDKAT LNVDKSSNTA 60
 YMQLSSLTSE DSAVYYCARG LRHYWYFDVW GQGTTVTVSS K 101

SEQ ID NO: 101 moltype = AA length = 99
 FEATURE Location/Qualifiers
 source 1..99

-continued

mol_type = protein
organism = Mus musculus

SEQUENCE: 101
TILWREGPFS YRASKSVSTS GYSYMHWNQQ KPGQPPRLLI YLVSNLESGV PARFSGSGSG 60
TDFTLNIHPV EEEDAATYYC QHIRELTRSE EVPSWRSNK 99

SEQ ID NO: 102 moltype = AA length = 110
FEATURE Location/Qualifiers
source 1..110
mol_type = protein
organism = Mus musculus

SEQUENCE: 102
GGGLVKPGGS LKLSCAASGF TFSSYGMSWV RQTPEKRLEW VATISSGGSY TYYPDSVKGR 60
FTISRDNKN TLYLQMSSLR SEDTAMYCA SLASYFYDYW GQGTTLTVSS 110

SEQ ID NO: 103 moltype = AA length = 113
FEATURE Location/Qualifiers
source 1..113
mol_type = protein
organism = Mus musculus

SEQUENCE: 103
GARCDVQMIQ SPSSLSASLG DIVTMTQAS QGTSINLNWF QQKPGKAPKL LIYGASSLED 60
GVPSRFRSGSC FGTDFTLTIS SLEDEDMATY FCLQHSYLPP LTFGAGTKLE LKR 113

SEQ ID NO: 104 moltype = AA length = 111
FEATURE Location/Qualifiers
source 1..111
mol_type = protein
organism = Mus musculus

SEQUENCE: 104
GPGLVQPSQS LSITCTVSGF SLTTYDLHWV RQSPGKGLEW LGVIWSGGST DYNAAFISRL 60
SISKDNSKQ VFFKMNSLQA NDTAIYYCAR NYGSAWPAY WGQGTLVTVS A 111

SEQ ID NO: 105 moltype = AA length = 118
FEATURE Location/Qualifiers
source 1..118
mol_type = protein
organism = Mus musculus

SEQUENCE: 105
PASSSDVLMT QTPLSLPVSL GDQASISCRS SQSIVHSNGN TYLEWYLQKP GQSPKLLIYK 60
VSNRFSGVDP RFGSGSGGTD FTLKISRVEA EDLGVYCFQ GSHVPLTFGA GTKLELKR 118

SEQ ID NO: 106 moltype = AA length = 114
FEATURE Location/Qualifiers
source 1..114
mol_type = protein
organism = Mus musculus

SEQUENCE: 106
GFELKKPGET VKISCKASGY TFTAYSMHWV KQTPGKGLKW LGWINTETGE PTYTDDFKGR 60
FTFSLETSAR IAYLQINDLK NEDTATYFCA RRIYYFGRGG FDYWGQGTTV TVSS 114

SEQ ID NO: 107 moltype = AA length = 118
FEATURE Location/Qualifiers
source 1..118
mol_type = protein
organism = Mus musculus

SEQUENCE: 107
PASSSDVLMT QTPLSLPVRL GDQSSISCRS SQSIVHSNGN TYLEWYLQKP GQSPKLLIYK 60
VSNRFSGVDP RFGSGSGGTD FTLKISRVEP EDLGVYCFQ GSHVPYTSEG DQAEIKLA 118

SEQ ID NO: 108 moltype = AA length = 114
FEATURE Location/Qualifiers
source 1..114
mol_type = protein
organism = Mus musculus

SEQUENCE: 108
GGGLVQPGGS MRLSCVASGF TFSNSWPNWV RQSPEKGLEW VAEIRLTSDN YAIYYAESVK 60
GRFTISRDDS KSSVYLQMNH LRAEDTGIYY CTRPETARAT PAYWGQGTTV TVSS 114

SEQ ID NO: 109 moltype = AA length = 118
FEATURE Location/Qualifiers
source 1..118
mol_type = protein
organism = Mus musculus

SEQUENCE: 109
PASTSDVLMT QTPLSLPVSL GDQASISCRS SQSIVHSNGN TYLEWYLQKP GQSPKVLIIK 60

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VFNRFSGVDP RFSGSGSGTD FTLKISRVEA EDLGVYYCFQ GSHVPRTFGG GTKLNQTG 118
 SEQ ID NO: 110 moltype = AA length = 111
 FEATURE Location/Qualifiers
 source 1..111
 mol_type = protein
 organism = Mus musculus
 SEQUENCE: 110
 GPDLVKPGAS VKISCKASGY SFTAYMHVW QSHGKSLEW IGRVNPNNGG TTYNQKFKGK 60
 AILTVDKSSS TAYMELRSLT FEDSAVYCA RRIYGYFDY WQGTTVTVS S 111
 SEQ ID NO: 111 moltype = AA length = 104
 FEATURE Location/Qualifiers
 source 1..104
 mol_type = protein
 organism = Mus musculus
 SEQUENCE: 111
 AFFAVSLGQR ATISCKASQS VDYDGD SYMN WYQKPGQPP KLLIYVASNL ESGVPARFSG 60
 SSGTDFTLN IHPVEEDAA TYCQQSNEP PYTFGGGTLK EIKQ 104
 SEQ ID NO: 112 moltype = AA length = 106
 FEATURE Location/Qualifiers
 source 1..106
 mol_type = protein
 organism = Mus musculus
 SEQUENCE: 112
 GAELVKPGAS VKLSCTASGL NIRDIYMHVW KQRPEQGLEW IGKIDPANGN TKYDPKFKGK 60
 ATITADTSSN TAYVQLSSLT SEDTAVYCA GTGDYWGQGT TVTVSS 106
 SEQ ID NO: 113 moltype = AA length = 112
 FEATURE Location/Qualifiers
 source 1..112
 mol_type = protein
 organism = Mus musculus
 SEQUENCE: 113
 GTCGDIVMSQ SPSSLAVSAG EKVTMSCKSS QSLNSRTRK NYLAWVQHKK GQSPRLLIYW 60
 ASTRESGVDP RFTGSGSGTD FTLTISSVQA EDLAVYYCRQ SYNLVTFGAG PS 112
 SEQ ID NO: 114 moltype = AA length = 112
 FEATURE Location/Qualifiers
 source 1..112
 mol_type = protein
 organism = Mus musculus
 SEQUENCE: 114
 GPVLVKPGAS VKMSCKASGY TFTSYVMHVW KQKPGQGLEW IGYINPYNDG TKYNEKFKGK 60
 ATLTSCKSSS TAYMELSSLT SEDSAVYCA RRYYYGSSG YFDVWAQDHV RT 112
 SEQ ID NO: 115 moltype = AA length = 108
 FEATURE Location/Qualifiers
 source 1..108
 mol_type = protein
 organism = Mus musculus
 SEQUENCE: 115
 DVQITQSPSY LAASPGETIT INCRASKSIS KYLAWEQEK GKTNKKLIYS GSTLQSGIPS 60
 RFSGSGSGTD FTLTISSLEP EDFAMYCCQ HNEYPTTFGG GTKLEIKR 108
 SEQ ID NO: 116 moltype = AA length = 113
 FEATURE Location/Qualifiers
 source 1..113
 mol_type = protein
 organism = Mus musculus
 SEQUENCE: 116
 GPVLVKPGAS VKISCKASGY SFTGYFMNVW MQSHGKSLEW IGRINPYNGD TFYNQKFKGK 60
 ATLTVDKSSS TAHMELRSLA SEDSAVYCA RRIHYYYGSS YYAMDYWGQE PHH 113
 SEQ ID NO: 117 moltype = AA length = 108
 FEATURE Location/Qualifiers
 source 1..108
 mol_type = protein
 organism = Mus musculus
 SEQUENCE: 117
 DVQITQSPSY LAASPGETIT INCRASKSIS KYLAWEQEK GKTNKKLIYS GSTLQSGIPS 60
 RFSGSGSGTD FTLTISSLEP EDFAMYCCQ HNEYPTTFGG GTKLEIKR 108
 SEQ ID NO: 118 moltype = AA length = 113
 FEATURE Location/Qualifiers

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source                1..113
                      mol_type = protein
                      organism = Mus musculus

SEQUENCE: 118
GAGLVKPGAS VKLSCKASGY TFTEYIIHWV KQRSGQGLEW IGWFYPGSGS IKYNEKFKDK 60
AFLTADKSSS TVYMELSRLT SEDSAVYFCA RHEVYYDYDK SMLWTTGVKN LIR 113

SEQ ID NO: 119        moltype = AA length = 108
FEATURE              Location/Qualifiers
source                1..108
                      mol_type = protein
                      organism = Mus musculus

SEQUENCE: 119
SPSSLAVSVG EKVMTSCKSS QSLLYSSNQK NYLAWYQQKP GQSPKLLIYW ASTRESGVPD 60
RFTGSGSGTD FTLTISSVKA EDLAVYCCQ YSYPTYTFGG GTKLEIKR 108

SEQ ID NO: 120        moltype = AA length = 113
FEATURE              Location/Qualifiers
source                1..113
                      mol_type = protein
                      organism = Mus musculus

SEQUENCE: 120
GAELVRPGTS VKVSCASVY AFTNYLIEWV KQRPQGQLEW IGVINPKSGG TKYNEKFRGK 60
AFLTADKSSS TAYMQLSSLT SGDSAVYFCA ITGTDYWGQG TTLTVSSAKT TPP 113

SEQ ID NO: 121        moltype = AA length = 113
FEATURE              Location/Qualifiers
source                1..113
                      mol_type = protein
                      organism = Mus musculus

SEQUENCE: 121
QGTRCDIQMT QTTSSLSASL GDRVITISCSA SQGINNYLNW YQKQPDGTVK LLIYTTSSLR 60
SGVPSRFSGS GSGTDYSLTI SNLEPEDVAT YYCQYSKLP RTFGGGTKLE IKR 113

SEQ ID NO: 122        moltype = AA length = 121
FEATURE              Location/Qualifiers
source                1..121
                      mol_type = protein
                      organism = Homo sapiens

SEQUENCE: 122
EVQLVESGGG LVQPGGSLRL SCAASGFSLs SHSLGWVRQA PGKGLEWIGD IRSGGSAYYA 60
NWAKGRFTIS RDNSKNTLYL QMNSLRAEDT AVYYCTRNG PSDLTNRLDL WQGGLVTVS 120
S 121

SEQ ID NO: 123        moltype = AA length = 112
FEATURE              Location/Qualifiers
source                1..112
                      mol_type = protein
                      organism = Homo sapiens

SEQUENCE: 123
QVLTQSPSSL SASVGDRVTI NCQASQSLYN NENLAWFQQK PGKVPKRLIY GASTLASGVS 60
SRFSGSGSGT EFTLTISLQ CEDFAIYYCL GEFSCGSADC FAFGGGTVKE IK 112

SEQ ID NO: 124        moltype = AA length = 121
FEATURE              Location/Qualifiers
source                1..121
                      mol_type = protein
                      organism = Homo sapiens

SEQUENCE: 124
EVQLVESGGG LVQPGGSLRL SCAASGFSLs SHSLGWVRQA PGKGLEWIGD IRSGGSAYYA 60
NWAKGRFTIS RDNSKNTLYL QMNSLRAEDT AVYFCTRNG PSDLTNRLDL WQGGLVTVS 120
S 121

SEQ ID NO: 125        moltype = AA length = 112
FEATURE              Location/Qualifiers
source                1..112
                      mol_type = protein
                      organism = Homo sapiens

SEQUENCE: 125
QVLTQSPSSL SASVGDRVTI NCQASQSLYN NENLAWFQQK PGKVPKRLIY GASTLASGVS 60
SRFSGSGSGT EFTLTISLQ CEDFAIYYCL GEFSCGSADC FAFGGGTVKE IK 112

SEQ ID NO: 126        moltype = AA length = 121
FEATURE              Location/Qualifiers
source                1..121
                      mol_type = protein

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                                organism = Homo sapiens
SEQUENCE: 126
EVQLVESGGG LVQPGGSLRL SCAASGFSL SLS SHSLGWVRQA PGKGLEWIGD IRSGGSAYYA 60
NWAKGRFTIS RDNSKNTLYL QMNSLRAEDT AVYYCTRNG PSDLTNRLDL WQGTTLVTVS 120
S 121

SEQ ID NO: 127      moltype = AA length = 113
FEATURE            Location/Qualifiers
source             1..113
                   mol_type = protein
                   organism = Homo sapiens

SEQUENCE: 127
EQVLTQSPSS LSASVGDRVT INCQASQSLY NNENLAWFQQ KPGKVPKRLI YGASTLASGV 60
SSRFSGSGSG TEFTLTISNL QPEDFAIYYC LGEFSCGSAD CFAFGGGTKV EIK 113

SEQ ID NO: 128      moltype = AA length = 121
FEATURE            Location/Qualifiers
source             1..121
                   mol_type = protein
                   organism = Homo sapiens

SEQUENCE: 128
EVQLVESGGG LVQPGGSLRL SCAASGFSL SLS SHSLGWVRQA PGKGLEWIGD IRSGGSAYYA 60
NWAKGRFTIS RDNSKNTLYL QMNSLRAEDT AVYFCTRNG PSDLTNRLDL WQGTTLVTVS 120
S 121

SEQ ID NO: 129      moltype = AA length = 113
FEATURE            Location/Qualifiers
source             1..113
                   mol_type = protein
                   organism = Homo sapiens

SEQUENCE: 129
EIVLTQSPSS LSASVGDRVT INCQASQSLY NNENLAWFQQ KPGKVPKRLI YGASTLASGV 60
SSRFSGSGSG TEFTLTISNL QPEDFATYYC LGEFSCGSAD CFAFGGGTKV EIK 113

SEQ ID NO: 130      moltype = AA length = 121
FEATURE            Location/Qualifiers
source             1..121
                   mol_type = protein
                   organism = Homo sapiens

SEQUENCE: 130
EVQLVESGGG LVQPGGSLRL SCAASGFSL SLS SHSLGWVRQA PGKGLEWIGD IRSGGSAYYA 60
NWAKGRFTIS RDNSKNTLYL QMNSLRAEDT AVYYCTRNG PSDLTNRLDL WQGTTLVTVS 120
S 121

SEQ ID NO: 131      moltype = AA length = 113
FEATURE            Location/Qualifiers
source             1..113
                   mol_type = protein
                   organism = Homo sapiens

SEQUENCE: 131
EIVLTQSPSS LSASVGDRVT INCQASQSLY NNENLAWFQQ KPGKVPKRLI YGASTLASGV 60
SSRFSGSGSG TEFTLTISNL QPEDFATYYC LGEFSCGSAD CFAFGGGTKV EIK 113

SEQ ID NO: 132      moltype = AA length = 121
FEATURE            Location/Qualifiers
source             1..121
                   mol_type = protein
                   organism = Homo sapiens

SEQUENCE: 132
EVQLVESGGG LVQPGGSLRL SCAASGFSL SLS SHSLGWVRQA PGKGLEWIGD IRSGGSAYYA 60
NWAKGRFTIS RDNSKNTLYL QMNSLRAEDT AVYFCTRNG PSDLTNRLDL WQGTTLVTVS 120
S 121

SEQ ID NO: 133      moltype = AA length = 113
FEATURE            Location/Qualifiers
source             1..113
                   mol_type = protein
                   organism = Homo sapiens

SEQUENCE: 133
EQVLTQSPSS LSASVGDRVT INCQASQSLY NNENLAWFQQ KPGKVPKRLI YGASTLASGV 60
SSRFSGSGSG TEFTLTISNL QPEDFAIYYC LGEFSCGSAD CFAFGGGTKV EIK 113

SEQ ID NO: 134      moltype = AA length = 5
FEATURE            Location/Qualifiers
source             1..5
                   mol_type = protein

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SEQUENCE: 134 GYDML	organism = Gallus gallus	5
SEQ ID NO: 135 FEATURE source	moltype = AA length = 17 Location/Qualifiers 1..17 mol_type = protein organism = Gallus gallus	
SEQUENCE: 135 GIGSTGGGTD YGAAVKG		17
SEQ ID NO: 136 FEATURE source	moltype = AA length = 19 Location/Qualifiers 1..19 mol_type = protein organism = Gallus gallus	
SEQUENCE: 136 VAGGCNSGYC RDSPGSIDA		19
SEQ ID NO: 137 FEATURE source	moltype = AA length = 10 Location/Qualifiers 1..10 mol_type = protein organism = Gallus gallus	
SEQUENCE: 137 SGGSRNYYG		10
SEQ ID NO: 138 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = Gallus gallus	
SEQUENCE: 138 DDQRPSN		7
SEQ ID NO: 139 FEATURE source	moltype = AA length = 11 Location/Qualifiers 1..11 mol_type = protein organism = Gallus gallus	
SEQUENCE: 139 SADSNTYEGS F		11
SEQ ID NO: 140 FEATURE source	moltype = AA length = 5 Location/Qualifiers 1..5 mol_type = protein organism = Mus musculus	
SEQUENCE: 140 DYNMD		5
SEQ ID NO: 141 FEATURE source	moltype = AA length = 17 Location/Qualifiers 1..17 mol_type = protein organism = Mus musculus	
SEQUENCE: 141 DINPNYDSTS YNQKFKG		17
SEQ ID NO: 142 FEATURE source	moltype = AA length = 11 Location/Qualifiers 1..11 mol_type = protein organism = Mus musculus	
SEQUENCE: 142 SRSYDYEFGA Y		11
SEQ ID NO: 143 FEATURE source	moltype = AA length = 11 Location/Qualifiers 1..11 mol_type = protein organism = Mus musculus	
SEQUENCE: 143 LSIVNRYHYM S		11

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SEQ ID NO: 144	moltype = AA length = 6	
FEATURE	Location/Qualifiers	
source	1..6	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 144		
EASITK		6
SEQ ID NO: 145	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 145		
QHNRGSFLP		9
SEQ ID NO: 146	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
source	1..5	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 146		
DYYMS		5
SEQ ID NO: 147	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 147		
RNKANGYTTE YSASVKG		17
SEQ ID NO: 148	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 148		
ARANWAFDY		9
SEQ ID NO: 149	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 149		
RASQSI SNYL H		11
SEQ ID NO: 150	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 150		
YASQSI S		7
SEQ ID NO: 151	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 151		
YASQSI S		7
SEQ ID NO: 152	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
source	1..5	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 152		
NYLIV		5
SEQ ID NO: 153	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	

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SEQUENCE: 153 VISPGSGGTN YNEKFKG	organism = Mus musculus	17
SEQ ID NO: 154 FEATURE source	moltype = AA length = 11 Location/Qualifiers 1..11 mol_type = protein organism = Mus musculus	
SEQUENCE: 154 EKIYDDYYEG Y		11
SEQ ID NO: 155 FEATURE source	moltype = AA length = 15 Location/Qualifiers 1..15 mol_type = protein organism = Mus musculus	
SEQUENCE: 155 TISCSASLGI GNYLN		15
SEQ ID NO: 156 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = Mus musculus	
SEQUENCE: 156 TSNLHSG		7
SEQ ID NO: 157 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = Mus musculus	
SEQUENCE: 157 HYSKLPLTF		9
SEQ ID NO: 158 FEATURE source	moltype = AA length = 5 Location/Qualifiers 1..5 mol_type = protein organism = Mus musculus	
SEQUENCE: 158 DYNMY		5
SEQ ID NO: 159 FEATURE source	moltype = AA length = 17 Location/Qualifiers 1..17 mol_type = protein organism = Mus musculus	
SEQUENCE: 159 YIYPGNGGTN YNQKFKG		17
SEQ ID NO: 160 FEATURE source	moltype = AA length = 11 Location/Qualifiers 1..11 mol_type = protein organism = Mus musculus	
SEQUENCE: 160 DYDDGGYAMD Y		11
SEQ ID NO: 161 FEATURE source	moltype = AA length = 15 Location/Qualifiers 1..15 mol_type = protein organism = Mus musculus	
SEQUENCE: 161 SVGETVTITC RASGN		15
SEQ ID NO: 162 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = Mus musculus	
SEQUENCE: 162 NAKTLAD		7

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SEQ ID NO: 163	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 163		
QHFVNIPWT		9
SEQ ID NO: 164	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
source	1..5	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 164		
DHSIH		5
SEQ ID NO: 165	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 165		
YISPGNGNIK YNEKFKG		17
SEQ ID NO: 166	moltype = AA length = 12	
FEATURE	Location/Qualifiers	
source	1..12	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 166		
SLGRGGPYYP DY		12
SEQ ID NO: 167	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 167		
RSSKSLHNSN GNTYLY		16
SEQ ID NO: 168	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 168		
RMSNLAS		7
SEQ ID NO: 169	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 169		
MQHREYPVT		9
SEQ ID NO: 170	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
source	1..5	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 170		
SYWIE		5
SEQ ID NO: 171	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 171		
EILPGSGSTN YNEKFKG		17
SEQ ID NO: 172	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	

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SEQUENCE: 172 YYWYFDVWAQ D	organism = Mus musculus	11
SEQ ID NO: 173 FEATURE source	moltype = AA length = 15 Location/Qualifiers 1..15 mol_type = protein organism = Mus musculus	
SEQUENCE: 173 SSKNLLHSNG ITYLY		15
SEQ ID NO: 174 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = Mus musculus	
SEQUENCE: 174 RVSNLAS		7
SEQ ID NO: 175 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = Mus musculus	
SEQUENCE: 175 AQLLELPYT		9
SEQ ID NO: 176 FEATURE source	moltype = AA length = 5 Location/Qualifiers 1..5 mol_type = protein organism = Mus musculus	
SEQUENCE: 176 SYDMS		5
SEQ ID NO: 177 FEATURE source	moltype = AA length = 17 Location/Qualifiers 1..17 mol_type = protein organism = Mus musculus	
SEQUENCE: 177 YISSGAGSTY YPDTVKG		17
SEQ ID NO: 178 FEATURE source	moltype = AA length = 11 Location/Qualifiers 1..11 mol_type = protein organism = Mus musculus	
SEQUENCE: 178 HFYRFDYWGQ G		11
SEQ ID NO: 179 FEATURE source	moltype = AA length = 15 Location/Qualifiers 1..15 mol_type = protein organism = Mus musculus	
SEQUENCE: 179 SAGDRITITC KASQS		15
SEQ ID NO: 180 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = Mus musculus	
SEQUENCE: 180 YASNRYT		7
SEQ ID NO: 181 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = Mus musculus	
SEQUENCE: 181 QQDDRFPLT		9

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SEQ ID NO: 182	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
source	1..5	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 182		
NYGMN		5
SEQ ID NO: 183	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 183		
WINTYTGEPT YADDFKG		17
SEQ ID NO: 184	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 184		
GAWFAYWAKD S		11
SEQ ID NO: 185	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 185		
SITCKASQDV GTAVA		15
SEQ ID NO: 186	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 186		
WASTRHT		7
SEQ ID NO: 187	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 187		
QQYSSYPLT		9
SEQ ID NO: 188	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
source	1..5	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 188		
DFWMN		5
SEQ ID NO: 189	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 189		
EIRLKSNNYA THYAESVKG		19
SEQ ID NO: 190	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
source	1..13	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 190		
LFYYDGTSG FAY		13
SEQ ID NO: 191	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	

-continued

SEQUENCE: 191 KSSQSLNSG DQKNYLT	organism = Mus musculus	17
SEQ ID NO: 192 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = Mus musculus	
SEQUENCE: 192 WASTRES		7
SEQ ID NO: 193 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = Mus musculus	
SEQUENCE: 193 QNDYDYPLT		9
SEQ ID NO: 194 FEATURE source	moltype = AA length = 5 Location/Qualifiers 1..5 mol_type = protein organism = Mus musculus	
SEQUENCE: 194 DYNMD		5
SEQ ID NO: 195 FEATURE source	moltype = AA length = 17 Location/Qualifiers 1..17 mol_type = protein organism = Mus musculus	
SEQUENCE: 195 DINPNYDSTS YNQKFKG		17
SEQ ID NO: 196 FEATURE source	moltype = AA length = 11 Location/Qualifiers 1..11 mol_type = protein organism = Mus musculus	
SEQUENCE: 196 SRSYDIEGFA Y		11
SEQ ID NO: 197 FEATURE source	moltype = AA length = 11 Location/Qualifiers 1..11 mol_type = protein organism = Mus musculus	
SEQUENCE: 197 RASGNIHNYL A		11
SEQ ID NO: 198 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = Mus musculus	
SEQUENCE: 198 NAKTLAD		7
SEQ ID NO: 199 FEATURE source	moltype = AA length = 8 Location/Qualifiers 1..8 mol_type = protein organism = Mus musculus	
SEQUENCE: 199 QHFWSTLT		8
SEQ ID NO: 200 FEATURE source	moltype = AA length = 5 Location/Qualifiers 1..5 mol_type = protein organism = Mus musculus	
SEQUENCE: 200 GYTMN		5

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SEQ ID NO: 201	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 201		
NPYNGGTSYN QKFKGK		16
SEQ ID NO: 202	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 202		
WGVWSAMDY		9
SEQ ID NO: 203	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 203		
KASQNVRTAV A		11
SEQ ID NO: 204	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 204		
LASNRDT		7
SEQ ID NO: 205	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 205		
LQHCNYPNE		9
SEQ ID NO: 206	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
source	1..5	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 206		
SYWMQ		5
SEQ ID NO: 207	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 207		
AIYPGDGDTR YTQKFKG		17
SEQ ID NO: 208	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 208		
ARGEYGNFYFA Y		11
SEQ ID NO: 209	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 209		
KASQDINSYL S		11
SEQ ID NO: 210	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	

-continued

SEQUENCE: 210 RANRLVD	organism = Mus musculus	7
SEQ ID NO: 211 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = Mus musculus	
SEQUENCE: 211 LQYDEFPLT		9
SEQ ID NO: 212 FEATURE source	moltype = AA length = 5 Location/Qualifiers 1..5 mol_type = protein organism = Mus musculus	
SEQUENCE: 212 DTYMH		5
SEQ ID NO: 213 FEATURE source	moltype = AA length = 17 Location/Qualifiers 1..17 mol_type = protein organism = Mus musculus	
SEQUENCE: 213 RIDPANGNTK YDPKFQG		17
SEQ ID NO: 214 FEATURE source	moltype = AA length = 14 Location/Qualifiers 1..14 mol_type = protein organism = Mus musculus	
SEQUENCE: 214 ARPIHYYYGS SLAY		14
SEQ ID NO: 215 FEATURE source	moltype = AA length = 11 Location/Qualifiers 1..11 mol_type = protein organism = Mus musculus	
SEQUENCE: 215 SVDSYGNSFM H		11
SEQ ID NO: 216 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = Mus musculus	
SEQUENCE: 216 RASNLRS		7
SEQ ID NO: 217 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = Mus musculus	
SEQUENCE: 217 QQSNEDPGR		9
SEQ ID NO: 218 FEATURE source	moltype = AA length = 5 Location/Qualifiers 1..5 mol_type = protein organism = Mus musculus	
SEQUENCE: 218 DYYMS		5
SEQ ID NO: 219 FEATURE source	moltype = AA length = 17 Location/Qualifiers 1..17 mol_type = protein organism = Mus musculus	
SEQUENCE: 219 RNKANGYTTE YSASVKG		17

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SEQ ID NO: 220	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 220		
ARANWAFDY		9
SEQ ID NO: 221	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 221		
RASQSISNYL H		11
SEQ ID NO: 222	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 222		
YASQSYS		7
SEQ ID NO: 223	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 223		
QQSNSWPYT		9
SEQ ID NO: 224	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
source	1..5	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 224		
DYYMS		5
SEQ ID NO: 225	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 225		
RNKANGYTTE YSASVKG		17
SEQ ID NO: 226	moltype = AA length = 12	
FEATURE	Location/Qualifiers	
source	1..12	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 226		
ARAPLLYYAM DY		12
SEQ ID NO: 227	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 227		
NVSTSGYSYM H		11
SEQ ID NO: 228	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 228		
LVSNNLES		7
SEQ ID NO: 229	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
source	1..8	
	mol_type = protein	

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SEQUENCE: 229 QHIRELTR	organism = Mus musculus	8
SEQ ID NO: 230 FEATURE source	moltype = AA length = 5 Location/Qualifiers 1..5 mol_type = protein organism = Mus musculus	
SEQUENCE: 230 SYWMH		5
SEQ ID NO: 231 FEATURE source	moltype = AA length = 17 Location/Qualifiers 1..17 mol_type = protein organism = Mus musculus	
SEQUENCE: 231 MIDPSNSETR LNQKFKD		17
SEQ ID NO: 232 FEATURE source	moltype = AA length = 12 Location/Qualifiers 1..12 mol_type = protein organism = Mus musculus	
SEQUENCE: 232 ARGLRHYWYF DV		12
SEQ ID NO: 233 FEATURE source	moltype = AA length = 11 Location/Qualifiers 1..11 mol_type = protein organism = Mus musculus	
SEQUENCE: 233 SVSTSGYSYM H		11
SEQ ID NO: 234 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = Mus musculus	
SEQUENCE: 234 LVSNLES		7
SEQ ID NO: 235 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = Mus musculus	
SEQUENCE: 235 QHIRELTRS		9
SEQ ID NO: 236 FEATURE source	moltype = AA length = 5 Location/Qualifiers 1..5 mol_type = protein organism = Mus musculus	
SEQUENCE: 236 SYGMS		5
SEQ ID NO: 237 FEATURE source	moltype = AA length = 14 Location/Qualifiers 1..14 mol_type = protein organism = Mus musculus	
SEQUENCE: 237 WVRQTPEKRL EWVA		14
SEQ ID NO: 238 FEATURE source	moltype = AA length = 11 Location/Qualifiers 1..11 mol_type = protein organism = Mus musculus	
SEQUENCE: 238 LASYYFDYWG Q		11

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SEQ ID NO: 239	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 239		
TMTCQASQGT SINLN		15
SEQ ID NO: 240	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 240		
GASSLED		7
SEQ ID NO: 241	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 241		
LQHSYLPPLT F		11
SEQ ID NO: 242	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
source	1..5	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 242		
TYDLH		5
SEQ ID NO: 243	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 243		
VIWSGGSTDY NAAFIS		16
SEQ ID NO: 244	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 244		
NYGYSAWFAY W		11
SEQ ID NO: 245	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 245		
ASISCRSSQS IVHSN		15
SEQ ID NO: 246	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 246		
KVSNRFS		7
SEQ ID NO: 247	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 247		
FQSSHVPLT		9
SEQ ID NO: 248	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
source	1..5	
	mol_type = protein	

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SEQUENCE: 248 AYSMH	organism = Mus musculus	5
SEQ ID NO: 249 FEATURE source	moltype = AA length = 17 Location/Qualifiers 1..17 mol_type = protein organism = Mus musculus	
SEQUENCE: 249 WINTETGEPT YTDDFKG		17
SEQ ID NO: 250 FEATURE source	moltype = AA length = 11 Location/Qualifiers 1..11 mol_type = protein organism = Mus musculus	
SEQUENCE: 250 RIYYFGRGGF D		11
SEQ ID NO: 251 FEATURE source	moltype = AA length = 15 Location/Qualifiers 1..15 mol_type = protein organism = Mus musculus	
SEQUENCE: 251 SSISCRSSQS IVHSN		15
SEQ ID NO: 252 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = Mus musculus	
SEQUENCE: 252 KVSNRFS		7
SEQ ID NO: 253 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = Mus musculus	
SEQUENCE: 253 FQGSHPYT		9
SEQ ID NO: 254 FEATURE source	moltype = AA length = 5 Location/Qualifiers 1..5 mol_type = protein organism = Mus musculus	
SEQUENCE: 254 NSWPN		5
SEQ ID NO: 255 FEATURE source	moltype = AA length = 19 Location/Qualifiers 1..19 mol_type = protein organism = Mus musculus	
SEQUENCE: 255 EIRLTSDNYA IYYAESVKG		19
SEQ ID NO: 256 FEATURE source	moltype = AA length = 11 Location/Qualifiers 1..11 mol_type = protein organism = Mus musculus	
SEQUENCE: 256 PETARATPAY W		11
SEQ ID NO: 257 FEATURE source	moltype = AA length = 15 Location/Qualifiers 1..15 mol_type = protein organism = Mus musculus	
SEQUENCE: 257 ASISCRSSQS IVHSN		15

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SEQ ID NO: 258	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 258		
KVFNRFPS		7
SEQ ID NO: 259	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 259		
FQGSHPRT		9
SEQ ID NO: 260	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
source	1..5	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 260		
AYYMH		5
SEQ ID NO: 261	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 261		
RVNPNNGGTT YNQKPKG		17
SEQ ID NO: 262	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 262		
RIYYGYFDYW G		11
SEQ ID NO: 263	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 263		
KASQSVDDYG DSYMN		15
SEQ ID NO: 264	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 264		
VASNLES		7
SEQ ID NO: 265	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 265		
QQSNEDPYT		9
SEQ ID NO: 266	moltype = AA length = 4	
FEATURE	Location/Qualifiers	
source	1..4	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 266		
DIYM		4
SEQ ID NO: 267	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	

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SEQUENCE: 267 KIDPANGNTK YDPKFQG	organism = Mus musculus	17
SEQ ID NO: 268 FEATURE source	moltype = AA length = 11 Location/Qualifiers 1..11 mol_type = protein organism = Mus musculus	
SEQUENCE: 268 TGDYWGQGT V		11
SEQ ID NO: 269 FEATURE source	moltype = AA length = 15 Location/Qualifiers 1..15 mol_type = protein organism = Mus musculus	
SEQUENCE: 269 TMSCKSSQSL LNSRT		15
SEQ ID NO: 270 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = Mus musculus	
SEQUENCE: 270 WASTRES		7
SEQ ID NO: 271 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = Mus musculus	
SEQUENCE: 271 RQSYNLVTF		9
SEQ ID NO: 272 FEATURE source	moltype = AA length = 5 Location/Qualifiers 1..5 mol_type = protein organism = Mus musculus	
SEQUENCE: 272 SYVMH		5
SEQ ID NO: 273 FEATURE source	moltype = AA length = 17 Location/Qualifiers 1..17 mol_type = protein organism = Mus musculus	
SEQUENCE: 273 YINPYNDGTK YNEKFKG		17
SEQ ID NO: 274 FEATURE source	moltype = AA length = 11 Location/Qualifiers 1..11 mol_type = protein organism = Mus musculus	
SEQUENCE: 274 RYYGSSGGY F		11
SEQ ID NO: 275 FEATURE source	moltype = AA length = 15 Location/Qualifiers 1..15 mol_type = protein organism = Mus musculus	
SEQUENCE: 275 RASKSISKYL AWYQE		15
SEQ ID NO: 276 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = Mus musculus	
SEQUENCE: 276 SGSTLQS		7

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SEQ ID NO: 277	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 277		
QQHNEYPYT		9
SEQ ID NO: 278	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
source	1..5	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 278		
GYFMN		5
SEQ ID NO: 279	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 279		
RINPYNGDTF YNQKFKG		17
SEQ ID NO: 280	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 280		
RIHYYYGSSY Y		11
SEQ ID NO: 281	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 281		
RASKSISKYL AWYQE		15
SEQ ID NO: 282	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 282		
SGSTLQS		7
SEQ ID NO: 283	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 283		
QQHNEYPWT		9
SEQ ID NO: 284	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
source	1..5	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 284		
EYIIH		5
SEQ ID NO: 285	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 285		
WFYPGSGSIK YNEKFKD		17
SEQ ID NO: 286	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	

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SEQUENCE: 286 HEVYYDYDKS M	organism = Mus musculus	11
SEQ ID NO: 287 FEATURE source	moltype = AA length = 15 Location/Qualifiers 1..15 mol_type = protein organism = Mus musculus	
SEQUENCE: 287 LYSSNQKNYL AWYQQ		15
SEQ ID NO: 288 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = Mus musculus	
SEQUENCE: 288 WASTRES		7
SEQ ID NO: 289 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = Mus musculus	
SEQUENCE: 289 QQYYSYPYT		9
SEQ ID NO: 290 FEATURE source	moltype = AA length = 5 Location/Qualifiers 1..5 mol_type = protein organism = Mus musculus	
SEQUENCE: 290 NYLIE		5
SEQ ID NO: 291 FEATURE source	moltype = AA length = 19 Location/Qualifiers 1..19 mol_type = protein organism = Mus musculus	
SEQUENCE: 291 VINPKSGGTK YNEKFRGKA		19
SEQ ID NO: 292 FEATURE source	moltype = AA length = 11 Location/Qualifiers 1..11 mol_type = protein organism = Mus musculus	
SEQUENCE: 292 TGTDYWGGT T		11
SEQ ID NO: 293 FEATURE source	moltype = AA length = 10 Location/Qualifiers 1..10 mol_type = protein organism = Mus musculus	
SEQUENCE: 293 VTISCSASQG		10
SEQ ID NO: 294 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = Mus musculus	
SEQUENCE: 294 YTSSLRS		7
SEQ ID NO: 295 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = Mus musculus	
SEQUENCE: 295 QQYSKLPRT		9

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SEQ ID NO: 296	moltype = AA length = 12	
FEATURE	Location/Qualifiers	
source	1..12	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 296		
FTSGEKEQVD EW		12
SEQ ID NO: 297	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 297		
AEQKRLKTVL ELQYVL		16
SEQ ID NO: 298	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 298		
VERVFQSNYF DSTHNN		16
SEQ ID NO: 299	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 299		
FQSMQTVFNM NAPVPP		16
SEQ ID NO: 300	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
source	1..5	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 300		
TNAMN		5
SEQ ID NO: 301	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 301		
RIRSKSNYYA TYYADSV		17
SEQ ID NO: 302	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 302		
DWDGFLYFDY W		11
SEQ ID NO: 303	moltype = AA length = 112	
FEATURE	Location/Qualifiers	
source	1..112	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 303		
GGGLVQPKGS LKLSCAASGF TFNTNAMNWV RQAPGKGLEW VARIRSKSNN YATYYADSVK		60
DRFTISRDDS QSMLYLQMNN LKTEDTAMYY CVRDWDGFLY FDYWAKHHLT LF		112
SEQ ID NO: 304	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 304		
SVSTSGYSYM HWNQQ		15
SEQ ID NO: 305	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	

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	mol_type = protein organism = Mus musculus	
SEQUENCE: 305 LVSNLES		7
SEQ ID NO: 306 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = Mus musculus	
SEQUENCE: 306 QHIRELTRS		9
SEQ ID NO: 307 FEATURE source	moltype = AA length = 104 Location/Qualifiers 1..104 mol_type = protein organism = Mus musculus	
SEQUENCE: 307 TQSPASLAVS LGQRATISYR ASKSVSTSGY SYMHWNQKQP GQPPRLLIYL VSNLESGVPA RFSGSGSGTD FTLNIHPVEE EDAATYYCQH IRELTRSEGG PSWK		60 104
SEQ ID NO: 308 FEATURE source	moltype = AA length = 14 Location/Qualifiers 1..14 mol_type = protein organism = Homo sapiens	
SEQUENCE: 308 PPVNEPETLK QQNQ		14
SEQ ID NO: 309 FEATURE source	moltype = AA length = 12 Location/Qualifiers 1..12 mol_type = protein organism = Homo sapiens	
SEQUENCE: 309 ETLKQQNQYQ AS		12
SEQ ID NO: 310 FEATURE source	moltype = AA length = 709 Location/Qualifiers 1..709 mol_type = protein organism = Homo sapiens	
SEQUENCE: 310 MPSATSHSGS GSKSGGPPPP SGSSGSEAAA GAGAAAPASQ HPATGTGAVQ TEAMKQILGV IDKKLRNLEK KKGKLDYQE RMNKGRLNQ DQLDAVSKYQ EVTNNLEFAK ELQRSFMALS QDIQTIKKT ARREQLMREE AEQKRLKTVL ELQYVLDKLG DDEVRTDLKQ GLNGVPILSE EELSLLDEFY KLVDPERDMS LRLNEQYEHY SIHLWDLLEG KEKPVCGTTY KVLKEIVERV FQSNYFDSTH NHQNGLCSEE EAASAPAVED QVPEAEPEPA EEYTEQSEVE STEYVNRQFM AETQFTSGEK EQVDEWTVET VEVVNSLQQQ PQAASPSVPE PHSLTPVAQA DPLVRRQRVQ DLMAQMGGPY NFIQDSMLDF ENQTLDPPIV SAQPMNPTQN MDMPQLVCPV VHSERLAQP NQVPVQPEAT QVPLVSTSE GYTASQPLYQ PSHATEQRPQ KEPIDQIQAT ISLNTDQTTA SSSLPAASQP QVFQAGTSKP LHSSGINVNA APFQSMQTVF NMNAPVPPVN EPETLKQQNQ YQASYNQSFY SQPHQVEQTE LQQEQLQTVV GTYHGSPPQS HQVTGNHQQP PQQNTGFPRS NQPYYNRSGV SRGGSRGARG LMNGYRGPAN GFRGGYDGYR PSFSNTPNSG YTQSQFSAPR DYSGYQRDGY QQNFKRSGSQ SGPRGAPRGR GGPFRPNRGM PQMNTQQVN	60 120 180 240 300 360 420 480 540 600 660 709	
SEQ ID NO: 311 FEATURE source	moltype = AA length = 694 Location/Qualifiers 1..694 mol_type = protein organism = Homo sapiens	
SEQUENCE: 311 MPSATSHSGS GSKSGGPPPP SGSSGSEAAA GAGAAAPASQ HPATGTGAVQ TEAMKQILGV IDKKLRNLEK KKGKLDYQE RMNKGRLNQ DQLDAVSKYQ EVTNNLEFAK ELQRSFMALS QDIQTIKKT ARREQLMREE AEQKRLKTVL ELQYVLDKLG DDEVRTDLKQ GLNGVPILSE EELSLLDEFY KLVDPERDMS LRLNEQYEHY SIHLWDLLEG KEKPVCGTTY KVLKEIVERV FQSNYFDSTH NHQNGLCSEE EAASAPAVED QVPEAEPEPA EEYTEQSEVE STEYVNRQFM AETQFTSGEK EQVDEWTVET VEVVNSLQQQ PQAASPSVPE PHSLTPVAQA DPLVRRQRVQ DLMAQMGGPY NFIQDSMLDF ENQTLDPPIV SAQPMNPTQN MDMPQLVCPV VHSERLAQP NQVPVQPEAT QVPLVSTSE GYTASQPLYQ PSHATEQRPQ KEPIDQIQAT ISLNTDQTTA SSSLPAASQP QVFQAGTSKP LHSSGINVNA APFQSMQTVF NMNAPVPPVN EPETLKQQNQ YQASYNQSFY SQPHQVEQTE LQQEQLQTVV GTYHGSPPQS HQVTGNHQQP PQQNTGFPRS NQPYYNRSGV SRGGSRGARG LMNGYRGPAN GFRGGYDGYR PSFSNTPNSG YTQSQFSAPR DYSGYQRDGY QQNFKRSGSQ SGPRGAPRGN ILWW	60 120 180 240 300 360 420 480 540 600 660 694	

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SEQ ID NO: 312 moltype = AA length = 448
 FEATURE Location/Qualifiers
 source 1..448
 mol_type = protein
 note = Canis lupus familiaris
 organism = Canis lupus

SEQUENCE: 312

MALSQDIQKT	IKKTARREQL	MREEAEQKRL	KTVLELQYVL	DKLGDEVRT	DLKQGLNGVP	60
ILSEEEILLS	DEFYKLADPE	RDMSLRLENEQ	YEHASIHLD	LLEGKEKSVC	GTTYKALKEI	120
VERVPQSNYF	DSTHNHNGNL	CEEEEAASAP	TVEDQVAEAE	PEPAEEYTEQ	SEVESTYEVN	180
RQFMAETQFS	SGEKEQVDEW	TVETVEVVNS	LQQQPQAASP	SVPEPHSLTP	VAQADPLVRR	240
QVRVDLMAQM	QGPYNFIQDS	MLDFENQTL	PAIVSAQPMN	PTQNMMPQL	VCPVHSESR	300
LAQPNQVPVQ	PEATQVPLVS	STSEGYTASQ	PLYQPSHATE	QRQKEPIDQ	IQATISLNTD	360
QTTASSSLPA	ASQPQVFQAG	TSKPLHSSGI	NVNAAPFQSM	QTVFNMNAPV	PPVNEPETLK	420
QQNQYQASYN	QSFSSQPHQV	EQTEGCRK				448

SEQ ID NO: 313 moltype = AA length = 717
 FEATURE Location/Qualifiers
 source 1..717
 mol_type = protein
 note = Canis lupus familiaris
 organism = Canis lupus

SEQUENCE: 313

MPSATSLSGS	GSKSSGPPPP	SGSSGSEAAA	AAGAAGAAGA	GAAAPASQHP	ATGTGAVQTE	60
AMKQILGVID	KKLRNLEKKK	GKLDDYQERM	NKGERLNQDQ	LDAVSKYQEV	TNNLEFAKEL	120
QRSFMALSQD	IQKTIKKTAR	REQLMREEAE	QKRLKTVLEL	QYVLDKLGDD	EVRTDLKQGL	180
NGVPILSEEE	LSLLDEFYKL	ADPERDMSLR	LNEQYEHASI	HLWDLLEGKE	KSVCGTTYKA	240
LKEIVERVFP	SNYFDSSTHH	QNGLCSEEEA	ASAPTVEDQV	AEAEPEPAEE	YTEQSEVEST	300
EYVNRQFMAE	TQFSSGEKEQ	VDEWTVETVE	VVNSLQQQPQ	AASPSVPEPH	SLTPVAQADP	360
LVRQRVQDL	MAQMGGPYNF	IQDSMLDFEN	QTLDPAIVSA	QPMNPTQNM	MPQLVCPVH	420
SESRLAQPQ	VPVQPEATQV	PLVSSSTSEGY	TASQPLYQPS	HATEQRPOKE	PIDQIQATIS	480
LNTDQTTASS	SLPAASQPQV	FQAGTSKPLH	SSGINVNAAP	FQSMQTVFNM	NAPVPPVNEP	540
ETLKQQNQYQ	ASYNQSFSSQ	PHQVEQTDLQ	QEQLQTVVGT	YHGSQDQPHQ	VTGNHQPPQ	600
QNTGFPRSSQ	PYYNSRGVSR	GGSRGARGLM	NGYRGFANGF	RGYDGYRPS	FSNTPNSGYT	660
QSQFSAPRDY	SGYQRDGYQQ	NFKRSGSGSG	PRGAPRGRGG	PPRPNRGMPQ	MNTQQQVN	717

SEQ ID NO: 314 moltype = AA length = 702
 FEATURE Location/Qualifiers
 source 1..702
 mol_type = protein
 note = Canis lupus familiaris
 organism = Canis lupus

SEQUENCE: 314

MPSATSLSGS	GSKSSGPPPP	SGSSGSEAAA	AAGAAGAAGA	GAAAPASQHP	ATGTGAVQTE	60
AMKQILGVID	KKLRNLEKKK	GKLDDYQERM	NKGERLNQDQ	LDAVSKYQEV	TNNLEFAKEL	120
QRSFMALSQD	IQKTIKKTAR	REQLMREEAE	QKRLKTVLEL	QYVLDKLGDD	EVRTDLKQGL	180
NGVPILSEEE	LSLLDEFYKL	ADPERDMSLR	LNEQYEHASI	HLWDLLEGKE	KSVCGTTYKA	240
LKEIVERVFP	SNYFDSSTHH	QNGLCSEEEA	ASAPTVEDQV	AEAEPEPAEE	YTEQSEVEST	300
EYVNRQFMAE	TQFSSGEKEQ	VDEWTVETVE	VVNSLQQQPQ	AASPSVPEPH	SLTPVAQADP	360
LVRQRVQDL	MAQMGGPYNF	IQDSMLDFEN	QTLDPAIVSA	QPMNPTQNM	MPQLVCPVH	420
SESRLAQPQ	VPVQPEATQV	PLVSSSTSEGY	TASQPLYQPS	HATEQRPOKE	PIDQIQATIS	480
LNTDQTTASS	SLPAASQPQV	FQAGTSKPLH	SSGINVNAAP	FQSMQTVFNM	NAPVPPVNEP	540
ETLKQQNQYQ	ASYNQSFSSQ	PHQVEQTDLQ	QEQLQTVVGT	YHGSQDQPHQ	VTGNHQPPQ	600
QNTGFPRSSQ	PYYNSRGVSR	GGSRGARGLM	NGYRGFANGF	RGYDGYRPS	FSNTPNSGYT	660
QSQFSAPRDY	SGYQRDGYQQ	NFKRSGSGSG	PRGAPRGNIL	WW		702

SEQ ID NO: 315 moltype = AA length = 679
 FEATURE Location/Qualifiers
 source 1..679
 mol_type = protein
 note = Canis lupus familiaris
 organism = Canis lupus

SEQUENCE: 315

MPSATSLSGS	GSKSSGPPPP	SGSSGSEAAA	AAGAAGAAGA	GAAAPASQHP	ATGTGAVQTE	60
AMKQILGVID	KKLRNLEKKK	GKLDDYQERM	NKGERLNQDQ	LDAVSKYQEV	TNNLEFAKEL	120
QRSFMALSQD	IQKTIKKTAR	REQLMREEAE	QKRLKTVLEL	QYVLDKLGDD	EVRTDLKQGL	180
NGVPILSEEE	LSLLDEFYKL	ADPERDMSLR	LNEQYEHASI	HLWDLLEGKE	KSVCGTTYKA	240
LKEIVERVFP	SNYFDSSTHH	QNGLCSEEEA	ASAPTVEDQV	AEAEPEPAEE	YTEQSEVEST	300
EYVNRQFMAE	TQFSSGEKEQ	VDEWTVETVE	VVNSLQQQPQ	AASPSVPEPH	SLTPVAQADP	360
LVRQRVQDL	MAQMGGPYNF	IQDSMLDFEN	QTLDPAIVSA	QPMNPTQNM	MPQLVCPVH	420
SESRLAQPQ	VPVQPEATQV	PLVSSSTSEGY	TASQPLYQPS	HATEQRPOKE	PIDQIQATIS	480
LNTDQTTASS	SLPAASQPQV	FQAGTSKPLH	SSGINVNAAP	FQSMQTVFNM	NAPVPPVNEP	540
ETLKQQNQYQ	ASYNQSFSSQ	PHQVEQTDLQ	QEQLQTVVGT	YHGSQDQPHQ	VTGNHQPPQ	600
QNTGFPRSSQ	PYYNSRGVSR	GGSRGARGLM	NGYRGFANGF	RGYDGYRPS	FSNTPNSGYT	660
QSQFSAPRDY	SGYQRGCRK					679

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SEQ ID NO: 316 moltype = AA length = 717
 FEATURE Location/Qualifiers
 source 1..717
 mol_type = protein
 note = Canis lupus familiaris
 organism = Canis lupus

SEQUENCE: 316

MPSATSLSGS	GSKSSGPPPP	SGSSGSEAAA	AAGAAGAAGA	GAAAPASQHP	ATGTGAVQTE	60
AMKQILGVID	KKLRNLEKKK	GKLDDYQERM	NKGERLNQDQ	LDAVSKYQEV	TNNLEFAKEL	120
QRSFMALSQD	IQKTIKKTAR	REQLMREEAE	QKRLKTVLEL	QYVLDKLGDD	EVRTDLKQGL	180
NGVPILSEEE	LSLLDEFYKL	ADPERDMSLR	LNEQYEHASI	HLWDLLEGKE	KSVCGTTYKA	240
LKEIVERVFQ	SNYFDSSTHH	QNGLCSEEEA	ASAPTVEDQV	AEAEPPEPAE	YTEQSEVEST	300
EYVNRQFMAE	TQFSSGKEQ	VDEWTVETVE	VVNSLQQQPQ	AASPSVPEPH	SLTPVAQADP	360
LVRQRVQDL	MAQMGGPYNF	IQDSMLDFEN	QTLDPAIVSA	QPMNPTQNMD	MPQLVCPVH	420
SESRLAQPNI	VPVQPEATQV	PLVSSSTSEGY	TASQPLYQPS	HATEQRPQKE	PIDQIQATIS	480
LNTDQTTASS	SLPAASQPQV	FQAGTSKPLH	SSGINVNAA	FQSMQTVFNM	NAPVPPVNEP	540
ETLKQQNQYQ	ASYNQSFSSQ	PHQVEQTDLQ	QEQLQTVVGT	YHGSQDQPHQ	VTGNHQPPQ	600
QNTGFPRSSQ	PYYNSRGVSR	GGSRGARGLM	NGYRGPFANG	RGGYDGYRPS	FSNTPNSGYT	660
QSQFSAPRDY	SGYQRDGYQQ	NFKRSGSQSG	PRGAPRGRG	PPRPNRGMPQ	MNTQQQVN	717

SEQ ID NO: 317 moltype = AA length = 708
 FEATURE Location/Qualifiers
 source 1..708
 mol_type = protein
 organism = Bos taurus

SEQUENCE: 317

MPSATSHSGS	GSKSSGPPPP	SGSSGNEAGA	GAAAPASQHP	MTGTGAVQTE	AMKQILGVID	60
KKLRNLEKKK	GKLDDYQERM	NKGERLNQDQ	LDAVSKYQEV	TNNLEFAKEL	QRSFMALSQD	120
IQKTIKKTAR	REQLMREEAE	QKRLKTVLEL	QYVLDKLGDD	EVRTDLKQGL	NGVPILSEEE	180
LSLLDEFYKL	ADPERDMSLR	LNEQYEHASI	HLWDLLEGKE	KPVCCTTYKA	LKEIVERVFQ	240
SNYFDSSTHH	QNGLCSEEEA	ASAPTVEDQA	AEAEPPEVVE	YTEQNEVEST	EYVNRQFMAE	300
TQFSSGKEQ	VDDWTVETVE	VVNSLQQQPQ	AASPSVPEPH	SLTPVAQADP	LVRQRVQDL	360
MAQMGGPYNF	IQDSMLDFEN	QTLDPAIVSA	QPMNPAQNMD	IPQLVCPVH	SESRLAQPNI	420
VSVQPEATQV	PLVSSSTSEGY	TASQPLYQPS	HATDQRPQKE	PIDQIQATIS	LNTDQTTASS	480
SLPAASQPQV	FQAGTSKPLH	SSGINVNAA	FQSMQTVFNM	NAPVPPVNEP	ETLKQQNQYQ	540
ASYNQSFSSQ	PHQVEQTELQ	QEQLQTVVGT	YHGSQDQPHQ	VTGNHQPPQ	QNTGFPRSSQ	600
PYYNSRGVSR	GGSRGARGLM	NGYRGPFANG	RGGYDGYRPS	FSTNTPNSGY	TQSQFSAPRD	660
YSGYQRDGYQ	QNFKRSGSQS	GPRGAPRGRG	GPPRPNRGMP	QMNTQQQVN		708

SEQ ID NO: 318 moltype = AA length = 638
 FEATURE Location/Qualifiers
 source 1..638
 mol_type = protein
 organism = Equus caballus

SEQUENCE: 318

MEGKLDDYQE	RMNKGERLNQ	DQLDAVSKYQ	EVTNNLEFAK	ELQRSFMALS	QDIQKTIKKT	60
ARREQLMREE	AEQKRLKTVL	ELQYVLDKLG	DEEVRTDLKQ	GLNGVPILSE	EELSLDEFY	120
KLADPVRDMS	LRLNEQYEH	SIHLWDLLEG	KEKSVCGTTY	KALREIVERV	FQSNYFDSTH	180
NHQNGLCEEE	EATSAPTAED	QGAEEPEPEP	EYTEQSEVE	STEYVNRQFM	AEAQFSGEKE	240
QVDEWTVETV	EVVNSLQQQP	QAASPSVPEP	HSLTPVAQAD	PLVRQRVQD	LMAQMGGPYN	300
FQDSMLDFE	NQTLDPAIVS	AQPMNPAQNM	DMPQLVCPVH	HAESRLAQPNI	QVPVQPEATQ	360
VPLVSSSTSEG	YTASQPLYQP	SHATEQRPQK	EPTDQIQATI	SLNTDQTTAS	SSLPAASQPQ	420
VFQAGTSKPL	HSSGINVNAA	PFQSMQTVFN	MNAPVPPVNE	PETLKQQNQY	QASYNQSFSS	480
PPHQVEQTEL	PQEQLQTVVG	TYHASQDQPH	QVTGNHQPPQ	QNTGFPRSSQ	QPYNSRGVSR	540
RGGSRGARGL	MNGYRGPFANG	FRGGYDGYRP	SFSNTPNSGY	TQSQFSAPRD	YSGYQRDGYQ	600
QNFKRSGSQS	GPRGAPRGRG	GPPRPNRGMP	QMNTQQQVN			638

SEQ ID NO: 319 moltype = AA length = 707
 FEATURE Location/Qualifiers
 source 1..707
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 319

MPSATSHSGS	GSKSSGPPPP	SGSSGSEAAA	GAAAPASQHP	ATGTGAVQTE	AMKQILGVID	60
KKLRNLEKKK	GKLDDYQERM	NKGERLNQDQ	LDAVSKYQEV	TNNLEFAKEL	QRSFMALSQD	120
IQKTIKKTAR	REQLMREEAE	QKRLKTVLEL	QYVLDKLGDD	DVRTDLKQGL	SGVPILSEEE	180
LSLLDEFYKL	VDPERDMSLR	LNEQYEHASI	HLWDLLEGKE	KPVCCTTYKA	LKEIVERVFQ	240
SNYFDSSTHH	QNGLCSEEEA	ASAPTVEDQV	AEAEPPEPAE	YTEQSEVEST	EYVNRQFMAE	300
TQFSSGKEQ	VDEWTVETVE	VVNSLQQQPQ	AASPSVPEPH	SLTPVAQSDP	LVRQRVQDL	360
MAQMGGPYNF	IQDSMLDFEN	QTLDPAIVSA	QPMNPTQNMD	MPQLVCPQVH	SESRLAQSNI	420
VPVQPEATQV	PLVSSSTSEGY	TASQPLYQPS	HATEQRPQKE	PMDQIQATIS	LNTDQTTASS	480
SLPAASQPQV	FQAGTSKPLH	SSGINVNAA	FQSMQTVFNM	NAPVPPANEP	ETLKQQSQYQ	540
ATYNQSFSSQ	PHQVEQTELQ	QDQLQTVVGT	YHGSQDQPHQ	VPGNHQPPQ	QNTGFPRSSQ	600
PYYNSRGVSR	GGSRGARGLM	NGYRGPFANG	RGGYDGYRPS	FSNTPNSGY	QSQFTAPRDY	660
SGYQRDGYQQ	NFKRSGSQSG	PRGAPRGRG	PPRPNRGMPQ	MNTQQQVN		707

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SEQ ID NO: 320 moltype = AA length = 707
FEATURE Location/Qualifiers
source 1..707
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 320

MPSATSHSGS	GSKSSGPPPP	SGSSGSEAAA	GAAAPASQHP	ATGTGAVQTE	AMKQILGVID	60
KKLRNLEKKK	GKLDYQERM	NKGERLNQDQ	LDAVSKYQEV	TNNLEFAKEL	QRSFMALSQD	120
IQKTIKKTAR	REQLMREEAE	QKRLKTVLEL	QYVLDKLGDD	DVRTDLKQGL	SGVPILSEEE	180
LSLLDEFYKL	VDPERDMSLR	LNEQYEHASI	HLWDLLEGKE	KPVCCTTYKA	LKEIVERVFQ	240
SNYFDSSTHSH	QNGLCEEEEA	ASAPTVEDQV	AEAEPEPAEE	YTEQSEVEST	EYVNRQFMAE	300
TQFSSGEKEQ	VDEWTVETVE	VVNSLQQQPQ	AASPSVPEPH	SLTPVAQSDP	LVRQRVQDL	360
MAQMCGPYNF	IQDSMLDFEN	QTLDPAIVSA	QPMNPTQNM	MPQLVCPQVH	SESRLAQSNQ	420
VPVQPEATQV	PLVSTSEGY	TASQPLYQPS	HATEQRPQKE	PMDQIQATIS	LNTDQTTASS	480
SLPAASQPVQ	FQAGTSKPLH	SSGINVNAAP	FQSMQTVFNM	NAPVPPANEP	ETLKQSSQYQ	540
ATYNQSFSSQ	PHQVEQTELQ	QDQLQTVVGT	YHGSQDQPHQ	VPGNHQPPQ	QNTGFPRSSQ	600
PYYNSRGVSR	GGSRGARGLM	NGYRGPANGF	RGGYDGYRPS	FSNTPNSGYS	QSQTAPRDY	660
SGYQRDGYQQ	NFKRSGSQSG	PRGAPRGRGG	PPRPNRGMFQ	MNTQQVN		707

SEQ ID NO: 321 moltype = AA length = 698
FEATURE Location/Qualifiers
source 1..698
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 321

MPSATSHSGS	GSKSSGPPPP	SGSSGSEAAA	GAAAPASQHP	ATGTGAVQTE	AMKQILGVID	60
KKLRNLEKKK	GKLDYQERM	NKGERLNQDQ	LDAVSKYQEV	TNNLEFAKEL	QRSFMALSQD	120
IQKTIKKTAR	REQLMREEAE	QKRLKTVLEL	QYVLDKLGDD	DVRTDLKQGL	SGVPILSEEE	180
LSLLDEFYKL	VDPERDMSLR	LNEQYEHASI	HLWDLLEGKE	KPVCCTTYKA	LKEIVERVFQ	240
SNYFDSSTHSH	QNGLCEEEEA	ASAPTVEDQV	AEAEPEPAEE	YTEQSEVEST	EYVNRQFMAE	300
TQFSSGEKEQ	VDEWTVETVE	VVNSLQQQPQ	AASPSVPEPH	SLTPVAQSDP	LVRQRVQDL	360
MAQMCGPYNF	IQTLDPAIVS	AQPMNPTQNM	DMPQLVCPQV	HSESRLAQSN	QVPVQPEATQ	420
VPLVSTSEGY	YTASQPLYQP	SHATEQRPQK	EPMDQIQATI	SLNTDQTTAS	SSLPAAASQPQ	480
VQAGTSKPL	HSSGINVNAA	PFQSMQTVFN	MNAPVPPANE	PETLKQSSQY	QATYNQSFSS	540
QPHQVEQTEL	QDQLQTVVGT	TYHGSQDQPH	QVPGNHQPPQ	QNTGFPRSSQ	QPYNSRGVSR	600
RGSRGARGL	MNGYRGPANG	FRGGYDGYRP	SFSNTPNSGY	SQSQTAPRD	YSQYQRDGYQ	660
QNFKRSGSQS	GPRGAPRGRG	GPPRPNRGMF	QMNTQQVN			698

SEQ ID NO: 322 moltype = AA length = 692
FEATURE Location/Qualifiers
source 1..692
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 322

MPSATSHSGS	GSKSSGPPPP	SGSSGSEAAA	GAAAPASQHP	ATGTGAVQTE	AMKQILGVID	60
KKLRNLEKKK	GKLDYQERM	NKGERLNQDQ	LDAVSKYQEV	TNNLEFAKEL	QRSFMALSQD	120
IQKTIKKTAR	REQLMREEAE	QKRLKTVLEL	QYVLDKLGDD	DVRTDLKQGL	SGVPILSEEE	180
LSLLDEFYKL	VDPERDMSLR	LNEQYEHASI	HLWDLLEGKE	KPVCCTTYKA	LKEIVERVFQ	240
SNYFDSSTHSH	QNGLCEEEEA	ASAPTVEDQV	AEAEPEPAEE	YTEQSEVEST	EYVNRQFMAE	300
TQFSSGEKEQ	VDEWTVETVE	VVNSLQQQPQ	AASPSVPEPH	SLTPVAQSDP	LVRQRVQDL	360
MAQMCGPYNF	IQDSMLDFEN	QTLDPAIVSA	QPMNPTQNM	MPQLVCPQVH	SESRLAQSNQ	420
VPVQPEATQV	PLVSTSEGY	TASQPLYQPS	HATEQRPQKE	PMDQIQATIS	LNTDQTTASS	480
SLPAASQPVQ	FQAGTSKPLH	SSGINVNAAP	FQSMQTVFNM	NAPVPPANEP	ETLKQSSQYQ	540
ATYNQSFSSQ	PHQVEQTELQ	QDQLQTVVGT	YHGSQDQPHQ	VPGNHQPPQ	QNTGFPRSSQ	600
PYYNSRGVSR	GGSRGARGLM	NGYRGPANGF	RGGYDGYRPS	FSNTPNSGYS	QSQTAPRDY	660
SGYQRDGYQQ	NFKRSGSQSG	PRGAPRGNIL	WW			692

SEQ ID NO: 323 moltype = AA length = 692
FEATURE Location/Qualifiers
source 1..692
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 323

MPSATSHSGS	GSKSSGPPPP	SGSSGSEAAA	GAAAPASQHP	ATGTGAVQTE	AMKQILGVID	60
KKLRNLEKKK	GKLDYQERM	NKGERLNQDQ	LDAVSKYQEV	TNNLEFAKEL	QRSFMALSQD	120
IQKTIKKTAR	REQLMREEAE	QKRLKTVLEL	QYVLDKLGDD	DVRTDLKQGL	SGVPILSEEE	180
LSLLDEFYKL	VDPERDMSLR	LNEQYEHASI	HLWDLLEGKE	KPVCCTTYKA	LKEIVERVFQ	240
SNYFDSSTHSH	QNGLCEEEEA	ASAPTVEDQV	AEAEPEPAEE	YTEQSEVEST	EYVNRQFMAE	300
TQFSSGEKEQ	VDEWTVETVE	VVNSLQQQPQ	AASPSVPEPH	SLTPVAQSDP	LVRQRVQDL	360
MAQMCGPYNF	IQDSMLDFEN	QTLDPAIVSA	QPMNPTQNM	MPQLVCPQVH	SESRLAQSNQ	420
VPVQPEATQV	PLVSTSEGY	TASQPLYQPS	HATEQRPQKE	PMDQIQATIS	LNTDQTTASS	480
SLPAASQPVQ	FQAGTSKPLH	SSGINVNAAP	FQSMQTVFNM	NAPVPPANEP	ETLKQSSQYQ	540
ATYNQSFSSQ	PHQVEQTELQ	QDQLQTVVGT	YHGSQDQPHQ	VPGNHQPPQ	QNTGFPRSSQ	600
PYYNSRGVSR	GGSRGARGLM	NGYRGPANGF	RGGYDGYRPS	FSNTPNSGYS	QSQTAPRDY	660
SGYQRDGYQQ	NFKRSGSQSG	PRGAPRGNIL	WW			692

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SEQ ID NO: 324      moltype = AA  length = 702
FEATURE            Location/Qualifiers
source              1..702
                   mol_type = protein
                   organism = Gallus gallus

SEQUENCE: 324
MPSATNGTMA SSSGKAGPGG NEQAPAAAAA APQASGGGSI SVQTEAMKQI LGVIDKKLRN 60
LEKKSKLDD YQERMNKGGER LNQDQLDAVS KYQEVNMLE FAKELQRSFM ALSQDIQKTI 120
KKTARREQLM REEAEQKRLK TVLELQFILD KLGDDVRSR LKQGSNGVPV LTEEELTMLD 180
EFYKLVYPER DMNMLNEQY EQASVHLWDL LEGKEKPVCG TTYKALKEVV ERILQTSYFD 240
STHNHQNGLC EEEEAAPTPA VEDTVAEAE DPAAEFTEPT EVESLEYVNR QFMAETQFSS 300
SEKEQVDEWT VETVEVNSL QQQTQATSP VPEPHTLTV AQADPLVRRQ RVQDLMAQMQ 360
GPYNFMQDSM LEFENQTLDP AIVSAQPMNP AQNLDMQMV CPPVHTESRL AQPQVVPVQP 420
EATQVPLVSS TSEGYTASQP MYQPSHTTEQ RPQKESIDQI QASMSLNADQ TPSSSSLPTA 480
SQPQVFQAGS SKPLHSSGIN VNAAPFQSMQ TVFNMNAPVP PVNEPEALKQ QNQYQASYNQ 540
SFSNQPHQVE QSDLQQEQLQ TVVGTYHGSP DQTHQVAGNH QPPQQTNTGF PRNSQPYNS 600
RGVSRGSGRG TRGLMNGYRG PANGFRGGYD GYRPSFSNTP NSGYTQPQFN APRDYSNYQR 660
DGYQQNFKRG SGQSGPRGAP RGRGGPPRPN RGMPQMNAQQ VN 702

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1. A medicament for treatment and/or prevention of cancer, comprising an antibody or a fragment thereof having an immunological reactivity with CAPRIN-1 protein, and a MAPK (mitogen-activated protein kinase) pathway inhibitor together or separately in combination.

2. The medicament according to claim 1, wherein the MAPK pathway inhibitor comprises a MEK inhibitor, a RAS inhibitor, a RAF inhibitor, an ERK inhibitor, or a combination of any two or more thereof.

3. The medicament according to claim 2, wherein the MEK inhibitor is at least one selected from the group consisting of trametinib, cobimetinib, selumetinib, refametinib, SL-327, and their derivatives.

4. The medicament according to claim 2, wherein the RAS inhibitor is at least one selected from the group consisting of sotorasib, lonafarnib, salirasib, and their derivatives.

5. The medicament according to claim 2, wherein the RAF inhibitor is at least one selected from the group consisting of dabrafenib, ZM 336372, and their derivatives.

6. The medicament according to claim 2, wherein the ERK inhibitor is at least one selected from the group consisting of bosutinib, SCH772984, and their derivatives.

7. The medicament according to claim 1, wherein the antibody or the fragment thereof has an immunological reactivity with CAPRIN-1 protein having an amino acid sequence shown by any one of the even numbered SEQ ID NOs: 2 to 30, or an amino acid sequence having 80% or more sequence identity with the amino acid sequence.

8. The medicament according to claim 1, wherein the antibody or the fragment thereof has an immunological reactivity with an extracellular region of CAPRIN-1 protein present on a cancer cell surface.

9. The medicament according to claim 1, wherein the antibody or the fragment thereof has an immunological reactivity with a partial polypeptide of CAPRIN-1 protein, the partial polypeptide having an amino acid sequence shown by any one of SEQ ID NOs: 31 to 35, 296 to 299, 308 and 309, or an amino acid sequence having 80% or more sequence identity with the amino acid sequence.

10. The medicament according to claim 1, wherein the antibody is a monoclonal antibody or a polyclonal antibody.

11. The medicament according to claim 1, wherein the antibody or the fragment thereof is any one of the following (A) to (M):

(A) an antibody or a fragment thereof comprising a heavy-chain variable region comprising complementarity determining regions of SEQ ID NOs: 36, 37 and 38 (CDR1, CDR2 and CDR3, respectively) and a light-chain variable region comprising complementarity determining regions of SEQ ID NOs: 40, 41 and 42 (CDR1, CDR2 and CDR3, respectively), and having an immunological reactivity with CAPRIN-1 protein;

(B) an antibody or a fragment thereof comprising a heavy-chain variable region comprising complementarity determining regions of SEQ ID NOs: 44, 45 and 46 (CDR1, CDR2 and CDR3, respectively) and a light-chain variable region comprising complementarity determining regions of SEQ ID NOs: 48, 49 and 50 (CDR1, CDR2 and CDR3, respectively), and having an immunological reactivity with CAPRIN-1 protein;

(C) an antibody or a fragment thereof comprising a heavy-chain variable region comprising complementarity determining regions of SEQ ID NOs: 52, 53 and 54 (CDR1, CDR2 and CDR3, respectively) and a light-chain variable region comprising complementarity determining regions of SEQ ID NOs: 56, 57 and 58 (CDR1, CDR2 and CDR3, respectively), and having an immunological reactivity with CAPRIN-1 protein;

(D) an antibody or a fragment thereof comprising a heavy-chain variable region comprising complementarity determining regions of SEQ ID NOs: 60, 61 and 62 (CDR1, CDR2 and CDR3, respectively) and a light-chain variable region comprising complementarity determining regions of SEQ ID NOs: 64, 65 and 66 (CDR1, CDR2 and CDR3, respectively), and having an immunological reactivity with CAPRIN-1 protein;

(E) an antibody or a fragment thereof comprising a heavy-chain variable region comprising complementarity determining regions of SEQ ID NOs: 170, 171 and 172 (CDR1, CDR2 and CDR3, respectively) and a light-chain variable region comprising complementarity determining regions of SEQ ID NOs: 173, 174 and 175 (CDR1, CDR2 and CDR3, respectively), and having an immunological reactivity with CAPRIN-1 protein;

- (F) an antibody or a fragment thereof comprising a heavy-chain variable region comprising complementarity determining regions of SEQ ID NOs: 176, 177 and 178 (CDR1, CDR2 and CDR3, respectively) and a light-chain variable region comprising complementarity determining regions of SEQ ID NOs: 179, 180 and 181 (CDR1, CDR2 and CDR3, respectively), and having an immunological reactivity with CAPRIN-1 protein;
- (G) an antibody or a fragment thereof comprising a heavy-chain variable region comprising complementarity determining regions of SEQ ID NOs: 182, 183 and 184 (CDR1, CDR2 and CDR3, respectively) and a light-chain variable region comprising complementarity determining regions of SEQ ID NOs: 185, 186 and 187 (CDR1, CDR2 and CDR3, respectively), and having an immunological reactivity with CAPRIN-1 protein;
- (H) an antibody or a fragment thereof comprising a heavy-chain variable region comprising complementarity determining regions of SEQ ID NOs: 188, 189 and 190 (CDR1, CDR2 and CDR3, respectively) and a light-chain variable region comprising complementarity determining regions of SEQ ID NOs: 191, 192 and 193 (CDR1, CDR2 and CDR3, respectively), and having an immunological reactivity with CAPRIN-1 protein;
- (I) an antibody or a fragment thereof comprising a heavy-chain variable region comprising complementarity determining regions of SEQ ID NOs: 146, 147 and 148 (CDR1, CDR2 and CDR3, respectively) and a light-chain variable region comprising complementarity determining regions of SEQ ID NOs: 149, 150 and 151 (CDR1, CDR2 and CDR3, respectively), and having an immunological reactivity with CAPRIN-1 protein;
- (J) an antibody or a fragment thereof comprising a heavy-chain variable region comprising complementarity determining regions of SEQ ID NOs: 272, 273 and 274 (CDR1, CDR2 and CDR3, respectively) and a light-chain variable region comprising complementarity determining regions of SEQ ID NOs: 275, 276 and 277 (CDR1, CDR2 and CDR3, respectively), and having an immunological reactivity with CAPRIN-1 protein;
- (K) an antibody or a fragment thereof comprising a heavy-chain variable region comprising complementarity determining regions of SEQ ID NOs: 290, 291 and 292 (CDR1, CDR2 and CDR3, respectively) and a light-chain variable region comprising complementarity determining regions of SEQ ID NOs: 293, 294 and 295 (CDR1, CDR2 and CDR3, respectively), and having an immunological reactivity with CAPRIN-1 protein;
- (L) an antibody or a fragment thereof comprising a heavy-chain variable region comprising complementarity determining regions of SEQ ID NOs: 301, 302 and 303 (CDR1, CDR2 and CDR3, respectively) and a light-chain variable region comprising complementarity determining regions of SEQ ID NOs: 305, 306 and 307 (CDR1, CDR2 and CDR3, respectively), and having an immunological reactivity with CAPRIN-1 protein; and
- (M) an antibody or a fragment thereof comprising a heavy-chain variable region comprising complementarity determining regions of SEQ ID NOs: 134, 135 and 136 (CDR1, CDR2 and CDR3, respectively) and a light-chain variable region comprising complementarity determining regions of SEQ ID NOs: 137, 138 and 139 (CDR1, CDR2 and CDR3, respectively), and having an immunological reactivity with CAPRIN-1 protein.
12. The medicament according to claim 1, wherein the antibody or the fragment thereof is any one of the following (a) to (al):
- (a) an antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 39 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 43;
 - (b) an antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 47 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 51;
 - (c) an antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 55 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 59;
 - (d) an antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 63 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 67;
 - (e) an antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 68 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 69;
 - (f) an antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 70 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 71;
 - (g) an antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 72 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 73;
 - (h) an antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 74 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 75;
 - (i) an antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 76 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 77;
 - (j) an antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 78 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 79;
 - (k) an antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 80 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 81;

- [illegible]

(ak) an antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 132 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 133; and

(al) an antibody or a fragment thereof comprising a heavy-chain variable region comprising the amino acid sequence of SEQ ID NO: 300 and a light-chain variable region comprising the amino acid sequence of SEQ ID NO: 304.

13. The medicament according to claim **1**, wherein the antibody is a human antibody, a humanized antibody, a chimeric antibody or a single chain antibody.

14. The medicament according to claim **1**, wherein the cancer is a cancer expressing CAPRIN-1 protein on a cell membrane surface.

15. The medicament according to claim **1**, wherein the cancer is melanoma, lung cancer, thyroid cancer, colon cancer, prostate cancer, ovarian cancer, pancreatic cancer, kidney cancer, breast cancer, gastric cancer, bile duct cancer, renal cell cancer, Hodgkin's lymphoma, head and neck cancer, mesothelioma, colorectal cancer, esophageal cancer, gastroesophageal cancer, hepatocellular cancer, glioblastoma, urothelial cancer, urinary bladder cancer, uterus can-

cer, primary central nervous system lymphoma, primary testicular lymphoma, biliary tract cancer, brain tumor, leukemia, lymphoma, liver cancer, sarcoma, fibrosarcoma, mastocytoma, adrenal cortex cancer, Ewing's tumor, multiple myeloma, testicle cancer, basal cell cancer, Paget's disease or skin cancer.

16. An agent increasing drug efficacy of a pharmaceutical composition for treatment and/or prevention of cancer comprising an antibody or a fragment thereof having an immunological reactivity with CAPRIN-1 protein as an active ingredient, wherein the agent comprises a MAPK pathway inhibitor as an active ingredient.

17. An agent increasing drug efficacy of a pharmaceutical composition for treatment and/or prevention of cancer comprising a MAPK pathway inhibitor as an active ingredient, wherein the agent comprises an antibody or a fragment thereof having an immunological reactivity with CAPRIN-1 protein as an active ingredient.

18. A method for treating and/or preventing cancer, comprising administering an antibody or a fragment thereof having an immunological reactivity with CAPRIN-1 protein, and a MAPK pathway inhibitor together or separately to a subject.

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